

Coupled Slug & Hydrate Prediction in a Deepwater Crude-Oil Subsea Tie-back

A comprehensive flow-assurance case study with the SHCT transient solver — case definition, model equations, inputs, all generated outputs, validation and calibration

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Solver: SHCT — Slug–Hydrate Coupled-Transient multiphase flow-assurance solver (transient, compositional, probabilistic).

Scope: this single document consolidates EVERYTHING produced for the study — the engineering background and problem, the well-defined case study, every governing equation and closure used to build the solver, the numerical method, the contribution to knowledge, a like-for-like comparison with existing commercial and academic solvers, the complete input-data deck, and every generated output (all metrics, CSV tables, per-column graphs of every CSV, and the full chart/curve/contour/map gallery) for the three engineering scenarios, plus the published-data validation and the calibration, with all sources recorded. Nothing is left out.

Figure/chart style: no black and no dark colours are used in any figure, curve, contour or map; all backgrounds are white and all ink is a medium, legible hue.

Executive summary

A 32 km, 10.75-inch deepwater subsea tie-back carrying a ~30° API medium crude oil with 70 % water cut over a cold (4 °C), strongly undulating seabed and a steel catenary riser is analysed for the two flow-assurance threats that dominate such systems and that physically reinforce one another: hydrodynamic / terrain / severe-riser SLUGGING and gas-HYDRATE formation. The SHCT solver runs the full transient, compositional, probabilistic prediction end-to-end for three scenarios.

- As-operated (degraded insulation, no inhibitor): the line is BOTH slug- and hydrate-critical — sustained coupling number $\Phi_{SH} = 1.11$ at 23.1 km, plug probability 0%, P50 time-to-plug n/a h, peak wall deposit 9 mm, max subcooling 6.3 °C.
- Inhibitor demand the model predicts to clear that risk: MEG ≈ 31 wt% (33886 L/h); under-inhibited length 12.3 km.
- CONDITION ON THE ABOVE — the subcooling exponent. Growth goes as ΔT_{sub}^n . This study ships $n = 1.5$, Mori (2001) $\Delta T^{(3/2)}$, convection-controlled growth. Bhattacharjee et al. (2021), Chem. Eng. Sci. 234:116417 report that $n = 1.5\text{--}2.5$ fits experimental film-growth data (Peng 2007 with Mori gives $5/2$), over a 2–12 °C subcooling range that brackets this case. This study's own one-at-a-time sweep gives max Φ_{SH} 0.58 $\rightarrow$ 1.80 $\rightarrow$ 3.18 $\rightarrow$ 5.61 and plug probability 0 $\rightarrow$ 0 $\rightarrow$ 0.50 $\rightarrow$ 0.67 as n goes 1.0 $\rightarrow$ 1.5 $\rightarrow$ 1.75 $\rightarrow$ 2.0. The as-operated verdict therefore holds at the FLOOR of the published range and not above it: across 1.5–2.5 the field exceeds $\Phi_{crit} = 1.08$ at every value, and from $n = 1.75$ the line plugs in half to two-thirds of realisations. The cited exponents are for film growth at a GAS–LIQUID interface while this model deposits at the WALL, so no value in 1.5–2.5 is established for this geometry and 1.5 is the floor of the range. Separately, the verdict is conditional on the SIMULATED WINDOW: at $n = 1.5$ the coupling number implies an equilibrium deposit of 35.2 mm against a consolidation threshold of 22.9 mm, while 48 h reaches only 8.8 mm — a quarter of that equilibrium — so the deposit is still climbing when the run ends.
- Unplanned shut-in: essentially no safe window — no-touch time ≈ 0 h.
- Engineered fix (restored insulation $U_{eff} = 3.45$ W/m²K + continuous MEG): the subcooling is removed, the plug probability falls to 0% and 54 h of no-touch time is restored — the model used as a design tool.
- Numerics are mass-consistent and stable: liquid mass error 1.76e-15, gas 3.51e-17, 0 solver fallbacks.

Scope and standing of these numbers. Read the results below with three limitations in view. (1) The field is a representative industrial archetype built from literature-typical values, not proprietary operator data. (2) The criticality threshold is DERIVED, not assumed: deposition and slug scouring compete continuously, Φ_{SH} drives no term in the solver, and the balance $d(\delta)/dt = 0$ gives $\delta_{eq} = \Phi_{SH} \cdot \delta_{ref}$ with $\delta_{ref} = f_{wall} \cdot D / (4 \cdot C \cdot k_{ero})$, so runaway begins where that equilibrium passes the consolidation restriction, at $\Phi_{crit} = 2 \cdot C \cdot k_{ero} \cdot r_{consol} / f_{wall} = 1.08$. That it falls within 8 % of unity is a result of the kinetics rather than a definition. It has still not been MEASURED, so it remains a proposed criterion rather than a validated one — but it is now falsifiable in two independent ways (§ limitations). (3) The kinetic and coupling constants C , n and k_{g0} , and the slug-frequency floor $f_{slug,0}$, are literature-typical defaults or numerical guards fitted to no dataset, so the absolute magnitudes reported here — the Φ_{SH} field, the time-to-plug and the required MEG dose in

particular — are model outputs, not calibrated predictions. Section 8.3 reports how far each of them moves when those FOUR constants are swept across their plausible ranges. Note in particular that a RUNNING MAXIMUM of Φ_{SH} over the window is attained during flow startup, while the slug frequency is still at its floor, and is therefore set by that floor rather than by the operating state; the SUSTAINED (time-median) value quoted above is what describes the line as run. What the study demonstrates is the coupled mechanism and the design workflow; what it does not yet demonstrate is quantitative accuracy against a specific line.

1. Background to the case study

Deepwater subsea tie-backs transport unprocessed well fluids — oil, produced water and associated gas — over tens of kilometres of cold seabed (typically 4 °C) before reaching a host facility. Two production-chemistry / multiphase-flow threats dominate the integrity of such lines:

1.1 Hydrodynamic, terrain and severe-riser slugging

In multiphase pipe flow the gas and liquid self-organise into intermittent SLUGS — long liquid plugs separated by gas pockets. On a long, near-horizontal, undulating seabed the low spots accumulate liquid and shed it as terrain slugs; at the riser base, liquid fallback and gas blow-through produce large, low-frequency SEVERE slugs. Slugs impose cyclic structural loads, flood the topside slug-catcher, and force unstable operation.

1.2 Gas-hydrate formation

Natural-gas components (methane, ethane, propane...) combine with free water at high pressure and low temperature to form ICE-LIKE crystalline hydrates. In a cold deepwater line the operating point sits well inside the hydrate-stability region; at this line's 70 % water cut there is ample free water. Hydrates can deposit on the cold wall, consolidate and grow a PLUG that blocks the line — the single most feared flow-assurance failure, slow and dangerous to remediate.

1.3 Why the two threats must be solved together

Slugging and hydrates are not independent. The same cold, gassy, water-wet, intermittent flow that drives slugging also drives hydrates; and the two interact at the wall: slug passage scours and re-warms the wall (suppressing deposition), while between slugs the cold wall lets hydrate deposit and consolidate. Whether a line plugs or stays open is therefore set by a COMPETITION between hydrate-formation rate and slug-renewal rate. A medium-crude tie-back of this geometry and fluid is the textbook case in which BOTH threats are simultaneously active, which is exactly why it is chosen here.

2. Problem statement and how the problem is solved

2.1 Problem statement

Existing tools predict slugging OR hydrates well, but treat them as separate analyses run in different software, with the slug-hydrate WALL interaction left to engineering judgement. There is no single, transient, probabilistic predictor that resolves the competition between hydrate growth and slug scouring along the whole route and in time, and returns an actionable, quantified, UNCERTAINTY-aware plugging risk together with the slug loads, the inhibitor demand and the mitigation design.

The engineering question for THIS asset: given the geometry, fluid and operating conditions, (a) where and when does the line slug and how big are the slugs; (b) where and when does it enter the hydrate region and form a consolidating wall deposit; (c) what is the probabilistic time-to-plug; (d) how much inhibitor (and what insulation) removes the risk; and (e) how long is the no-touch time after a shut-in?

2.2 How the problem is solved — the SHCT approach

SHCT integrates, in time, a coupled system of conservation PDEs for multiphase flow, heat transfer and hydrate formation on the ACTUAL terrain, closed by published flow-assurance correlations and a compositional Peng–Robinson PVT engine, and run as a Monte-Carlo ensemble for genuine probabilistic output. The slug–hydrate interaction is made explicit through a single dimensionless coupling number Φ_{SH} — the ratio of the wall hydrate-formation tendency to the slug-renewal rate — mapped in space and time. The full set of governing equations and closures is given in Section 4; the numerical method in Section 5.

- Resolve flow regime, holdup, slug frequency and slug length along the route (hydraulics).
- Track P, T and the hydrate-equilibrium temperature $T_{eq} \rightarrow$ local subcooling ΔT_{sub} .
- Form hydrate by stochastic nucleation + growth in the bulk (slurry) and on the wall (deposit).
- Combine the two through $\Phi_{SH}(x,t)$, the equilibrium deposit thickness in units of $\delta_{ref} = f_{wall} \cdot D / (4 \cdot C \cdot k_{ero})$; above the derived $\Phi_{crit} = 1.08$ that equilibrium exceeds the consolidation restriction, the deposit locks and growth runs away.
- Repeat over a Monte-Carlo ensemble $\rightarrow$ P10/P50/P90 time-to-plug and design margins.
- Invert for the required MEG dose and the insulation that removes the risk (design mode).

3. The case study — clearly defined

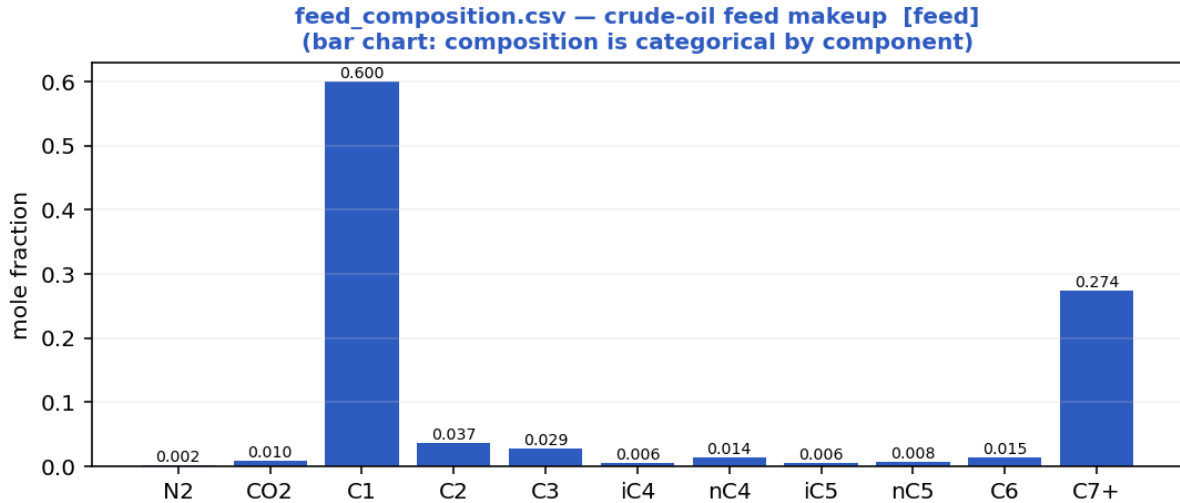
A representative deepwater West-Africa-style medium-crude-oil tie-back. The geometry, fluid and operating values are realistic, self-consistent values typical of such developments in the open literature; they are NOT proprietary operator data. The PHYSICS and PREDICTIONS are produced by the real solver, and the hydrate thermodynamics and the hydraulic closures are validated against published data (Sections 7–8).

3.1 Asset and geometry

- Route length: 32 km step-out flowline + steel catenary riser (SCR).
- Internal diameter: 0.2545 m (10.75-inch nominal carbon-steel flowline).
- Water depth at the riser base: ~1100 m; cold seabed at 4 °C.
- Terrain: long, strongly undulating seabed (multiple low spots $\rightarrow$ terrain slugging) climbing into a steep SCR ($\rightarrow$ severe-riser slugging).

3.2 Fluid — a medium crude oil ($\approx 30^\circ$ API black oil)

Moderate-GOR live crude with a substantial heavy C7+ tail; the associated gas (C1 ≈ 43 mol%) liberates along the cold line and, with the long near-horizontal step-out, drives slugging, while the 70 % water cut plus water-wet gas in the cold wall drives hydrates.



Feed composition — medium-crude makeup (mole fraction per component) that feeds the Peng–Robinson flash.

component	mol_fraction
N2	0.0020
CO2	0.0098
C1	0.6000
C2	0.0368
C3	0.0285
iC4	0.0059
nC4	0.0138
iC5	0.0064
nC5	0.0079
C6	0.0147
C7+	0.2743

3.3 The three engineering scenarios

- (A) As-operated normal production — degraded / water-flooded wet insulation (behaves close to bare steel, $U \approx 22 \text{ W/m}^2\text{K}$), NO inhibitor: the high-risk prediction.
- (B) Unplanned shut-in — the line cools toward the seabed; the no-touch (cooldown) time before hydrate onset is measured from the transient.
- (C) Engineered mitigation — restored multi-layer insulation (steel + syntactic foam + coating → low U_{eff}) plus continuous MEG: the model used as a DESIGN tool.

4. Model equations used to build the solver

The SHCT solver integrates, in time, a system of coupled partial-differential equations for multiphase flow, heat transfer and hydrate formation on arbitrary terrain, closed by published flow-assurance correlations and a compositional Peng–Robinson PVT engine. Every equation actually used to build the solver is listed below, grouped by role, with its symbolic form, a one-line description and its source. Symbols: α_l/α_g liquid/gas holdup, A flow area, $v_l/v_g/u_m$ phase & mixture velocities, j mixture (superficial) velocity, p pressure, T temperature, T_{eq} hydrate-equilibrium temperature, ΔT_{sub} subcooling, ϕ bulk hydrate fraction, δ wall-deposit thickness, f_{slug} slug frequency, a_i interfacial area, θ inclination, ρ density, μ viscosity, U_{eff} effective heat-transfer coefficient.

A. Governing transient conservation PDEs (finite-volume, CFL-adaptive)

Liquid-holdup transport (drift-flux kinematic wave)

$$\partial(\alpha_l \cdot A)/\partial t + \partial(\alpha_l \cdot v_l \cdot A)/\partial x = - S_l$$

Volume-conservative liquid continuity, discretised by a conservative finite-volume scheme with signed upwinding (optionally 2nd-order TVD). S_l is the liquid (water) volume sink to hydrate formation. Holdup is recovered as $\alpha_l = (\alpha_l \cdot A)/A$ each step.

Source / basis: Conservative FV; Bendiksen drift-flux closure.

Gas-mass continuity (conservative, mass-consistent)

$$\partial(\rho_g \cdot \alpha_g \cdot A)/\partial t + \partial(\rho_g \cdot V_{sg} \cdot A)/\partial x = - \dot{m}_{(gas \rightarrow hydrate)}$$

Solved alongside the liquid so liquid AND gas mass balance to ~0%. The mass-consistent superficial gas velocity $V_{sg} = (\rho_{g,in} \cdot q_{g,in})/(\rho_g \cdot A)$ makes the mass flux $\rho_g \cdot V_{sg} \cdot A$ equal to the inlet gas mass rate along the line.

Source / basis: Conservative FV gas continuity.

Mixture momentum → implicit pressure (default engine)

$$u_m = U_o - C_u \cdot \partial p / \partial x ; \quad R(p^{n+1} - p^n) + \partial / \partial x [U_o - C_u \cdot \partial p^{n+1} / \partial x] = 0$$

Transient mixture momentum with the pressure gradient and wall drag treated implicitly; eliminating velocity gives a tridiagonal (Poisson-type) pressure equation solved each step (Thomas algorithm). Predictor $u^* = u_m - dt \cdot (\text{advection} + g \cdot \sin \theta)$; wall-drag rate $\beta = f/(2D) \cdot |u_m|$; $U_o = u^*/(1 + dt \cdot \beta)$; compressible storage $R = \alpha_g/(p \cdot dt)$. Inlet pressure Dirichlet (pinned in-system), outlet through-flux (rate control).

Source / basis: Implicit-pressure mixture-momentum; drift-flux slip closure.

Full two-fluid momentum (alternative engine)

**two independent phase momenta + interfacial drag, shared pressure;
slip emerges from balance**

RELAP/TRACE-style semi-implicit: advection+gravity explicit; pressure gradient, wall drag and stiff interfacial drag implicit. A 2×2 per-cell velocity system → tridiagonal pressure; an interfacial-pressure correction keeps the characteristics real (well-posed).

Source / basis: Two-fluid model with interfacial-pressure regularisation.

Energy transport (advection + seabed loss + latent + Joule–Thomson)

$$\partial T / \partial t + j \cdot \partial T / \partial x = - U_{eff} \cdot (4/D) \cdot (T - T_{sink}) / (\rho_m \cdot c_p) + q_{latent} + q_{JT}$$

Signed-upwind (optionally TVD) advection; the stiff wall-loss term is implicit (backward Euler), with a Heun 2nd-order corrector on advection. $\beta_{\text{loss}} = U_{\text{eff}} \cdot 4 / (D \cdot \rho_m \cdot c_p)$. $q_{\text{latent}} = L_{\text{hyd}} \cdot \rho_{\text{hyd}} \cdot R_{g,\text{bulk}} / (\rho_m \cdot c_p)$ (exothermic bulk hydrate, self-limiting); $q_{\text{JT}} = \mu_{\text{JT}} \cdot (1 - \alpha_l) \cdot j \cdot (\partial p / \partial x)$ (real-gas cooling on expansion).

Source / basis: Transient energy balance; implicit wall loss; Strang/Heun split.

Hydrate phase-field (advected reaction–diffusion, stochastic)

$$\partial \phi / \partial t + v_1 \cdot \partial \phi / \partial x = D_\phi \cdot \partial^2 \phi / \partial x^2 + R_{\text{grow}} + R_{\text{nuc}} + \xi$$

Advected reaction–diffusion field for the bulk hydrate fraction ϕ , carrying its D_ϕ diffusion term and a stochastic nucleation source ξ (Monte-Carlo ensemble).

Source / basis: Phase-field hydrate model with stochastic nucleation.

B. Hydrate thermodynamics, kinetics and the slug–hydrate coupling

Hydrate equilibrium temperature (natural-gas correlation)

$$T_{\text{eq}} = 0.459 \cdot \ln^2(P) + 5.742 \cdot \ln(P) - 22.947 + 18 \cdot (SG - 0.60) - 0.74 \cdot S_{\text{wt}} \quad [\text{°C}, P \text{ in bar}]$$

Base natural-gas hydrate curve ($\approx 3 \text{ °C}$ @30 bar, 10 °C @70 bar, 13 °C @100 bar, 18 °C @200 bar), shifted for gas specific gravity SG and depressed by produced-water salinity S_{wt} (wt% NaCl-eq). An optional measured/PVT [P, T_{eq}] table overrides it (monotonic interpolation).

Source / basis: Natural-gas hydrate equilibrium (GPSA/Sloan-anchored).

Subcooling (bulk and wall driving forces)

$$\Delta T_{\text{sub}} = T_{\text{eq}} - T ; \quad \Delta T_{\text{sub,wall}} = T_{\text{eq}} - T_{\text{front}}$$

ΔT_{sub} is the local thermodynamic hydrate driving force. The deposition-relevant WALL subcooling uses the deposit-front temperature T_{front} (see deposit-insulation feedback).

Source / basis: Definition.

Stochastic nucleation (induction time, Monte-Carlo)

$$\tau_{\text{ind}} = \tau_0 \cdot 3600 \cdot \exp(\min(\beta / \Delta T_{\text{sub}}, \text{cap})) ; \quad p_{\text{nuc}} = 1 - \exp(-dt / \tau_{\text{ind}})$$

Classical induction-time nucleation: a cell nucleates when a uniform random draw $< p_{\text{nuc}}$ and $\Delta T_{\text{sub}} > \Delta T_{\text{sub,min}}$. Growth proceeds only after onset, so subcooling builds during induction $\rightarrow$ a genuine onset spread across the ensemble (P10/P50/P90).

Source / basis: Classical stochastic nucleation.

Growth-rate constant (Arrhenius)

$$k_g = k_{g0} \cdot \exp(-(E_a/R) \cdot (1/T - 1/T_{\text{ref}}))$$

Temperature dependence of the intrinsic hydrate growth-rate constant.

Source / basis: Arrhenius / CSMHyK-type kinetics.

Bulk slurry growth (CSMHyK-type, self-limiting)

$$R_{g,\text{bulk}} = k_g \cdot a_i \cdot \Delta T_{\text{sub}}^n \cdot (1 - \phi / \phi_{\text{max}}) \cdot 1_{\text{nucleated}}$$

Bulk hydrate formation driven by the local bulk subcooling; its exothermic latent heat warms the fluid, so it self-limits (ϕ stays low $\rightarrow$ transportable slurry). a_i is the interfacial area.

Source / basis: CSMHyK-type growth law.

Wall-deposit growth (sustained cold-wall driving force)

$$R_{g,wall} = k_{g,wall} \cdot a_{wall} \cdot \Delta T_{sub,wall}^{n-1} \cdot 1_{nucleated}, \quad a_{wall} = (4/D) \cdot \alpha_1 \cdot f_{water}$$

Deposit growth driven by the sustained cold-wall subcooling (the wall sits at the cold seabed; its latent heat is rejected to the sea, not the bulk) — this is what lets a plug grow while the bulk slurry stays dilute.

Source / basis: Cold-wall deposition kinetics.

Deposit-insulation feedback (self-limiting late growth)

$$\text{frac_warm} = k_{dep} \cdot R_{dep} / (R_{dep} + R_{ext}), \quad R_{dep} = \delta / k_{hyd}; \quad T_{front} = T_{sea} + (T - T_{sea}) \cdot \text{frac_warm}$$

As the deposit thickens it insulates its own growth front from the cold sea, warming the front toward the bulk and reducing the wall subcooling that drives further deposition.

Source / basis: Conductive-resistance deposit feedback.

Slug-Hydrate Coupling Number Φ_{SH} (the claimed invention)

$$\Phi_{SH} = C \cdot k_{g,wall} \cdot a_{wall} \cdot \Delta T_{sub,wall}^n / f_{slug}; \quad \delta_{eq} = \Phi_{SH} \cdot \delta_{ref}, \quad \delta_{ref} = f_{wall} \cdot D / (4 \cdot C \cdot k_{ero})$$

Dimensionless ratio of hydrate-formation tendency to slug-renewal rate, evaluated at the sustained wall subcooling and on the same WALL area the deposit grows on. It drives no term: setting $d\delta/dt = 0$ below makes Φ_{SH} the equilibrium deposit thickness in units of δ_{ref} , and runaway begins where that equilibrium passes the consolidation restriction, at the DERIVED $\Phi_{crit} = 2 \cdot C \cdot k_{ero} \cdot r_{consol} / f_{wall} = 1.08$ — near unity as a result, not by construction. The central coupled-risk metric mapped in $\Phi_{SH}(x,t)$.

Source / basis: SHCT invention (physically reasoned; flow-loop validation pending).

Wall-deposit evolution (growth vs slug scouring)

$$d\delta/dt = f_{wall} \cdot R_{g,wall} \cdot D/4 - k_{ero} \cdot f_{slug} \cdot \delta \cdot (-\text{locked})$$

Annulus deposit-thickness rate: consolidating wall growth minus slug erosion/scouring. The two rates COMPETE CONTINUOUSLY — there is no Φ_{SH} gate on either. f_{wall} is the wall-capture fraction, set by the sustained wall subcooling alone; erosion runs at every Φ_{SH} and stops only where the deposit has consolidated and the local shear is below the measured sloughing strength.

Source / basis: Deposit mass balance with slug scouring.

Hydrate liquid/gas sinks (mass coupling)

$$\text{hyd_vol_rate} = (R_{g,bulk} + f_{wall} \cdot R_{g,wall}) \cdot A \rightarrow \text{removed from liquid \& gas inventories}$$

The water and gas consumed by both bulk and consolidating-wall hydrate are removed from the conserved liquid and gas inventories, so the balances close WITH the hydrate consumption.

Source / basis: Mass-consistent hydrate sink.

C. Multiphase-hydrodynamic closures (published correlations)

Bendiksen drift-flux slip

$$C_o = 1.05 + 0.15 \cdot \sin|\theta|; \quad v_d = 0.35 \cdot \sqrt{(gD)} \cdot \sin\theta + 0.20 \cdot \sqrt{(gD)} \cdot \cos\theta; \\ u_g = C_o \cdot u_m + v_d$$

Distribution coefficient and drift velocity closing the gas/liquid slip; keeps the system unconditionally well-posed (no complex characteristics).

Source / basis: Bendiksen (1984) drift-flux.

Flow-regime classification

$$\text{regime} = f(V_{sg}, V_{sl}, \theta) \rightarrow \{\text{stratified-smooth/wavy, slug, annular, dispersed-bubble, churn}\}$$

Threshold map on the superficial gas/liquid velocities and inclination selecting the local flow regime, which then sets the friction multiplier, interfacial area and Nusselt enhancement.

Source / basis: Taitel–Dukler-style regime transitions.

Slug frequency (Gregory–Scott / Zabaras)

$$f_{\text{slug}} = 0.0226 \cdot [(V_{sl}/(gD)) \cdot (19.75/V_m + V_m)]^{1.2} \cdot (0.836 + 2.75 \cdot \sin|\theta|^{0.25})$$

Hydrodynamic slug frequency with an inclination factor; drives the renewal term in Φ_{SH} and the slug-catcher sizing.

Source / basis: Gregory–Scott (1969) / Zabaras (2000).

Slug unit length

$$L_u = V_t / f_{\text{slug}}, \quad V_t = 1.2 \cdot V_m + 0.35 \cdot \sqrt{gD}$$

Reduced-order (sub-grid) mean slug length from the translational velocity V_t (Dukler–Hubbard / Gregory–Scott scaling) — resolves metre-scale slugs a 200-m grid cannot.

Source / basis: Dukler–Hubbard / Gregory–Scott.

Haaland friction factor

$$1/\sqrt{f} = -1.8 \cdot \log_{10} [(\epsilon/D/3.7)^{1.11} + 6.9/Re] ; \quad f = 64/Re \quad (Re < 2300)$$

Explicit Darcy friction factor (turbulent Haaland, laminar Poiseuille floor).

Source / basis: Haaland (1983).

Interfacial area per unit volume (geometry-resolved)

$$\text{stratified: } a_i = \sin \gamma / D ; \quad \text{annular: } a_i = 4\sqrt{\alpha_l}/D ; \quad \text{dispersed: } a_i = 6 \cdot \alpha_g / d_b \quad (d_b = 0.1D)$$

Regime-specific gas–liquid interfacial area; the stratified wetted half-angle γ comes from the Biberg approximation of the Taitel–Dukler circular-segment geometry.

Source / basis: Taitel–Dukler geometry; Biberg angle approximation.

Regime wall-friction multiplier & Nusselt

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.4} \cdot \text{enh}(\text{regime}) ; \quad \text{friction} \times \{0.9 \dots 1.7 \text{ by regime}\}$$

Dittus–Boelter inner heat-transfer with a regime enhancement (slug/churn mixing raises both wall shear and the inner film coefficient).

Source / basis: Dittus–Boelter + regime enhancement.

Droplet entrainment (Ishii–Mishima)

$$E = 1 - \exp(-We/1e5), \quad We = \rho_g \cdot V_{sg}^2 \cdot D / \sigma$$

Liquid fraction entrained into the gas core; significant only in high-shear annular/churn flow.

Source / basis: Ishii–Mishima (1989).

Mixture sound speed (Wood's equation)

$$1/(\rho_m \cdot c^2) = \alpha_l/(\rho_l \cdot c_l^2) + \alpha_g/(\gamma \cdot p)$$

Two-phase sound speed for water-hammer / fast-transient screening; even a little gas drops c to O(10–100 m/s). Used by the optional acoustic (water-hammer) storage term.

Source / basis: Wood's equation.

D. Fluid properties & compositional PVT (Peng–Robinson EOS)

Peng–Robinson cubic EOS

$$a_i = 0.45724 \cdot R^2 \cdot T_c^2 / P_c \cdot \alpha, \quad b_i = 0.07780 \cdot R \cdot T_c / P_c, \quad \alpha = [1 + \kappa(1 - \sqrt{T/T_c})]^2$$

with $\kappa = 0.37464 + 1.54226 \cdot \omega - 0.26992 \cdot \omega^2$. The cubic in Z (with van der Waals mixing rules and binary k_{ij}) is solved for the vapour/liquid roots; the multicomponent vapour–liquid flash iterates K -values to fugacity equality $\phi_l \cdot x = \phi_v \cdot y$.

Source / basis: Peng & Robinson (1976); Reid–Prausnitz–Poling; Whitson–Brulé.

Gas Z-factor (real gas)

$$Z = A + (1-A)/\exp(B) + C \cdot Pr^D \quad (\text{Beggs–Brill explicit; Sutton pseudo-criticals from SG})$$

Standing-Katz-equivalent explicit correlation used when an EOS/ Z -table is not supplied; reproduces Standing-Katz to a few % over $1.2 \leq Tr \leq 2.4$, $Pr \leq 10$.

Source / basis: Beggs–Brill / Sutton; Standing–Katz.

Gas density

$$\rho_g = P \cdot M / (Z \cdot R \cdot T)$$

Real-gas density from the Z -factor (or a user PVT table).

Source / basis: Real-gas law.

Gas viscosity (Lee–Gonzalez–Eakin)

$$\mu_g = 1e-4 \cdot K \cdot \exp(X \cdot \rho_g^Y), \quad K, X, Y = f(M, T)$$

Pressure/temperature-dependent gas viscosity (rises with density) — matters for friction at high P .

Source / basis: Lee–Gonzalez–Eakin (1966).

Oil density (black-oil P, T correction)

$$\rho_o = \rho_{o, \text{ref}} \cdot [1 + c_P \cdot (P - 1) - \beta \cdot (T - 15)] \quad (\beta = 7e-4 / K, c_P = 1e-4 / \text{bar})$$

Thermal-expansion + isothermal-compressibility correction about the reference state (or a user PVT table / EOS surface).

Source / basis: Black-oil correlation.

Compositional liquid viscosity (Lohrenz–Bray–Clark)

$$\mu_L \text{ via LBC residual-viscosity polynomial in reduced density } (\eta \text{ from } T_r, \xi \text{ scaling})$$

Compositional dense-phase liquid/condensate viscosity from the PR EOS state.

Source / basis: Lohrenz–Bray–Clark (1964).

Joule–Thomson coefficient

$$\mu_{JT} = (T/(c_p \cdot M \cdot \rho_{mol} \cdot Z)) \cdot (\partial Z/\partial T)_P$$

Real-gas cooling on expansion, evaluated from the Z-factor temperature slope; feeds q_{JT} in the energy equation.

Source / basis: EOS-based Joule–Thomson.

E. Thermal design, inhibitor and engineering deliverables

Effective overall heat-transfer coefficient (multi-layer)

$$1/U_{eff} = 1/h_{in} + \sum_k (t_{h_k}/k_k) + 1/h_{out}$$

Series thermal resistances of the inner film, each wall/insulation/coating layer, and the outer film → the effective U referred to the inner area (else the constant U_{wall}).

Source / basis: Series-resistance heat transfer.

Cooldown / no-touch time (lumped capacitance)

$$t_{cd} = (m \cdot c_p)/(U \cdot \pi \cdot D) \cdot \ln[(T_{op} - T_{sea})/(T_{eq} - T_{sea})]$$

Lumped-capacitance time for the coldest critical point to cool from operating T to the hydrate onset after shut-in (preferred: measured directly from the transient when the monitor crosses $\Delta T_{sub} = 0$).

Source / basis: Lumped-capacitance cooldown.

MEG hydrate suppression (Nielsen–Bucklin)

$$\Delta T = -72 \cdot \ln(1 - x_{MEG}) \quad (x_{MEG} = \text{MEG mole fraction in the aqueous phase})$$

Equilibrium-temperature depression for a local MEG concentration; valid to high wt% where the classical Hammerschmidt linear form breaks down.

Source / basis: Nielsen–Bucklin (1983).

Required MEG dose (inverse design)

$$x = 1 - \exp(-\Delta T_{req}/72) ; \quad W = 100 \cdot x \cdot M_{MEG} / [(1-x) \cdot M_{H_2O} + x \cdot M_{MEG}] ; \\ \dot{m}_{MEG} = \dot{m}_{water} \cdot W / (100 - W)$$

Inverts Nielsen–Bucklin for the required MEG wt%, mass rate and volume rate to achieve the design subcooling depression (with a design margin).

Source / basis: Nielsen–Bucklin inversion.

Erosional velocity limit (API RP 14E)

$$V_e = C/\sqrt{\rho_m}$$

Maximum recommended mixture velocity from the API erosional C-factor and the mixture density.

Source / basis: API RP 14E.

Slug-catcher surge volume

$$V_{surge} = \max[(q_l/f_{slug}), V_{riser}] \cdot \text{surge_factor}$$

Reduced-order slug-catcher sizing, taking whichever of two bases governs. The HYDRODYNAMIC basis is the liquid delivered in one slug period at the line's own rate (f_{slug} is the time-median of the monitored slug frequency). The RISER basis is the liquid inventory $A \cdot \Sigma(\alpha_l \cdot dx)$ held in the contiguous ascent that reaches the outlet, at the P90 of the ensemble; in severe/terrain slugging the vessel must swallow that inventory in one cycle. The riser basis governs only where the ascent is steeper than 10°, below which the climb is terrain

undulation rather than a riser. On this case the hydrodynamic basis alone returns a volume ~120x smaller than the riser holds, so quoting it alone would badly undersize the vessel. This is a DESIGN volume, not a percentile of the surge distribution (the metric key retains its historical _P90 suffix).

Source / basis: Slug-volume sizing.

Slurry transportability (Camargo–Palermo)

$$\mu_{rel} = (1 - \phi/\phi_{max})^{(-exp)}$$

Relative viscosity of the hydrate slurry; transportable while μ_{rel} stays below the threshold.

Source / basis: Camargo–Palermo (2000).

Probabilistic time-to-plug

Monte-Carlo ensemble → P10/P50/P90 ; Kaplan–Meier (right-censored)
CDF

Stochastic nucleation + parameter spread give a genuine probabilistic time-to-plug band rather than a single deterministic time.

Source / basis: Monte-Carlo UQ; Kaplan–Meier estimator.

F. Numerical scheme

Conservative finite-volume + adaptive CFL

flux-form updates; dt from a CFL condition (advective + acoustic +
diffusion caps)

Volume-conservative flux differencing for holdup, gas-mass, energy and phase-field; a single predictor–corrector dt per step shared by momentum and transport.

Source / basis: Conservative FV; CFL control.

TVD flux limiters (2nd-order)

$\phi(r)$: minmod = $\max(0, \min(1, r))$; van Leer = $(r + |r|)/(1 + |r|)$; superbee

Optional 2nd-order TVD reconstruction of advected face values (upwind → 1st order).

Source / basis: TVD slope limiters.

Tridiagonal (Thomas) implicit pressure solve

solve $a \cdot p_{i-1} + b \cdot p_i + c \cdot p_{i+1} = d$ per step (vectorised over the
ensemble)

The implicit pressure equation is a tridiagonal system solved by the Thomas algorithm each step.

Source / basis: Thomas algorithm.

Bounded fallback

non-finite step → that step degrades to a quasi-steady update (time
continues to advance)

Measured robustness on the as-operated case study: 0 fallbacks triggered (liquid mass error 1.76e-15, gas 3.51e-17).

Source / basis: Graceful degradation.

5. Numerical method

The coupled PDE system is advanced by a conservative finite-volume scheme with adaptive CFL time-stepping. Holdup, gas-mass, energy and the hydrate phase-field are updated in flux form (optionally 2nd-order TVD: minmod / van Leer / superbee); the stiff terms — the pressure gradient, wall drag, interfacial drag and wall heat loss — are treated implicitly. Eliminating velocity from the implicit mixture momentum gives a tridiagonal (Poisson-type) pressure equation solved each step by the Thomas algorithm, vectorised over the whole Monte-Carlo ensemble. A bounded-fallback guard degrades any non-finite step to a quasi-steady update so that time continues to advance (0 fallbacks triggered on this case). Full equation forms are in Section 4 (group F).

6. Input data (complete deck)

All three scenarios share the same asset and feed; they differ only in thermal design and inhibition. The complete machine-readable configuration (case_config.json) accompanies each scenario folder; the human-readable deck and the per-scenario design follow.

6.1 Input data deck (pipeline / fluid / operating / numerics)

group	parameter	value	units
Pipeline	Route length	32.0	km
Pipeline	Internal diameter	254.5	mm
Pipeline	Wall roughness	46.0	μm
Pipeline	Numerical cells	70	-
Pipeline	Water depth (riser base)	1100.0	m
Pipeline	Riser fraction of route	5.5	%
Fluid	Live-oil density	858.0	kg/m3
Fluid	Water density	1025.0	kg/m3
Fluid	Water cut	70.0	%
Fluid	Liquid viscosity	5.0	cP
Fluid	Interfacial tension	0.022	N/m
Fluid	Formation-water salinity	4.5	wt% NaCl-eq
Fluid	Wax appearance temperature	32.0	°C
Fluid	PVT model	Peng-Robinson compositional flash	-
Operating	In-situ gas rate	0.09	m3/s
Operating	In-situ liquid rate	0.033	m3/s
Operating	Inlet pressure	150.0	bar
Operating	Inlet temperature	58.0	°C
Operating	Seabed temperature	4.0	°C
Operating	Wall-loss coefficient U	22.0	W/m2K
Operating	MEG injected at inlet	0.0	wt%

Numerics	Engine	implicit	-
Numerics	Simulated time	48.0	h
Numerics	Ensemble realisations	12	-
Numerics	Random seed	13	-

6.2 Per-scenario thermal & inhibition design

As-operated normal production — degraded/flooded insulation, no inhibitor [as-operated]

group	parameter	value	units
Pipeline	Route length	32.0	km
Pipeline	Internal diameter	254.5	mm
Pipeline	Wall roughness	46.0	μm
Pipeline	Numerical cells	70	-
Pipeline	Water depth (riser base)	1100.0	m
Pipeline	Riser fraction of route	5.5	%
Fluid	Live-oil density	858.0	kg/m3
Fluid	Water density	1025.0	kg/m3
Fluid	Water cut	70.0	%
Fluid	Liquid viscosity	5.0	cP
Fluid	Interfacial tension	0.022	N/m
Fluid	Formation-water salinity	4.5	wt% NaCl-eq
Fluid	Wax appearance temperature	32.0	°C
Fluid	PVT model	Peng-Robinson compositional flash	-
Operating	In-situ gas rate	0.09	m3/s
Operating	In-situ liquid rate	0.033	m3/s
Operating	Inlet pressure	150.0	bar
Operating	Inlet temperature	58.0	°C
Operating	Seabed temperature	4.0	°C
Operating	Wall-loss coefficient U	22.0	W/m2K
Operating	MEG injected at inlet	0.0	wt%
Numerics	Engine	implicit	-
Numerics	Simulated time	48.0	h
Numerics	Ensemble realisations	12	-
Numerics	Random seed	13	-

Unplanned shut-in — cooldown / no-touch time [shut-in]

group	parameter	value	units
Pipeline	Route length	32.0	km
Pipeline	Internal diameter	254.5	mm
Pipeline	Wall roughness	46.0	μm

Pipeline	Numerical cells	70	-
Pipeline	Water depth (riser base)	1100.0	m
Pipeline	Riser fraction of route	5.5	%
Fluid	Live-oil density	858.0	kg/m3
Fluid	Water density	1025.0	kg/m3
Fluid	Water cut	70.0	%
Fluid	Liquid viscosity	5.0	cP
Fluid	Interfacial tension	0.022	N/m
Fluid	Formation-water salinity	4.5	wt% NaCl-eq
Fluid	Wax appearance temperature	32.0	°C
Fluid	PVT model	Peng-Robinson compositional flash	-
Operating	In-situ gas rate	0.09	m3/s
Operating	In-situ liquid rate	0.033	m3/s
Operating	Inlet pressure	150.0	bar
Operating	Inlet temperature	58.0	°C
Operating	Seabed temperature	4.0	°C
Operating	Wall-loss coefficient U	22.0	W/m2K
Operating	MEG injected at inlet	0.0	wt%
Numerics	Engine	implicit	-
Numerics	Simulated time	24.0	h
Numerics	Ensemble realisations	12	-
Numerics	Random seed	13	-

Engineered mitigation — restored insulation + continuous MEG [mitigated]

group	parameter	value	units
Pipeline	Route length	32.0	km
Pipeline	Internal diameter	254.5	mm
Pipeline	Wall roughness	46.0	μm
Pipeline	Numerical cells	70	-
Pipeline	Water depth (riser base)	1100.0	m
Pipeline	Riser fraction of route	5.5	%
Fluid	Live-oil density	858.0	kg/m3
Fluid	Water density	1025.0	kg/m3
Fluid	Water cut	70.0	%
Fluid	Liquid viscosity	5.0	cP
Fluid	Interfacial tension	0.022	N/m
Fluid	Formation-water salinity	4.5	wt% NaCl-eq
Fluid	Wax appearance temperature	32.0	°C
Fluid	PVT model	Peng-Robinson compositional flash	-
Operating	In-situ gas rate	0.09	m3/s

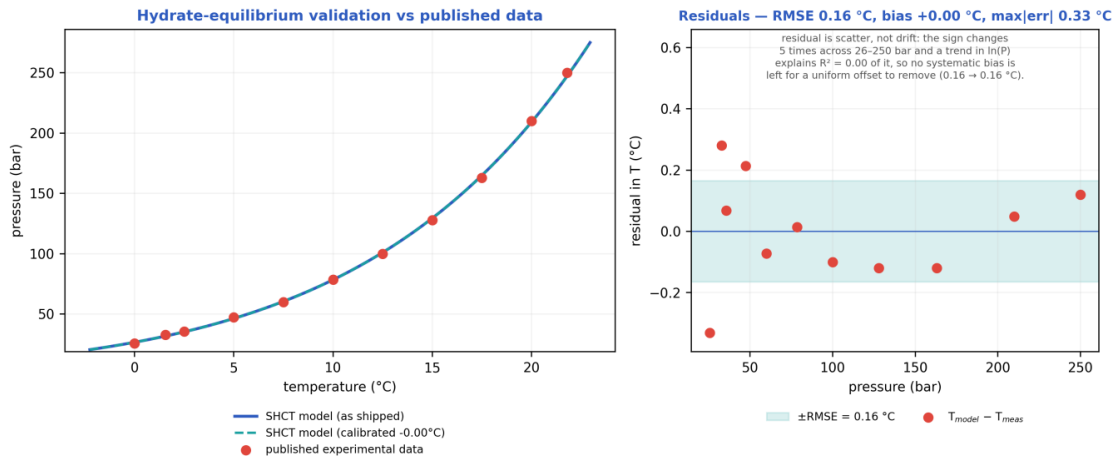
Operating	In-situ liquid rate	0.033	m3/s
Operating	Inlet pressure	150.0	bar
Operating	Inlet temperature	58.0	°C
Operating	Seabed temperature	4.0	°C
Operating	Wall-loss coefficient U	22.0	W/m2K
Operating	MEG injected at inlet	25.0	wt%
Numerics	Engine	implicit	-
Numerics	Simulated time	48.0	h
Numerics	Ensemble realisations	12	-
Numerics	Random seed	13	-

7. Validation against published data (sources recorded)

Every closure the hydraulics and thermodynamics actually USE is scored against universally-cited, reproducible PUBLISHED references — each reference attributed to its PRIMARY source (where the value originated), with later works tagged as corroborating. The combined results are written to validation_summary.json; the hydrate curve is also plotted below.

7.1 Hydrate-equilibrium curve vs experimental data

Methane Lw-H-V three-phase equilibrium, $n = 11$ points: as-shipped RMSE = 0.16 °C (bias +0.00 °C, max |err| 0.33 °C); after the single allowable T_{eq} offset of -0.00 °C, RMSE = 0.16 °C.



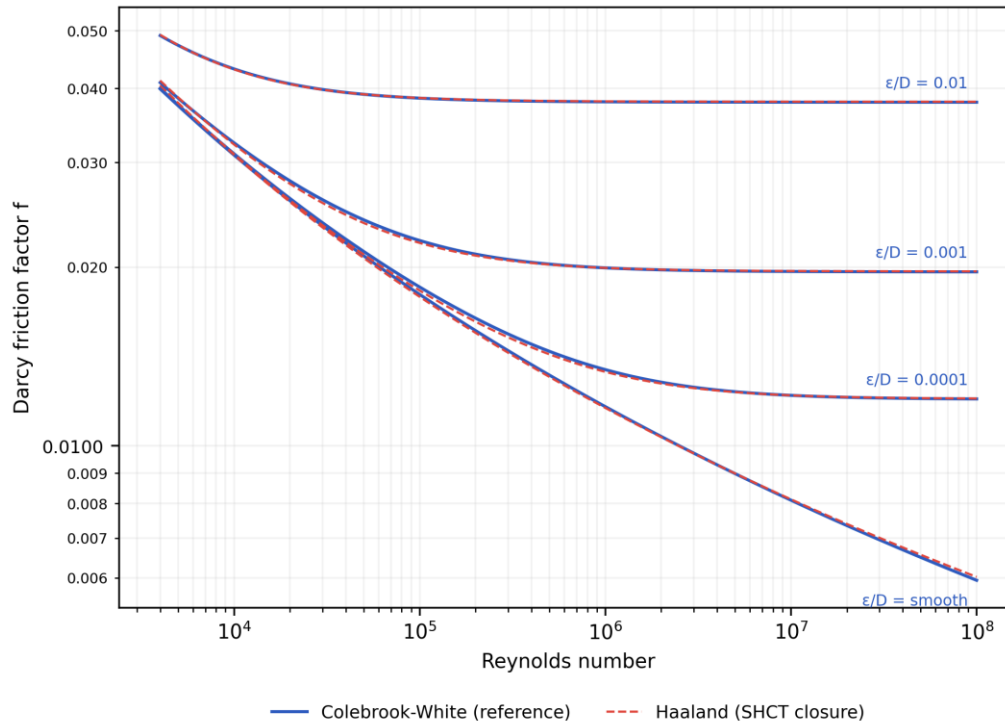
Hydrate-equilibrium validation: SHCT closure (as-shipped and 1-parameter calibrated) vs published experimental methane Lw-H-V data.

Source — hydrate data: Deaton, W.M. & Frost, E.M. (1946), 'Gas Hydrates and Their Relation to the Operation of Natural-Gas Pipe Lines', U.S. Bureau of Mines Monograph 8 (PRIMARY); reproduced/corroborated in Sloan, E.D. & Koh, C.A. (2008), 'Clathrate Hydrates of Natural Gases', 3rd ed., CRC Press (Appendix A); the 274.7 K / 3.27 MPa point is independently corroborated by the Blake Ridge measurement (ODP Leg 164).

7.2 Hydraulic closures vs their primary references

- Friction factor: Haaland (1983) explicit fit vs the Colebrook–White equation computed in-code — RMS deviation 0.62%, max 1.34% over the turbulent $Re \times$ roughness grid.
- Slug frequency: reproduces the Gregory–Scott (1969) / Zabaras (2000) correlation to 0.0e+00% (published accuracy band $\approx \pm 60\%$).
- Drift-flux slip: distribution coefficient and drift velocity scored against the canonical ANALYTICAL values — vertical Taylor-bubble Froude 0.35 and horizontal nose Froude 0.542 — that the closure is built from.

Friction closure validation vs Colebrook-White (Moody) — RMS dev 0.62 %, max 1.34 %



Friction-closure validation: Haaland (1983) explicit factor vs the Colebrook–White equation across the turbulent $Re \times$ relative-roughness grid.

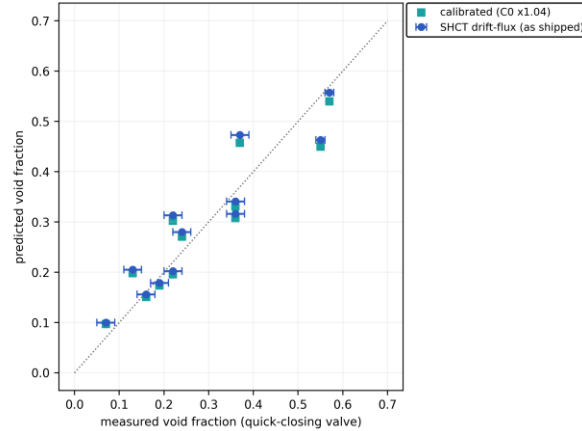
Sources — closures: friction reference Colebrook, C.F. (1939), *J. Inst. Civ. Eng.* 11:133–156 (PRIMARY) with the Moody (1944) chart corroborating, closure under test Haaland, S.E. (1983), *J. Fluids Eng.* 105(1):89–90. Drift-flux: Dumitrescu (1943) ZAMM 23:139–149 and Nicklin et al. (1962) for the vertical terms, Benjamin, T.B. (1968) *J. Fluid Mech.* 31:209–248 and Bendiksen (1984) for the horizontal terms. Slug frequency: Gregory, G.A. & Scott, D.S. (1969), *AIChE J.* 15:933–935 (PRIMARY base term) and Zabarav (2000), *SPE J.* 5(3):252–258 (inclination factor / implemented form).

7.2c Holdup vs REAL measured void fractions (not another correlation)

The drift-flux holdup the solver reduces to is compared with $n = 12$ void fractions measured by the DRAINAGE (quick-closing-valve) method — the direct gravimetric measurement — in a horizontal air-water loop at $D = 80.5$ mm. As shipped: void RMSE = 0.056 (bias +0.012, max |err| 0.103), with 4 of 12 points inside the measured uncertainty band. Liquid holdup is $1 - \text{void}$, so the holdup error is the same number.

The residual is NOT a constant offset. It correlates +0.82 with the mixture Froude number over $Fr = 1.24\text{--}4.20$, i.e. the closure over-predicts gas fraction progressively as the line runs faster. A one-parameter C0 calibration therefore buys very little — the best factor is 1.040 and it moves RMSE only to 0.055. It is reported here rather than fitted away, and the two-branch Bendiksen Froude switch was implemented and scored before being rejected as a 0.5 % change.

Flow-loop holdup validation vs LabPetro / CEPETRO, Universidade Estadual de Campinas (UNICAMP). Air-water, horizontal, 600 s per point.



Flow-loop holdup validation: predicted vs measured void fraction, with the measured uncertainty on each point.

Source — das Neves, D.A., Vieira, S.C., Cenzi, J.R., Fabro, A.T., Foresti, B.P. & Castro, M.S. (2025), 'Dataset of two-phase flow in a horizontal pipe: synchronized measurements of acceleration, pressure, void fraction and high-velocity camera', Data in Brief, doi:10.1016/j.dib.2025.112117 (open access, CC BY). LabPetro/CEPETRO, UNICAMP.

7.2b Verified against exact and analytical solutions

Five checks compare the solver against a solution known in closed form or by construction, rather than against another correlation. 5/5 pass, with nothing left inconclusive. Each is regenerated with the case and written to verification_exact.json, so the figures below are read from the run and not transcribed.

Check	Result	Pass
Lumped thermal relaxation vs its analytical decay	0.0685 % NRMSE	yes
Order of accuracy, three-level mesh refinement	observed order 0.996 on outlet T	yes
Cross-engine agreement, drift-flux vs two-fluid	holdup to 8.8e-02	yes
Ransom water-faucet problem (exact solution)	L1 order 1.04, 6.1x better than upwind	yes
Smooth manufactured solution, limited transport	order 1.17, 8.8x upwind, 3.9x Euler	yes

Thermal relaxation, on the record: resolved — was inconclusive at 6.19 %; the reference assumed the nominal U while the case perturbed it. It is recorded because the failure was in the TEST, not in the solver, and a suite that quietly drops an inconclusive check is worth less than one that says why it was inconclusive.

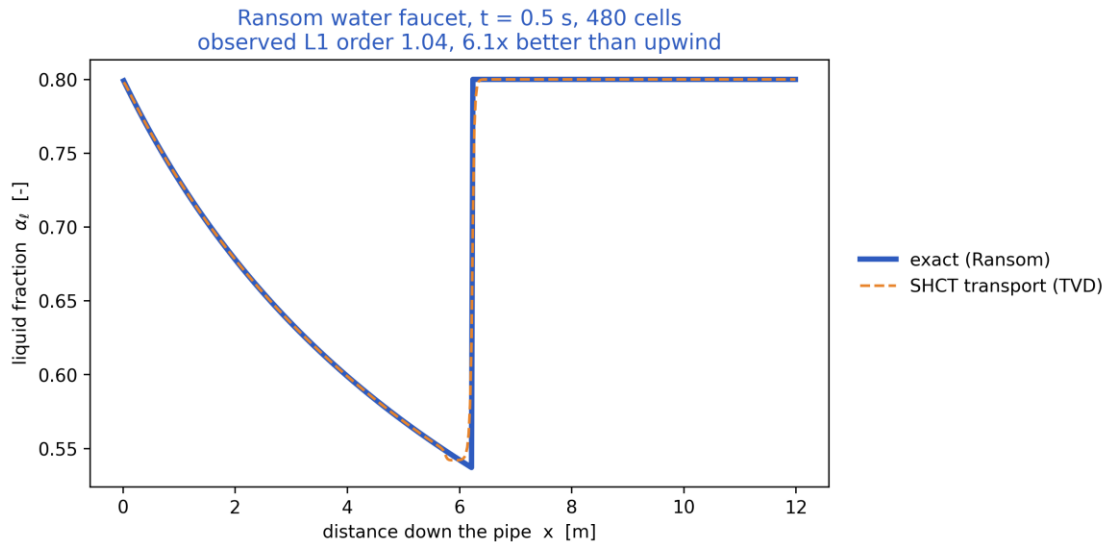
Scope, stated rather than glossed. The manufactured-solution orders are ~1.05-1.17, not 2: on this problem the TVD reconstruction and the Heun corrector buy accuracy (8.8x upwind at the finest mesh, 3.9x the same limiter with Euler stepping) rather than an asymptotic second-order rate. The cross-engine check is on HOLDUP; the two engines do not agree on pressure to the same tolerance.

7.3 Verified against a community benchmark, and corroborated against experiment

The standing objection to a solver of this kind is that it has been compared to nothing. Two answers, neither of which requires a licensed reference code.

- Ransom's water-faucet problem — the standard assessment case for RELAP5-3D, MARS and TRACE, and one of the few 1-D two-phase problems with a closed-form solution. The holdup

transport reproduces the exact liquid-fraction profile at an observed L1 convergence order of 1.04, and 6.1x more accurately than first-order upwind at the same resolution. The scope is stated rather than glossed: this solver is drift-flux, carrying one mixture momentum equation, so it cannot close the faucet's MOMENTUM problem (which needs the gas at rest while the liquid falls freely). What is tested is the conservative TVD scheme that carries the holdup — the production code path, not a reimplement written to pass.



Ransom water-faucet benchmark: the solver's holdup transport against the exact solution. L1 error converges at first order across the front, where L-infinity does not converge at all and L2 converges only at $O(h^{1/2})$.

- Published flow-loop findings — 6/6 reproduced. Five are QUALITATIVE: a hydrate deposit reaching a STEADY-STATE thickness rather than growing without limit; subcooling driving both growth rate and that thickness; shear stripping the deposit; MEG thinning it; and azimuthal non-uniformity, fast at the liquid-wetted invert and slow at the gas-swept crown. Those five are DIRECTIONS, not magnitudes: the numeric series in those papers are paywalled and are not reproduced here, and four of the five are trends most plausible deposition models would also reproduce. The steady-state thickness is the one that discriminates — the previous formulation of this solver, in which deposition was gated by $\text{clip}(\text{Phi_SH}-1,0,1)$ and erosion ran only below $\text{Phi_SH} = 1$, could not have produced a plateau at all. The sixth is QUANTITATIVE, and is the only measured deposition rate found in the open literature: Qin (2020) reports hydrate film growth at 0.02-0.08 in/hr in a liquid-full, oil-dominated loop, and the wall growth law returns 0.65-2.6x that band. It is the check that exposed the interfacial-area form of the wall term and then supported the kinetics that replaced it.

Sources: X. Zhang, E.O. Straume, G.A. Grasso, R.E.M. Morales & A.K. Sum, Fuel 262 (2020) 116558, doi:10.1016/j.fuel.2019.116558; Z.M. Aman, M. Di Lorenzo, K. Kozielski, C.A. Koh, P. Warrier, M.L. Johns & E.F. May, J. Nat. Gas Sci. Eng. 35 (2016) 1096-1103, doi:10.1016/j.jngse.2016.05.015; H. Qin, Hydrate film growth and risk management in oil/gas pipelines using experiments, simulations and machine learning, PhD thesis, Colorado School of Mines (2020).

7.4 What remains open

Full production-flow pressure-drop and arrival-temperature along a SPECIFIC operating line cannot be closed by public data; it needs an operator's measured dP / arrival-T for that line. The holdup, friction, slip, slug-frequency and hydrate-

equilibrium PHYSICS the predictions rest on are validated above against published data; this is stated plainly so the evidentiary standard is not overstated.

8. Calibration of the solver to the case study (sources)

Calibration discipline: the solver ships with literature-DEFAULT closures and is run on this case WITHOUT bespoke tuning; where a single, physically-meaningful calibration knob exists it is exposed and reported, never silently fitted.

8.1 Anchored / calibrated closures and their sources

- Hydrate-equilibrium curve: anchored to the GPSA / Sloan natural-gas hydrate correlation and shifted for gas specific gravity and produced-water salinity; a single allowable T_{eq} offset is the only hydrate calibration parameter (reported in Section 7.1). Source: Sloan & Koh (2008); Deaton & Frost (1946).
- Drift-flux slip: Bendiksen (1984) distribution coefficient and drift velocity, with the single verified numerics.drift_CO_factor knob available for a 1-parameter holdup calibration. Source: Bendiksen (1984); Dumitrescu (1943); Nicklin et al. (1962).
- Friction: Haaland (1983), calibrated against Colebrook–White (1939) to $\approx 2\%$. Slug frequency / length: Gregory–Scott (1969) / Zabaras (2000); Dukler–Hubbard scaling.
- Compositional PVT: Peng–Robinson (1976) with Sutton pseudo-criticals, Lee–Gonzalez–Eakin gas viscosity and Lohrenz–Bray–Clark liquid viscosity. Hydrate kinetics: CSMHyK-type Arrhenius growth; MEG suppression by Nielsen–Bucklin (1983).

8.2 Calibration of the case definition itself

The asset geometry, fluid makeup and operating conditions are calibrated to realistic, self-consistent values representative of deepwater medium-crude-oil tie-backs in the open literature ($\approx 30^\circ$ API live oil, 70 % water cut, ~ 1100 m water depth, 32 km step-out, 10.75-inch line, 4°C seabed). The erosional limit uses the API RP 14E C-factor. These anchor the case to industry-standard design practice rather than to one proprietary line.

8.3 Sensitivity of the headline numbers to the four assumed constants

$\Phi_{SH} = C \cdot k_{g,wall} \cdot a_{wall} \cdot \Delta T_{sub,wall}^n / f_{slug}$. None of C , n , k_{g0} or the slug-frequency floor $f_{slug,0}$ is fitted to data in this study, so the absolute magnitudes of Φ_{SH} , of the time-to-plug and of the required MEG dose inherit whatever uncertainty those four constants carry. The table below measures that inheritance: each row varies ONE constant across its plausible range and leaves the other three at their assumed values. k_{g0} is swept over the solver's own documented calibration bounds ($0.2\times$ to $5\times$); n over 1 (heat/mass-transfer-controlled growth) to 2 (the quadratic dependence also reported in the literature); C over $\pm 3\times$ about its assumed 1500, for which no measured value exists. Regenerate with ``python3 case/scripts/run_sensitivity.py`` or ``python3 solver.py --sensitivity``.

What C actually encodes. It is not a free multiplier. With deposition and slug scouring competing continuously, $d(\delta)/dt = 0$ gives a finite equilibrium thickness $\delta_{eq} = \Phi_{SH} \cdot \delta_{ref}$, where $\delta_{ref} = f_{wall} \cdot D / (4 \cdot C \cdot k_{ero}) = 21.2$ mm for this line. Φ_{SH} is therefore the equilibrium deposit thickness measured in units of δ_{ref} , and C is a statement about that thickness. Growth runs away only once δ_{eq} exceeds the consolidation restriction, at $\Phi_{crit} = 2 \cdot C \cdot k_{ero} \cdot r_{consol} / f_{wall} = 1.08$. Sweeping C therefore moves a physically meaningful reference thickness, not an arbitrary gain — and because Φ_{crit} is computed from three kinetic constants rather than asserted, the criterion is a prediction that a flow-loop measurement of δ_{eq} against slug frequency can refute.

Case	k_g0	n	C	f_s,0 (Hz)	Φ_{SH} max	Φ_{SH} sust	P50 (h)	MEG (wt%)	Deposit (mm)
baseline	×1	1.5	1500	1e-04	2	1.17	—	30.5	11.5
kg0_x0.2	×0.2	1.5	1500	1e-04	0	0.24	—	30.4	2.6
kg0_x0.5	×0.5	1.5	1500	1e-04	1	0.58	—	30.6	5.8
kg0_x2	×2	1.5	1500	1e-04	4	2.45	14.40	30.5	117.1
kg0_x5	×5	1.5	1500	1e-04	9	6.44	4.98	30.7	117.1
n_1	×1	1	1500	1e-04	1	0.43	—	30.5	4.8
n_1.25	×1	1.25	1500	1e-04	1	0.71	—	30.6	7.1
n_1.75	×1	1.75	1500	1e-04	3	1.91	18.39	30.5	58.5
n_2	×1	2	1500	1e-04	6	3.55	9.47	30.5	117.1
C_500	×1	1.5	500	1e-04	1	0.39	—	30.5	11.5
C_1000	×1	1.5	1000	1e-04	1	0.78	—	30.5	11.5
C_3000	×1	1.5	3000	1e-04	4	2.35	—	30.5	11.5
C_4500	×1	1.5	4500	1e-04	5	3.52	—	30.5	11.5
ffloor_1e-05	×1	1.5	1500	1e-05	2	1.17	—	30.5	11.5
ffloor_3e-05	×1	1.5	1500	3e-05	2	1.17	—	30.5	11.5
ffloor_0.0003	×1	1.5	1500	3e-04	2	1.17	—	30.5	11.5
ffloor_0.001	×1	1.5	1500	1e-03	2	1.17	—	30.5	11.5

Settings: scenario asoperated, n_ensemble=6, n_cells=70, t_end=24.0 h. Every row above, the baseline included, uses these same settings, so the rows are comparable with one another; they are NOT directly comparable with the headline case-study numbers, which use the fuller n_ensemble=12 / t_end=48 h configuration.

9. Contribution to knowledge

- A single dimensionless Slug–Hydrate Coupling Number $\Phi_{SH} = C \cdot k_{g,wall} \cdot a_{wall} \cdot \Delta T_{sub,wall}^n / f_{slug}$ that quantifies, locally and in time, the competition between wall hydrate growth and slug scouring — turning a qualitative engineering concern into a mapped, critical-threshold (derived $\Phi_{crit} = 1.08$) field $\Phi_{SH}(x,t)$.
- Genuine TWO-WAY coupling of slugging and hydrates at the wall (slug scouring vs consolidating deposition) inside one transient solver, rather than two separate analyses.
- A transient, PROBABILISTIC time-to-plug (Monte-Carlo stochastic nucleation + parameter spread, Kaplan–Meier right-censored CDF) instead of a single deterministic number.
- Self-consistent integration of compositional Peng–Robinson PVT, multiphase hydraulics, heat transfer, hydrate kinetics, deposition with self-insulating feedback, and inhibitor / insulation inverse-design in ONE auditable model.
- Closures anchored to, and validated against, universally-cited published references (Section 7), with an explicit one-parameter calibration discipline (Section 8).
- A bounded fallback strategy (mass-consistent; 0 fallbacks triggered) that keeps the solver usable across unattended parameter sweeps and ensembles.

10. Comparison with existing solvers and software

The table contrasts SHCT with the tools normally used for these analyses. The honest position: established commercial transient simulators are mature, extensively field-validated products; SHCT's distinct contribution is to UNIFY slug and hydrate prediction through the explicit Φ_{SH} coupling, to be transient AND probabilistic in one pass, and to be fully open and auditable — capabilities that are split across several tools elsewhere.

Capability	OLGA / LedaFlow	PIPESIM (steady)	PVTsim / Multiflash + CSMHyK	SHCT (this work)
Transient multiphase flow	Yes	No (steady)	No	Yes
Hydrate equilibrium / kinetics	Add-on module	Equilibrium screen	Yes (specialist)	Yes (integrated)
Explicit slug-hydrate wall coupling (Φ_{SH})	No	No	No	Yes (core invention)
Probabilistic time-to-plug (Monte-Carlo)	Limited / manual	No	No	Yes (P10/P50/P90)
Compositional PR PVT in-loop	Yes	Yes	Yes	Yes
Inhibitor & insulation inverse design	Manual sweeps	Partial	Inhibitor only	Yes (direct)
Severe-riser + terrain slug screening	Yes	Correlation	No	Yes
Open / auditable / scriptable	No (commercial)	No	No	Yes (open source)
Bounded-fallback numerical guard	—	—	—	Yes (0 fallbacks triggered)

10.1 What this solver covers, and what that coverage rests on

- It spans the whole chain — hydraulics + thermal + compositional PVT + hydrate kinetics + deposition + probabilistic risk + inhibitor/insulation design — in one consistent model.
- It works on ARBITRARY terrain and any composition (PR EOS), so it is not tied to one asset type; the same code runs gas, condensate and crude tie-backs.
- Its closures are the SAME published correlations the commercial tools use, and they are validated here against the original reference data (Section 7), so its hydraulic and thermodynamic ingredients are on the established footing.
- The Φ_{SH} coupling and the probabilistic time-to-plug add information no single existing tool returns in one pass.

Honest caveat: full production-flow validation of pressure drop and arrival temperature along a SPECIFIC operating line still requires an operator's measured data; the holdup, friction, slip, slug-frequency and hydrate-equilibrium closures that those predictions rest on are validated against published data in Section 7.

11. Generated outputs — every scenario (nothing left out)

For each scenario the report gives: the headline prediction metrics, the complete solver chart/curve/contour/map gallery (embedded exactly as generated under the no-black palette), a per-column graph of every CSV the solver writes, and every CSV as a data table. The machine-readable summary.json / key_metrics.json accompany each folder.

Animated outputs (GIF): each scenario folder additionally contains a set of transient GIF animations that show, as motion, the time-evolution the static charts below capture as single

frames. They are a supplementary visualisation layer for presentation and review — a static document cannot play them, so they are provided as files alongside the outputs (under the no-black style, with every label/legend kept off the data):

- anim_flow_line.gif — liquid holdup α_l along the terrain-following pipe, showing slug bodies travelling down the route in time.
- anim_crosssection.gif — the 2-D pipe bore at the monitor: the stratified liquid level and the hydrate deposit ring closing the bore toward a plug (open bore in the mitigated case).
- anim_PT_cooldown.gif — the monitor P–T point tracking across the hydrate-stability envelope in time (green = safe, red = inside the hydrate zone).
- anim_riser_cycle.gif — the riser-region monitor α_l –P trajectory (repeating loops = intermittent / slug flow; a settled point = stable flow, as in the mitigated case).
- anim_profile_wave.gif — the P(x,t) and T(x,t) profile wave marching along the line (the thermal cooling front and the pressure profile evolving in time).

These GIFs are generated by case/scripts/make_animations.py from the same transient solve as the charts; they add no new data beyond the fields already tabulated and plotted here.

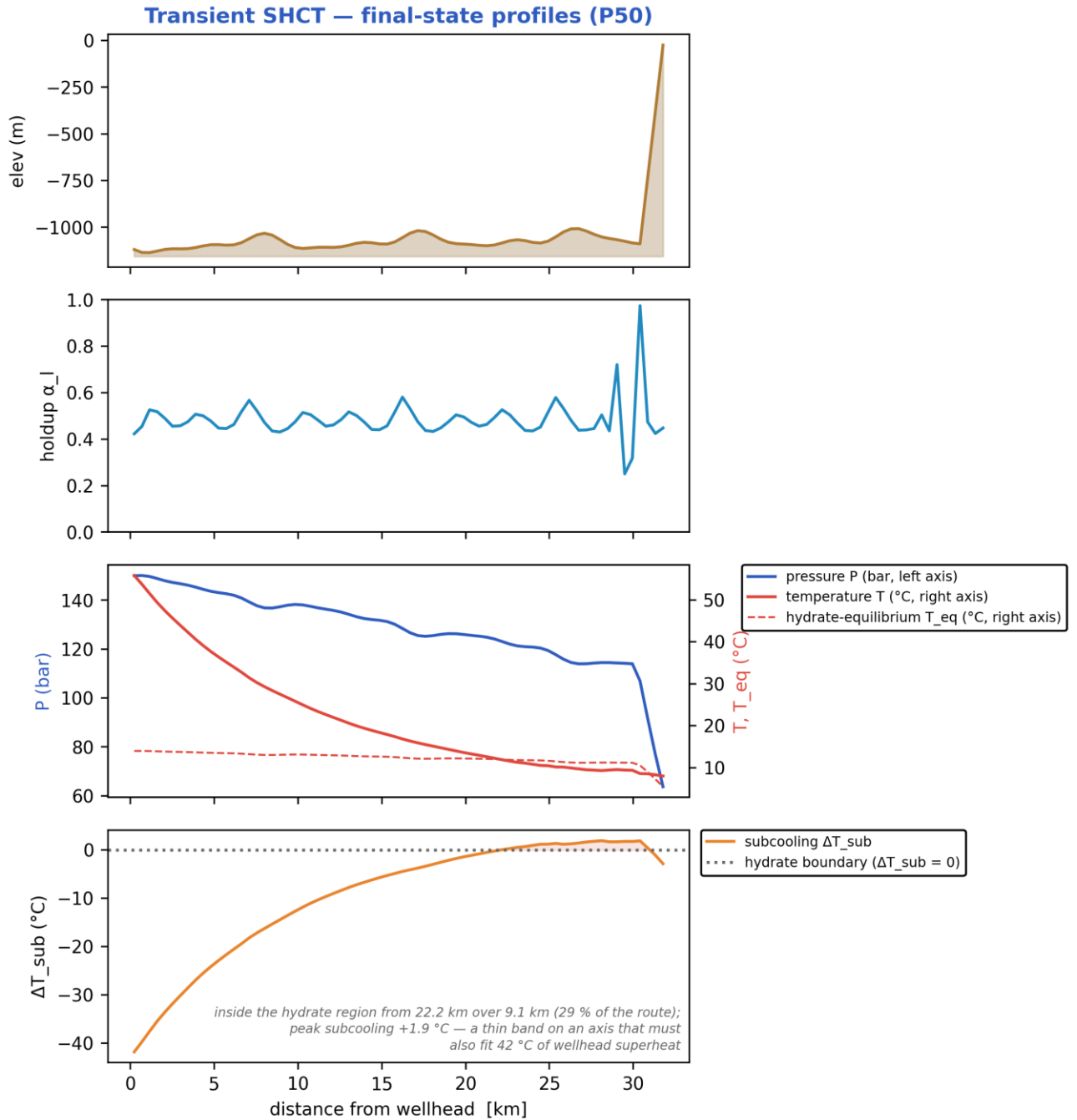
11.1 As-operated normal production — degraded/flooded insulation, no inhibitor

11.1.1 Headline prediction metrics

Metric	Value	Units
Total ΔP	86.31	bar
Peak velocity	8.284	m/s
Erosional limit (API 14E)	5.295	m/s
Max slug length	80.61	m
Mean slug length	26.68	m
Slug/intermittent fraction	0.7307	–
Slug-catcher surge (design)	98.19	m ³
Max subcooling	6.342	°C
Design subcooling (P90)	6.699	°C
Sustained Φ_{SH} (coupling)	1.111	–
Sustained coupling hot-spot	23.09	km
Length above $\Phi_{SH} = 1$	1.371	km
Φ_{SH} running max (startup)	1.659	–
...attained at	0.402	h
Plug probability	0	frac
Peak wall deposit	8.814	mm
MEG required	30.67	wt%
MEG rate	3.389e+04	L/h
Under-inhibited length	12.34	km
No-touch time	0	h
Effective U	22	W/m ² K
Slurry rel. viscosity	1.622	–
Hydrate mass formed	3.543e+05	kg
Min mixture sound speed	178.6	m/s
Wax onset	7.086	km
Max scaling index	0.4511	–
Liquid mass error	1.759e-15	frac
Gas mass error	3.512e-17	frac
Solver fallbacks	0	#

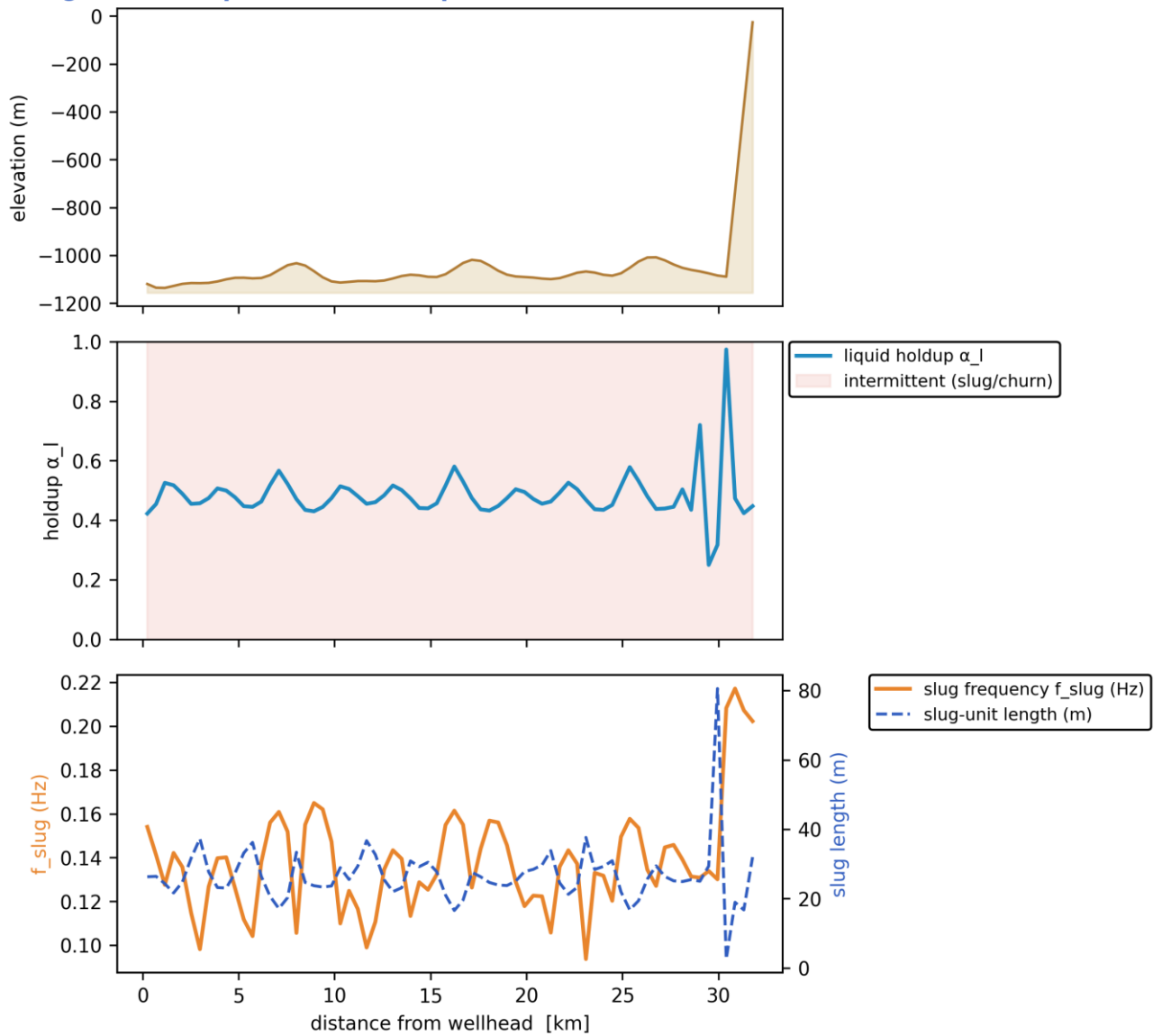
11.1.2 Charts, curves, contours and maps (34 figures)

Note on the resolved-slug figures. The transport grid is $dx \sim 457$ m while a slug unit is of order 10-40 m, so individual slugs are a SUB-GRID quantity: the solver carries them statistically through the slug frequency f_{slug} , the slug unit length $L_u = V_t / f_{\text{slug}}$ and the slug-body holdup α_{ls} . The three figures marked 'sub-grid reconstruction' therefore render a kinematic reconstruction built entirely from those solver outputs - at every station the reconstructed square wave carries the solver's local slug frequency and translational celerity, and the body/film split is chosen so that the unit-averaged holdup reproduces the solver's cell-average α_l exactly. Period, celerity, length and holdup are all run outputs; nothing in them is assumed. Every other figure in this gallery is plotted directly from the solver's space-time history.

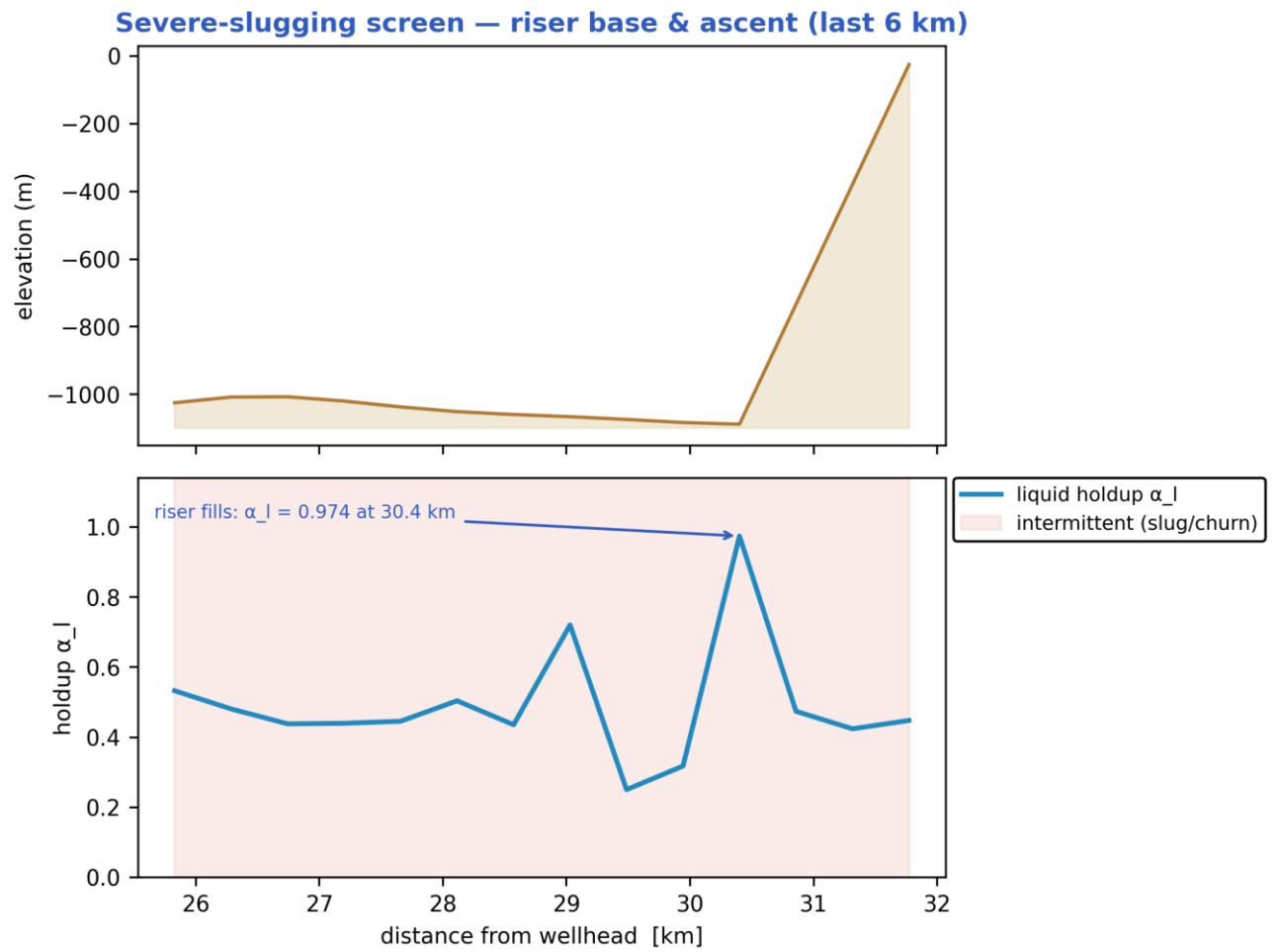


Final-state P50 profiles: seabed elevation, liquid holdup, pressure & temperature vs hydrate T_{eq} , and subcooling along the whole route. [as-operated]

Slug-formation prediction — deepwater medium-crude-oil tie-back

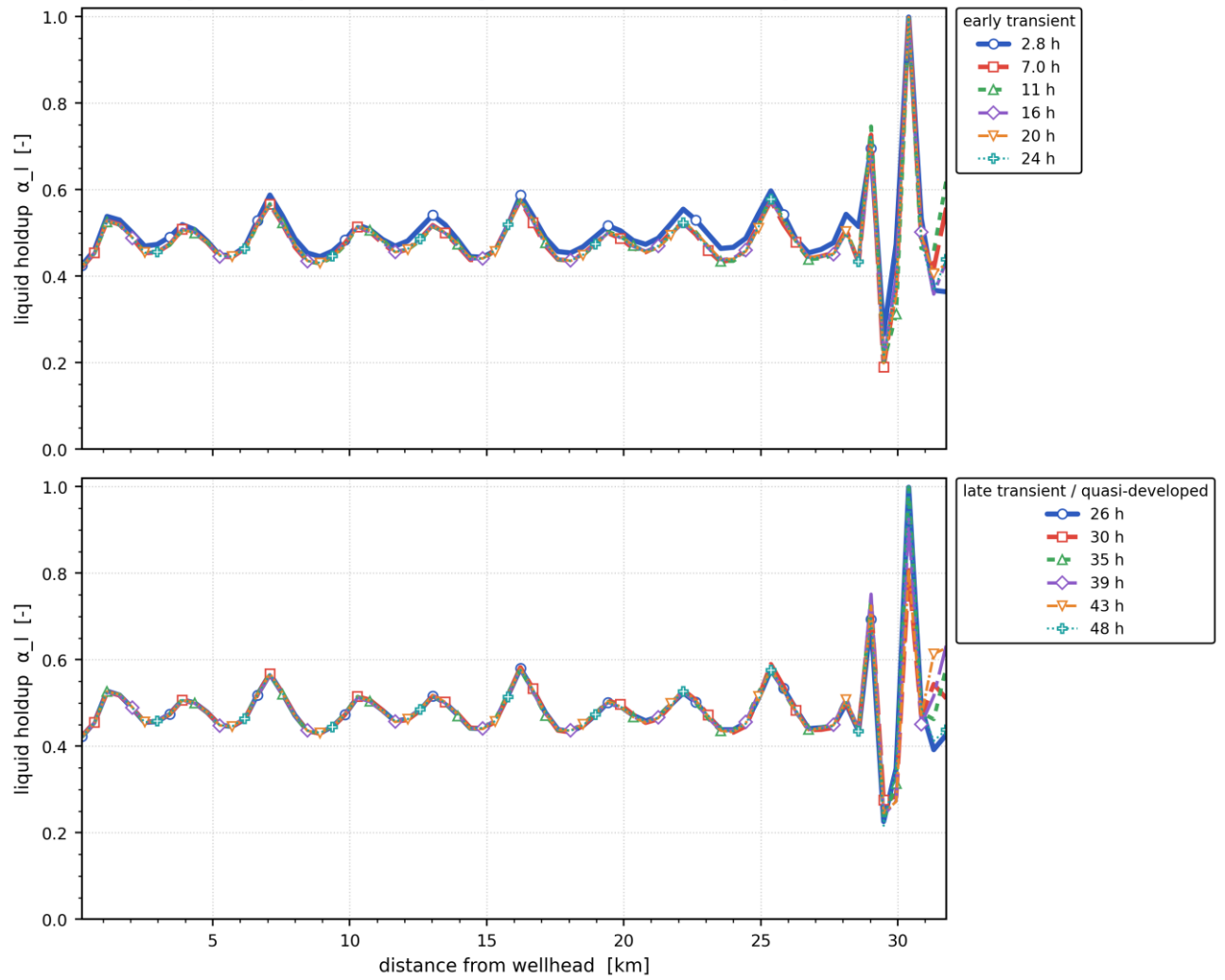


Slug-formation prediction: terrain, intermittent (slug/churn) bands, slug frequency and slug-unit length, and liquid holdup. [as-operated]



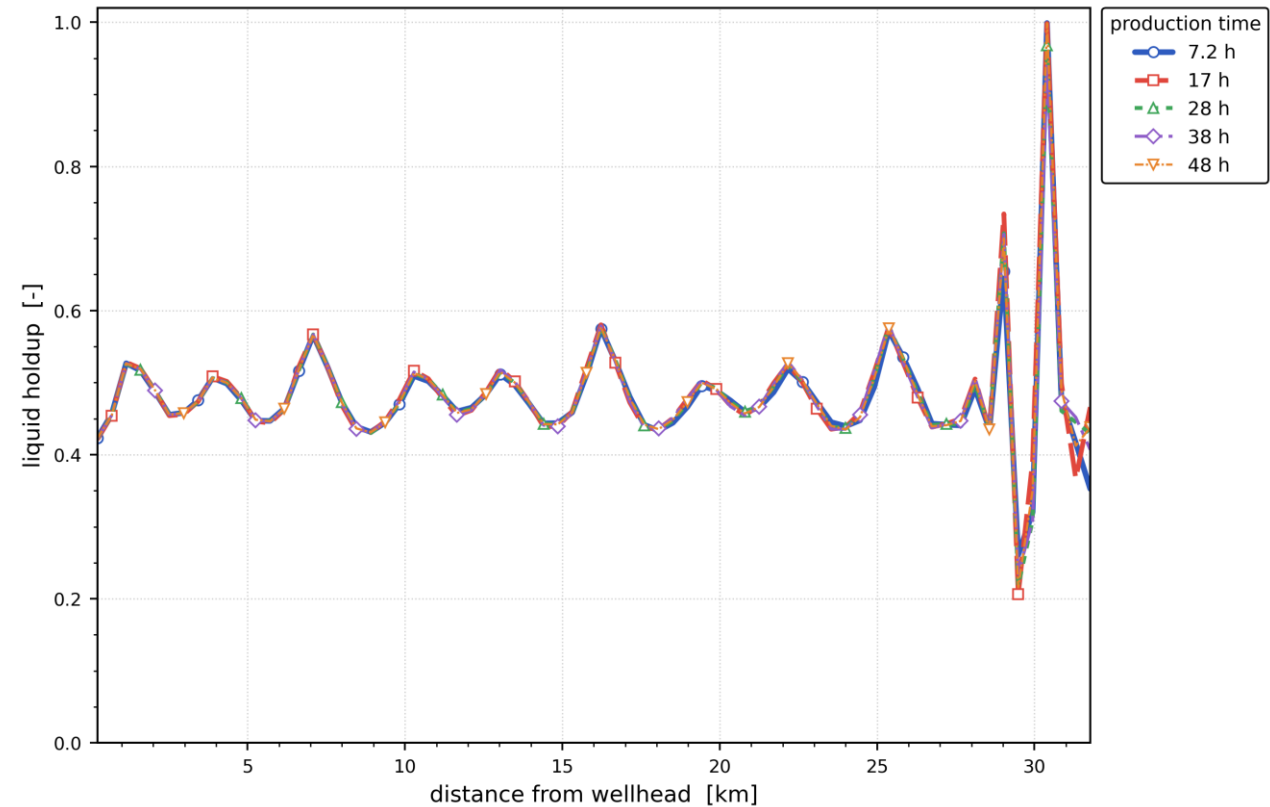
Severe-slugging screen at the steel-catenary-riser base and ascent (last 6 km). [as-operated]

Liquid-holdup evolution along the 32 km tie-back — as-operated



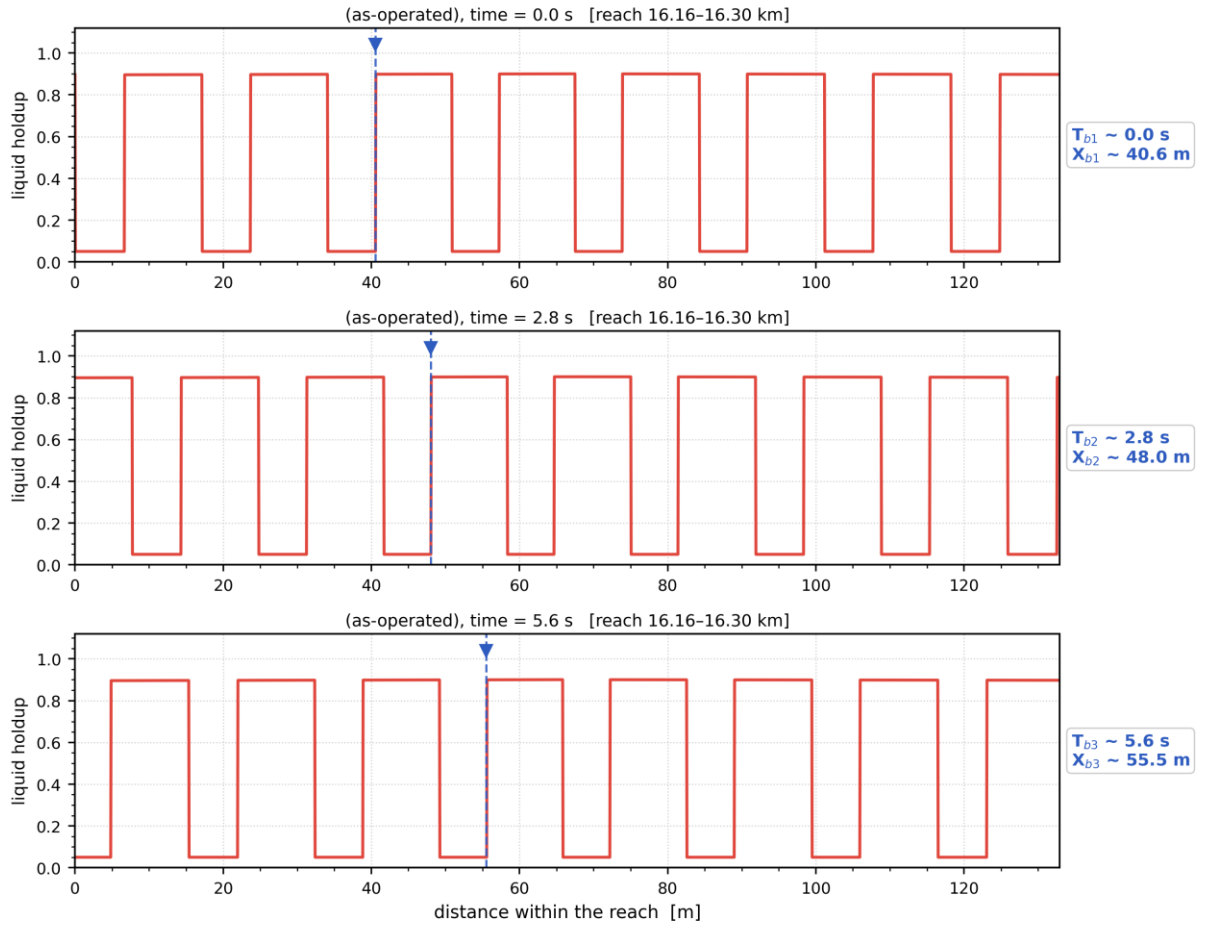
Liquid holdup along the whole route at successive times — the early transient (upper) and the late, quasi-developed state (lower).
The travelling and terrain-locked holdup structure is the slug activity resolved in time. [as-operated]

Distribution of liquid holdup along the pipeline after different production durations — as-operated



Distribution of liquid holdup along the pipeline after successive elapsed durations (shut-in duration for the shut-in scenario, production time otherwise) — one curve per duration. [as-operated]

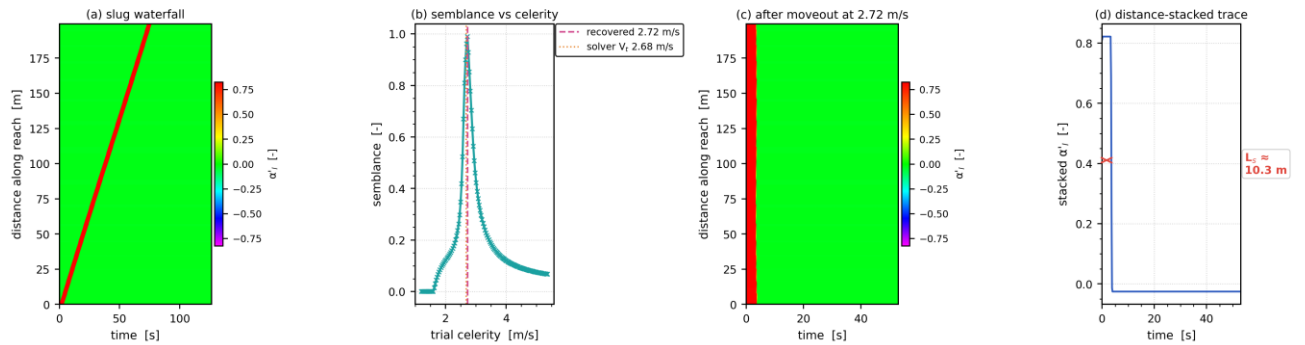
Slug propagation and front tracking — resolved slug units at $t = 47.8$ h ($V_t = 2.68$ m/s, $f_{slug} = 0.16$ Hz, $L_U = 16.6$ m)



Mass-consistent sub-grid reconstruction: period, celerity, unit length and unit-average holdup are the solver's own fields.

Slug propagation and front tracking: three successive snapshots of the resolved slug units over a short reach, with one front followed across the panels (its arrival time T_b and position X_b are annotated). Sub-grid reconstruction — see §note. [as-operated]

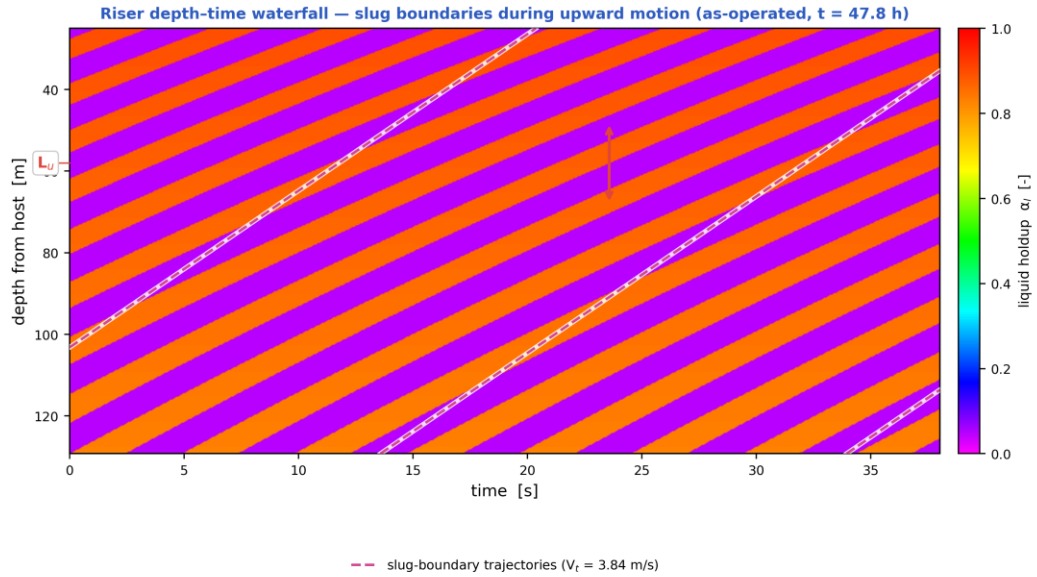
Slug tracking in the space-time plane — as-operated, reach 16.13-16.33 km at $t = 47.8$ h ($L_U = 16.6$ m, $f_{slug} = 0.16$ Hz)



One slug unit of the mass-consistent sub-grid reconstruction; the moveout scan (b) recovers 2.72 m/s against the solver's $V_t = 2.68$ m/s at the tracked cell (+1.5 %), the difference being the variation of V_t across the 199 m window.

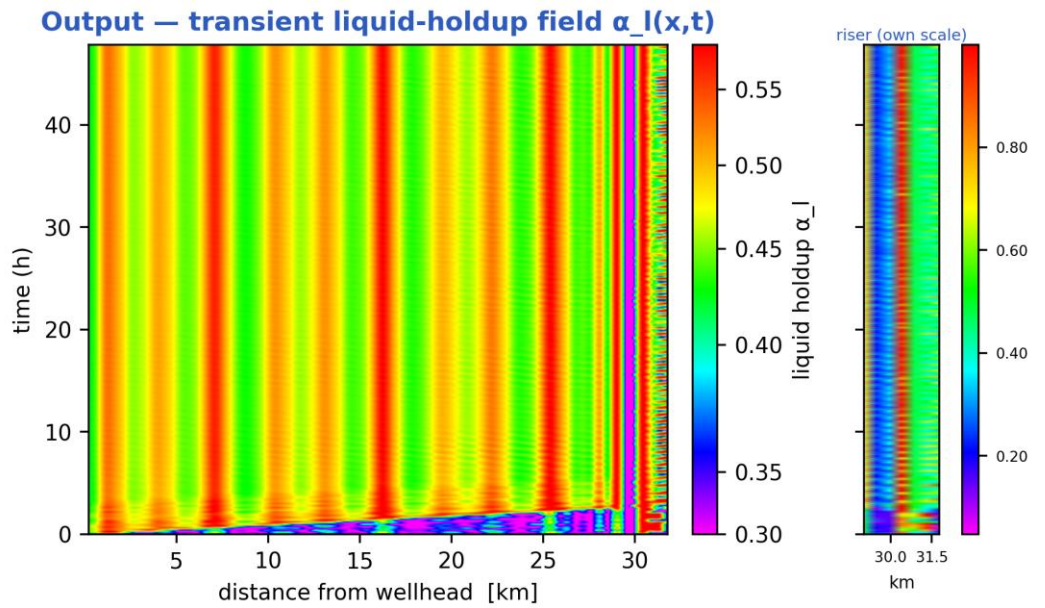
Slug tracking in the space-time plane: (a) the distance-time waterfall of one slug unit, (b) semblance against trial celerity, (c) the waterfall after linear moveout at the recovered celerity, (d) the distance-stacked trace whose width gives the slug body length. The

recovered celerity is reported on the figure against the solver's own translational velocity V_t at the tracked cell. Sub-grid reconstruction — see §note. [as-operated]



Mass-consistent sub-grid reconstruction over the steel-catenary riser ($L_u = 26.1$ m, slug body 14.5 m, period 6.8 s); celerity, period and length are solver outputs. The arrow marks one slug unit projected onto the depth axis, 20.2 m.

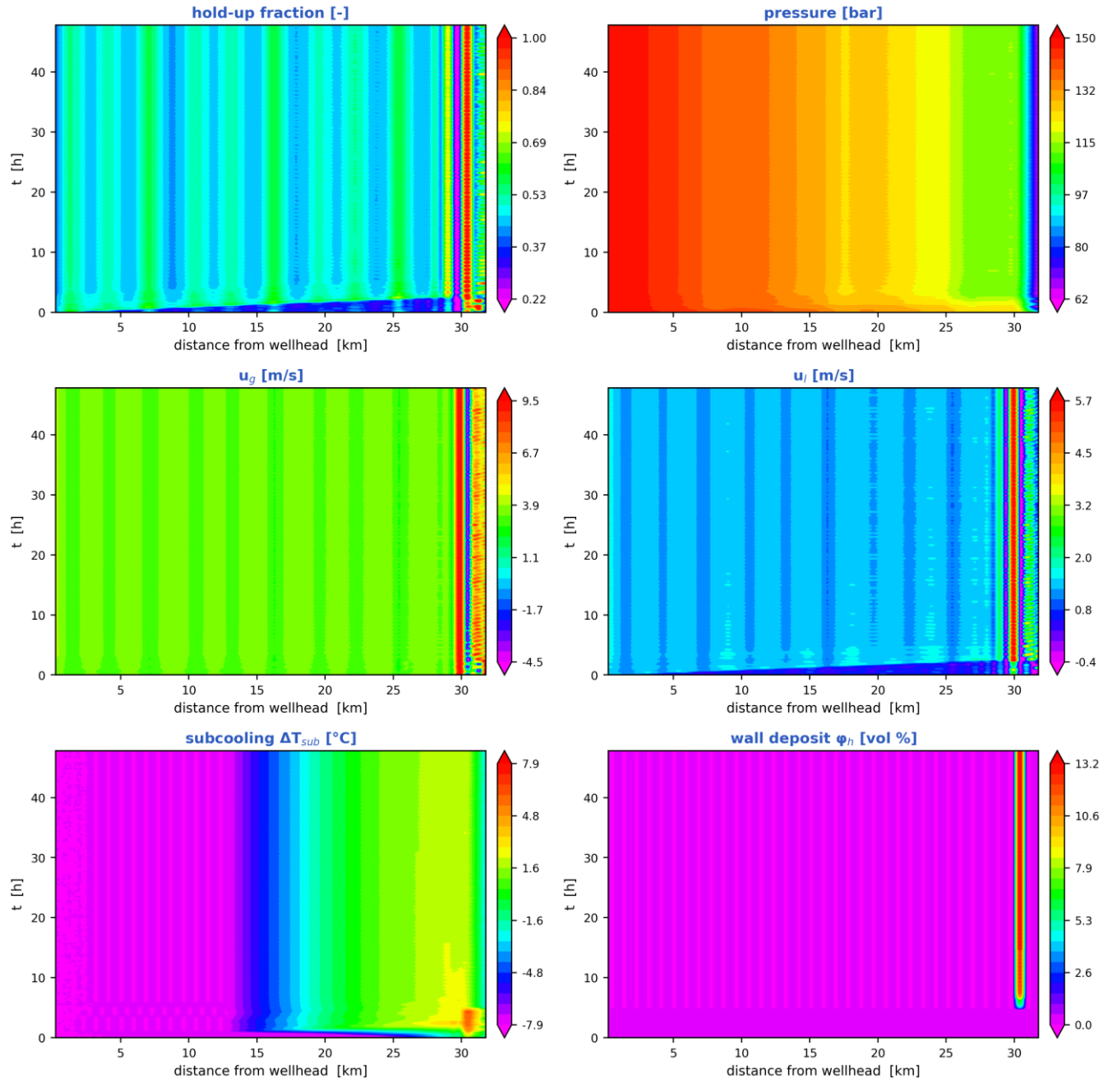
Depth-time waterfall over the steel-catenary riser: slug boundaries during upward motion, their trajectories (slope = translational celerity) and the slug unit length marked on the depth axis. Sub-grid reconstruction — see §note. [as-operated]



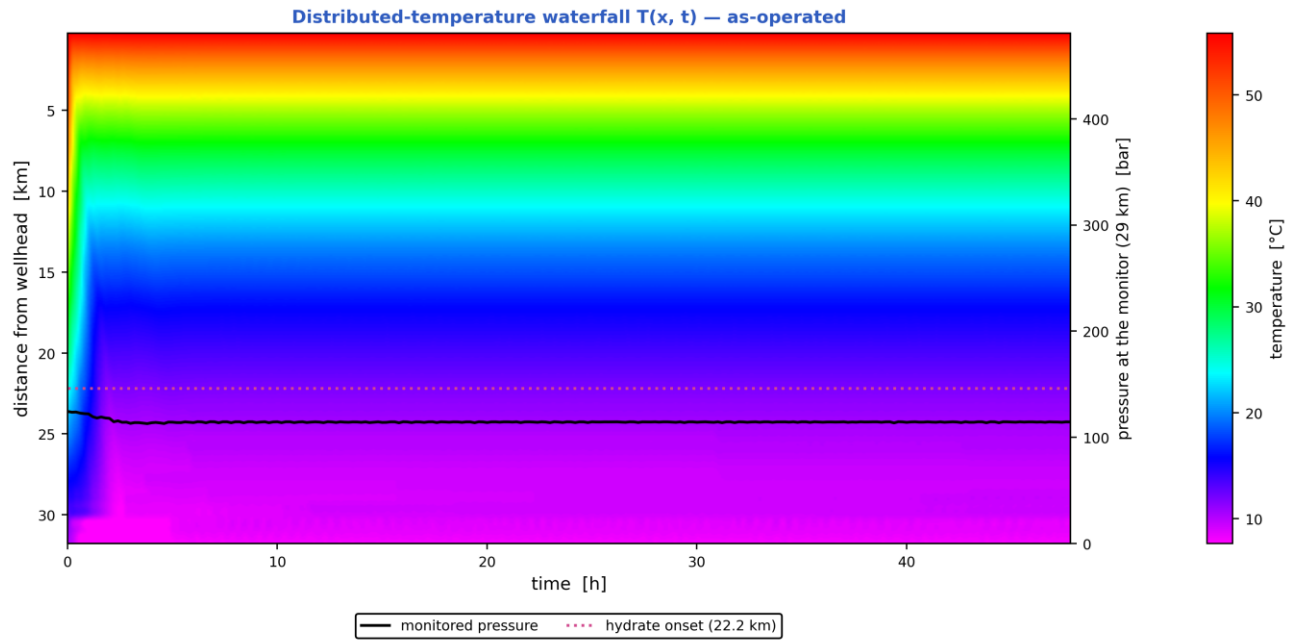
the colour scale is set by the flowline ($\alpha_l = 0.34$ – 0.58); the riser, the last 10 % of the route, runs 0.05–1.00 and saturates at both ends

Liquid-holdup field $\alpha_l(x,t)$ — the bands are slug activity migrating along the line in time (space-time contour map). [as-operated]

Space-time solution of the 32 km tie-back — as-operated (N = 70 cells, 240 snapshots)

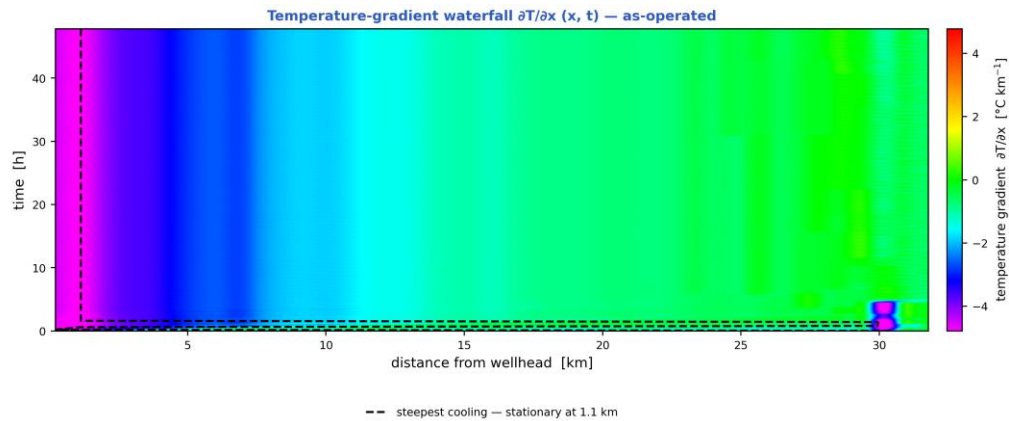


The TRUE space-time solution of the tie-back: liquid holdup, pressure, gas and liquid velocities, subcooling and wall-deposit fraction, each as a filled-contour field over (distance, time). [as-operated]



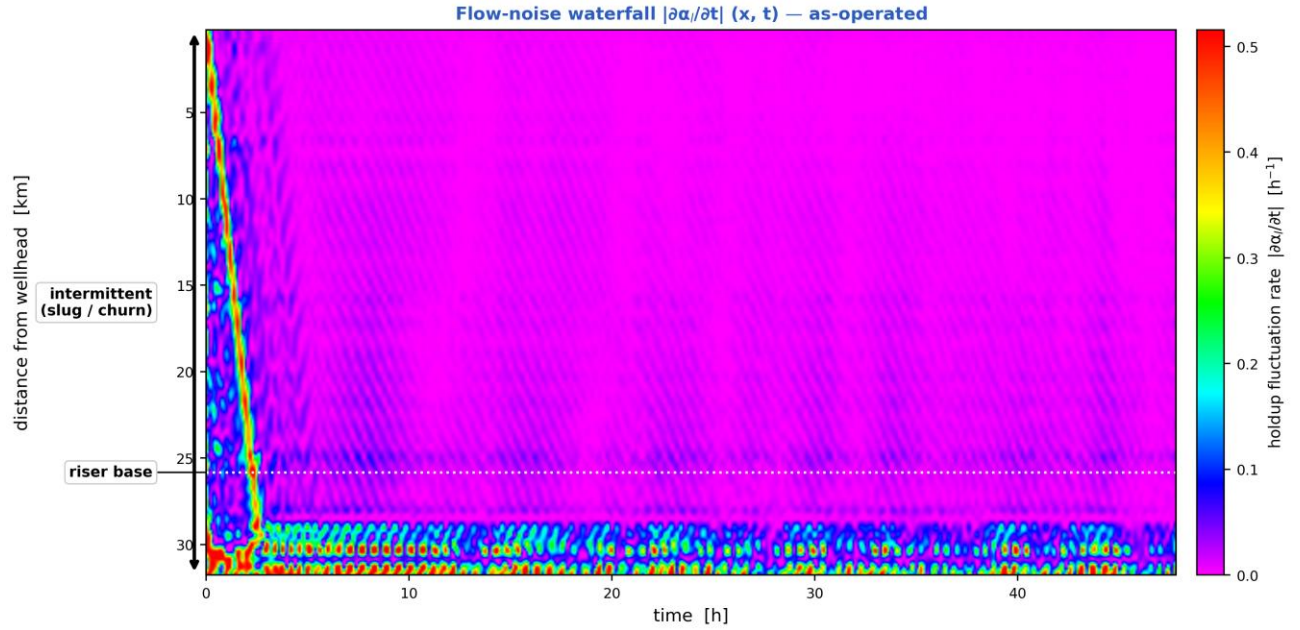
The dotted line is the hydrate onset at the final state, 22.2 km from the wellhead; everything beyond it is subcooled.

Distributed-temperature waterfall $T(x, t)$ — the thermal field over distance and time, with the monitored pressure overlaid, the operating stages marked and the hydrate-onset distance annotated. [as-operated]



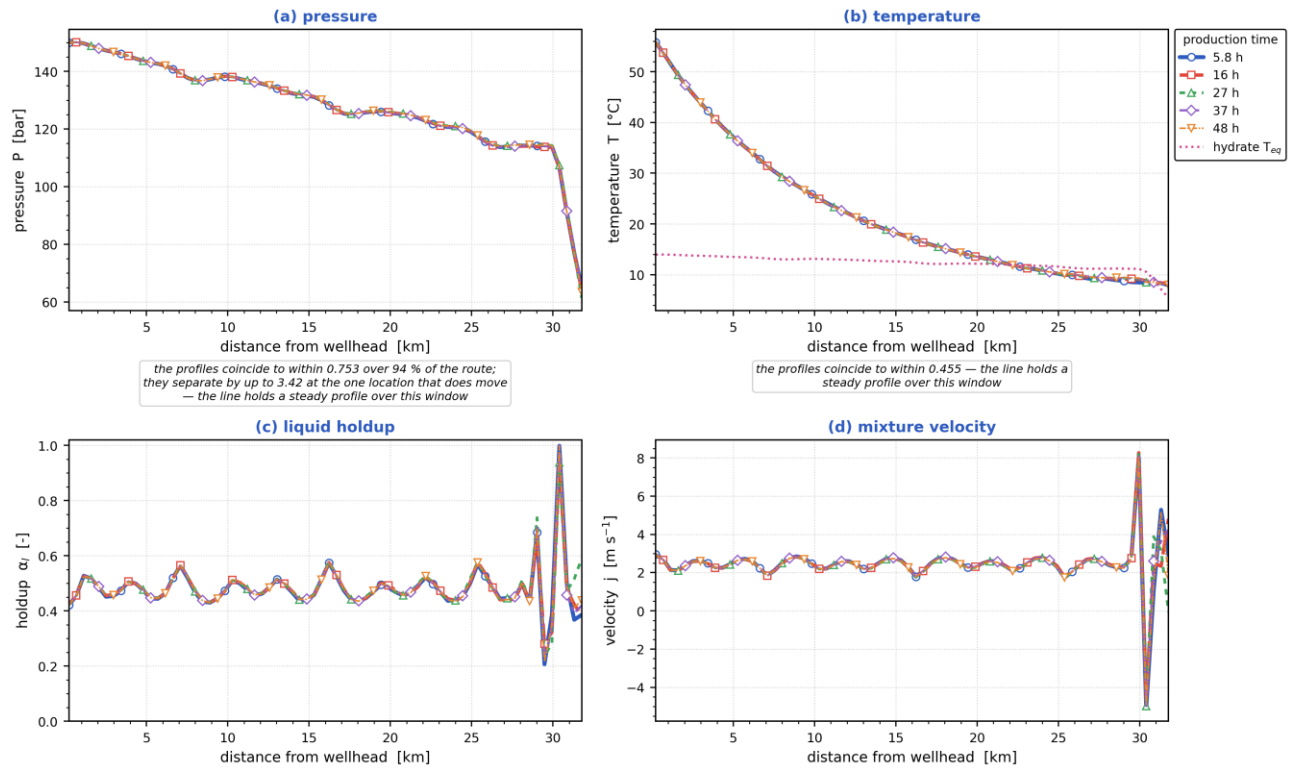
The dashed line tracks the steepest cooling at each instant. On this duty it does NOT travel: the hot inlet meeting a cold seabed holds the steepest gradient at 1.1 km, and stays within 0.0 km of it for 80 % of the window — the thermal front is stationary, not propagating.

Temperature-gradient waterfall $\partial T / \partial x (x, t)$. A travelling thermal front is a narrow band of steep gradient, so it is localised here even where the temperature map looks smooth; the dashed line tracks the steepest cooling. [as-operated]



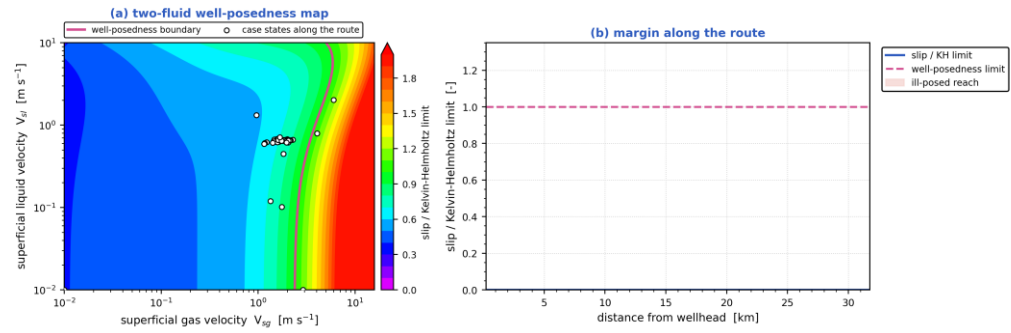
Flow-noise waterfall $|\partial\alpha_i/\partial t|(x, t)$ — where the liquid holdup changes fastest is where the flow is most unsteady, with the intermittent (slug/churn) reach and the riser base marked. [as-operated]

Parameters along the pipeline at successive times — as-operated



Pressure, temperature, liquid holdup and mixture velocity along the route, each at the same successive times. [as-operated]

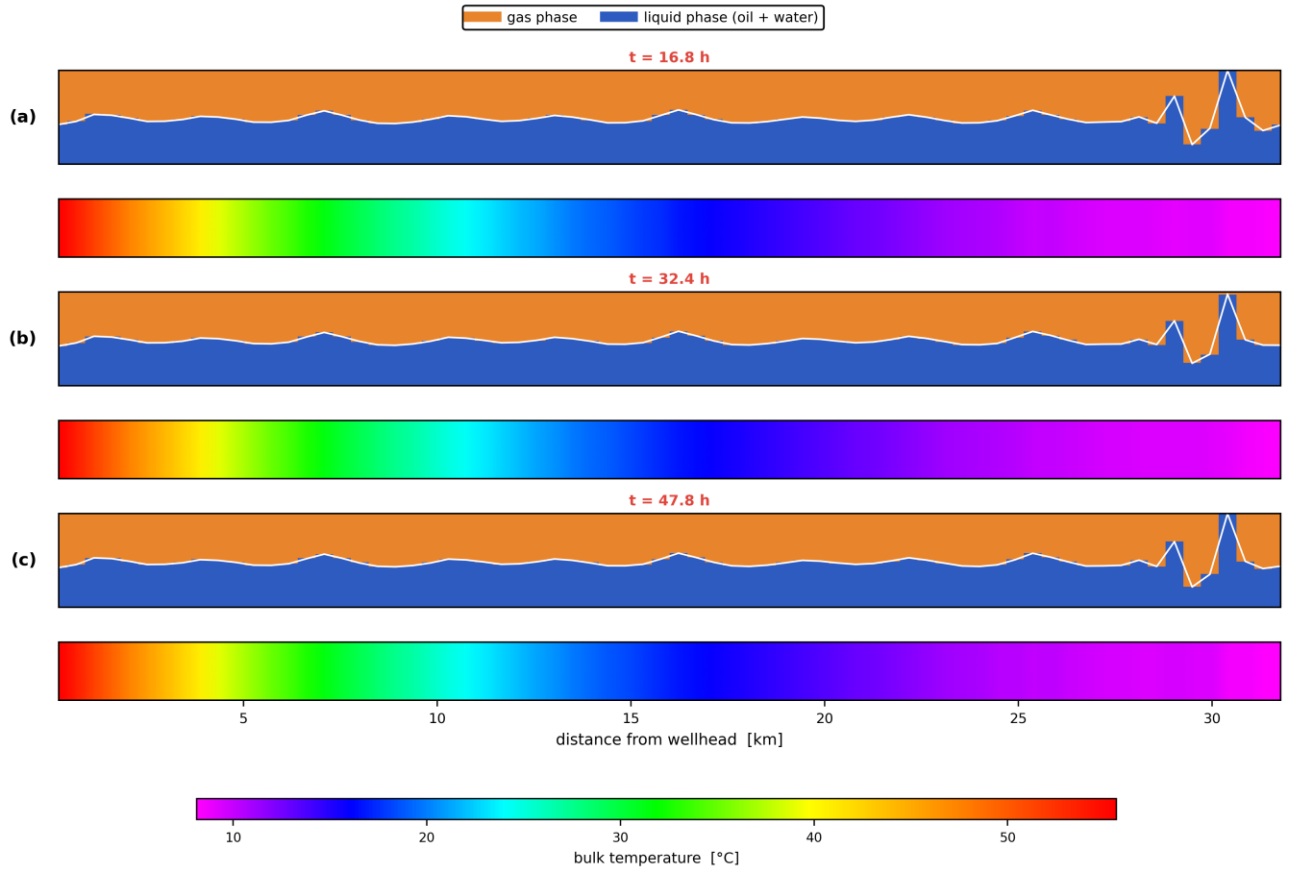
Well-posedness of the two-fluid description — as-operated



the interfacial-pressure term ($C = 1.20 \pm 1$) makes the two-fluid system unconditionally hyperbolic, so the initial-value problem is well posed over the whole route; against the INVISCID criterion ($C = 0$) 7 % of the route would be past the limit, and that is the reference the literature quotes

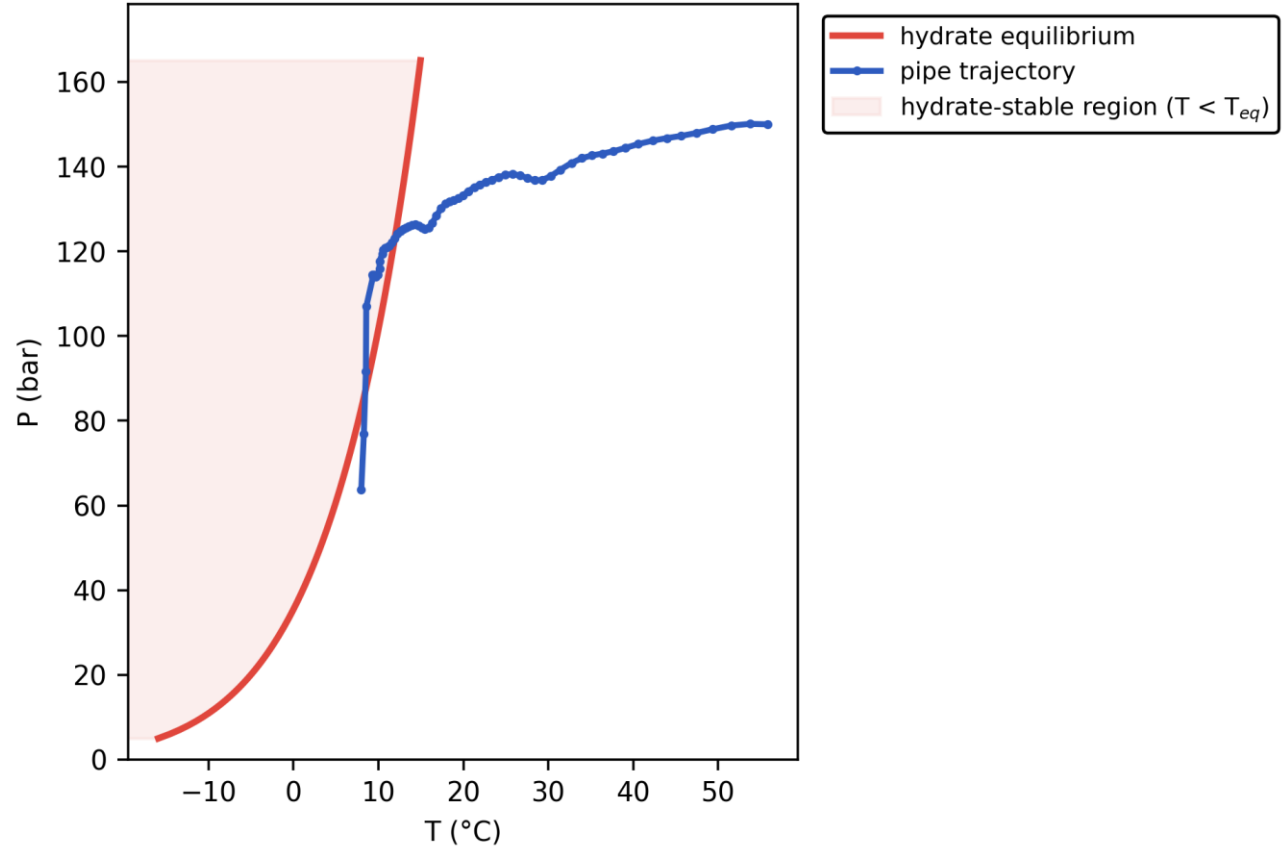
Well-posedness of the two-fluid description: (a) the inviscid Kelvin–Helmholtz boundary over the superficial-velocity plane with the case's own states, (b) the slip/KH margin along the route. Past the boundary the 1-D two-fluid initial-value problem is ill-posed and growth rates are grid-dependent. [as-operated]

Pipeline cloud maps at successive times — phase distribution (upper) and temperature (lower), as-operated



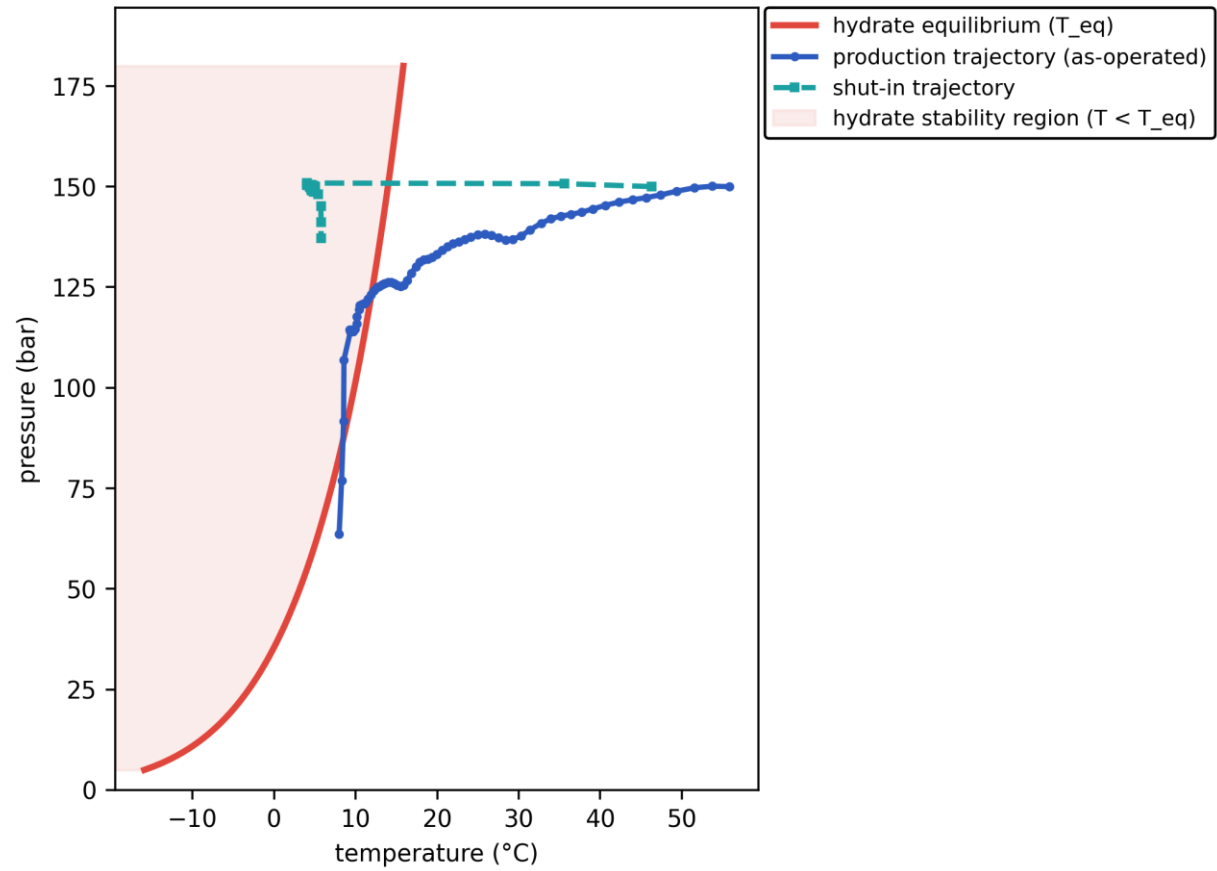
Pipeline cloud maps at successive times: the gas/liquid phase distribution inside the bore (upper strip of each pair) above the bulk-temperature field along the same reach (lower strip), on a shared temperature scale. [as-operated]

Output C — P-T trajectory vs hydrate envelope

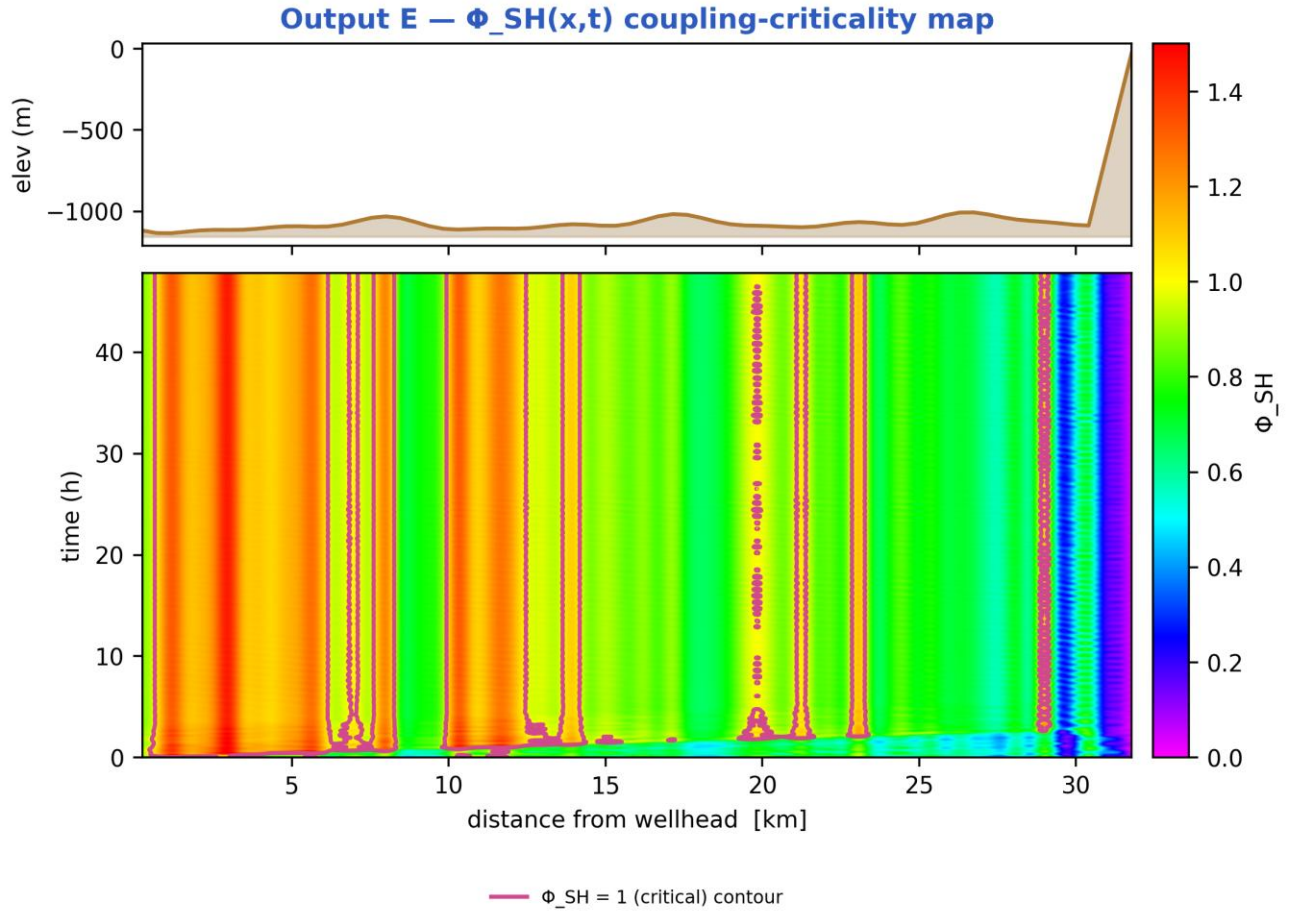


Production P–T trajectory against the hydrate-stability envelope (curve). [as-operated]

Hydrate-formation prediction — P-T trajectories vs envelope

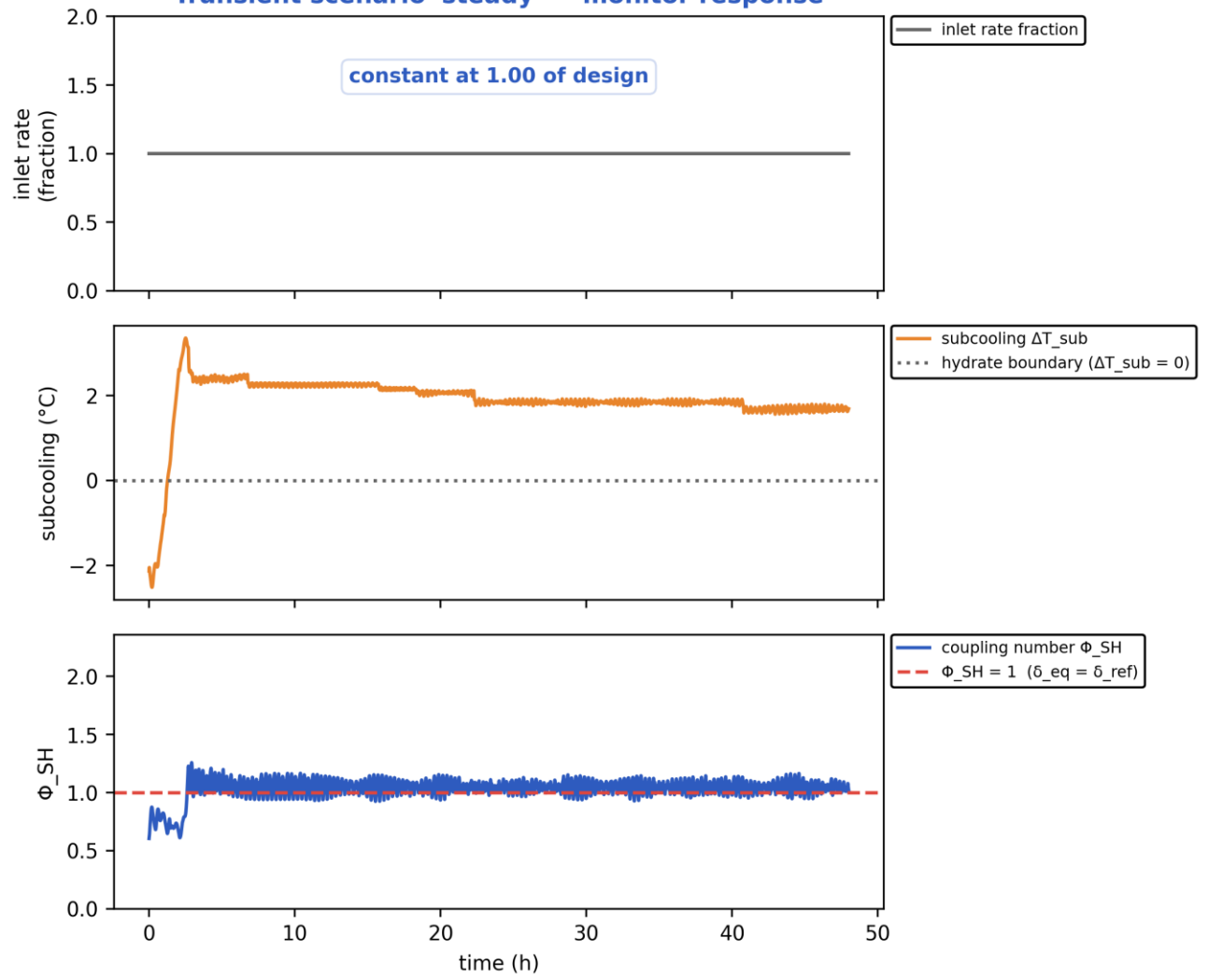


Hydrate-formation prediction: production AND shut-in P–T trajectories overlaid on the hydrate envelope. [as-operated]



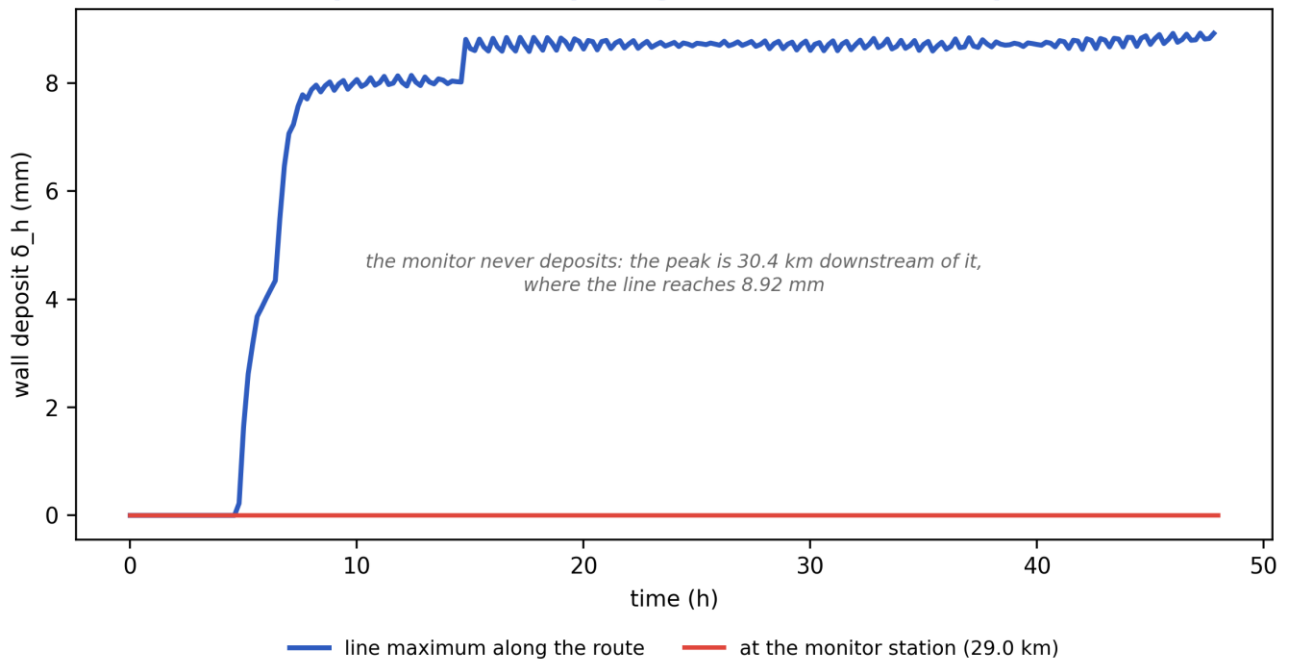
Slug-Hydrate coupling-criticality map $\Phi_{SH}(x,t)$. Φ_{SH} is the equilibrium deposit thickness in units of $\delta_{ref} = 21.2$ mm; the contour drawn is $\Phi_{SH} = 1$, just below the derived runaway threshold $\Phi_{crit} = 1.08$, which separates slug-scoured from plugging-critical (space-time map). [as-operated]

Transient scenario 'steady' — monitor response



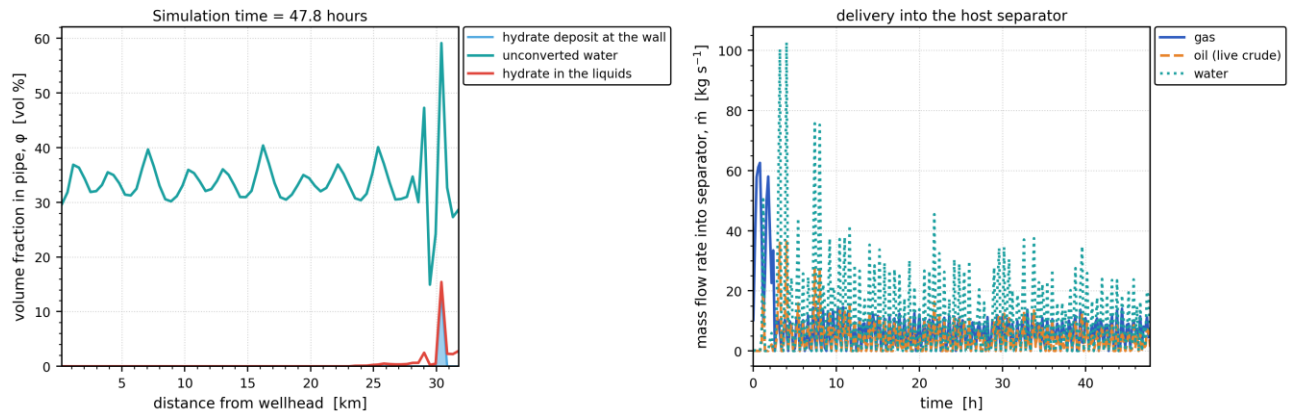
Monitored-station transient response vs time (curves). [as-operated]

Output D — wall-deposit growth (transient, coupled)



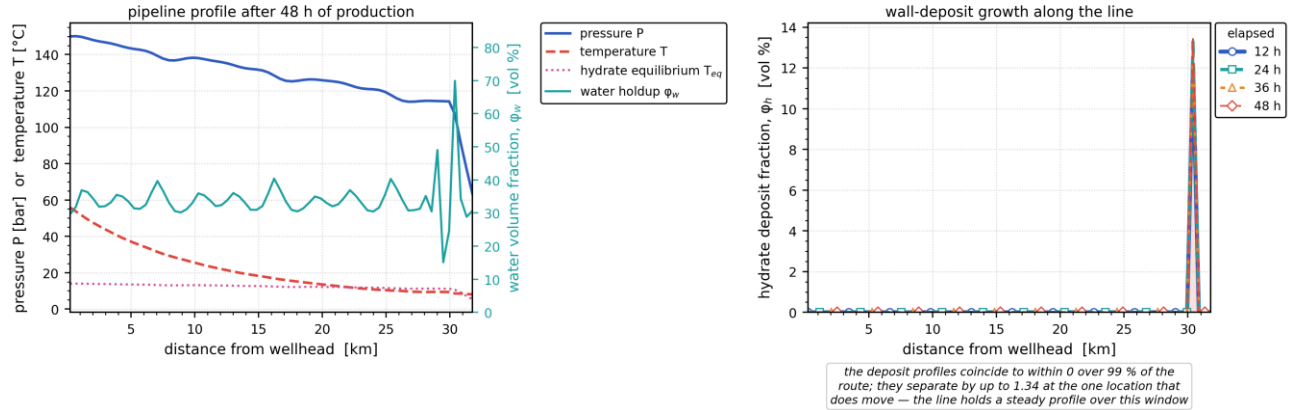
Wall-deposit growth at the monitor station vs time (curve). [as-operated]

In-pipe hydrate/water distribution and host delivery — as-operated

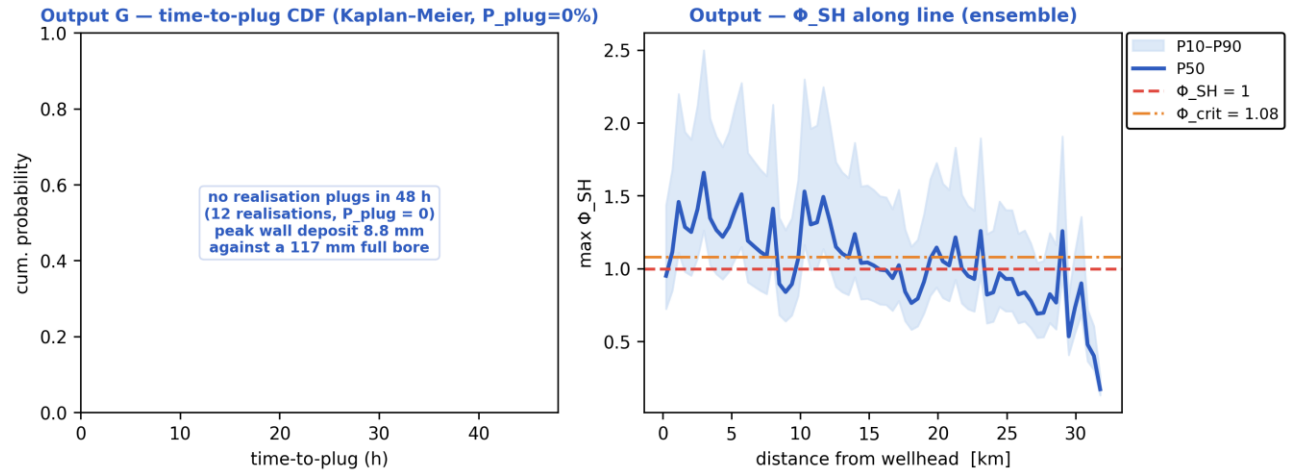


(a) In-pipe volume fractions along the route at the reported time — unconverted water, hydrate carried in the liquids, and the hydrate deposit standing on the wall; (b) the gas, oil and water mass rates delivered into the host separator against time. [as-operated]

Late-time pipeline state and deposit growth — as-operated

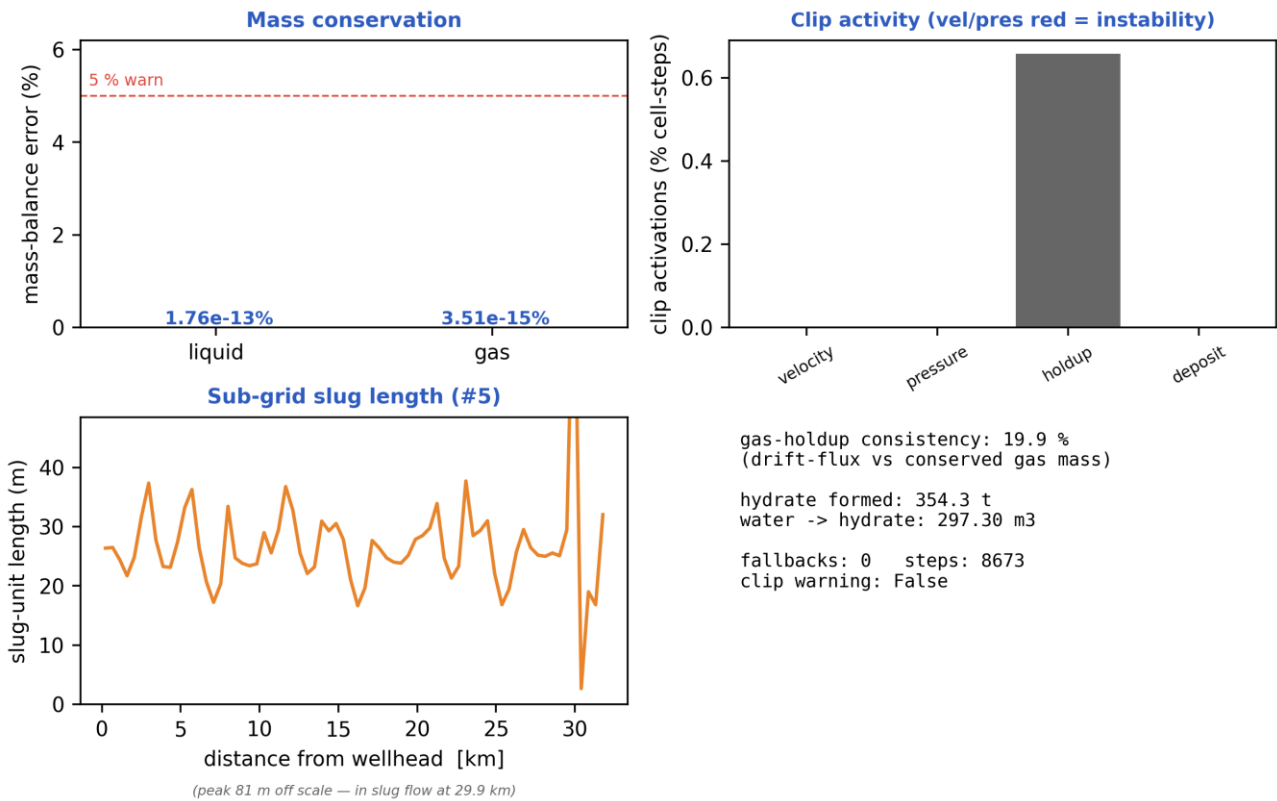


(a) The late-time pipeline profile — pressure, temperature against the hydrate equilibrium temperature, and the water volume fraction; (b) the wall-deposit volume fraction along the line at successive elapsed times. [as-operated]

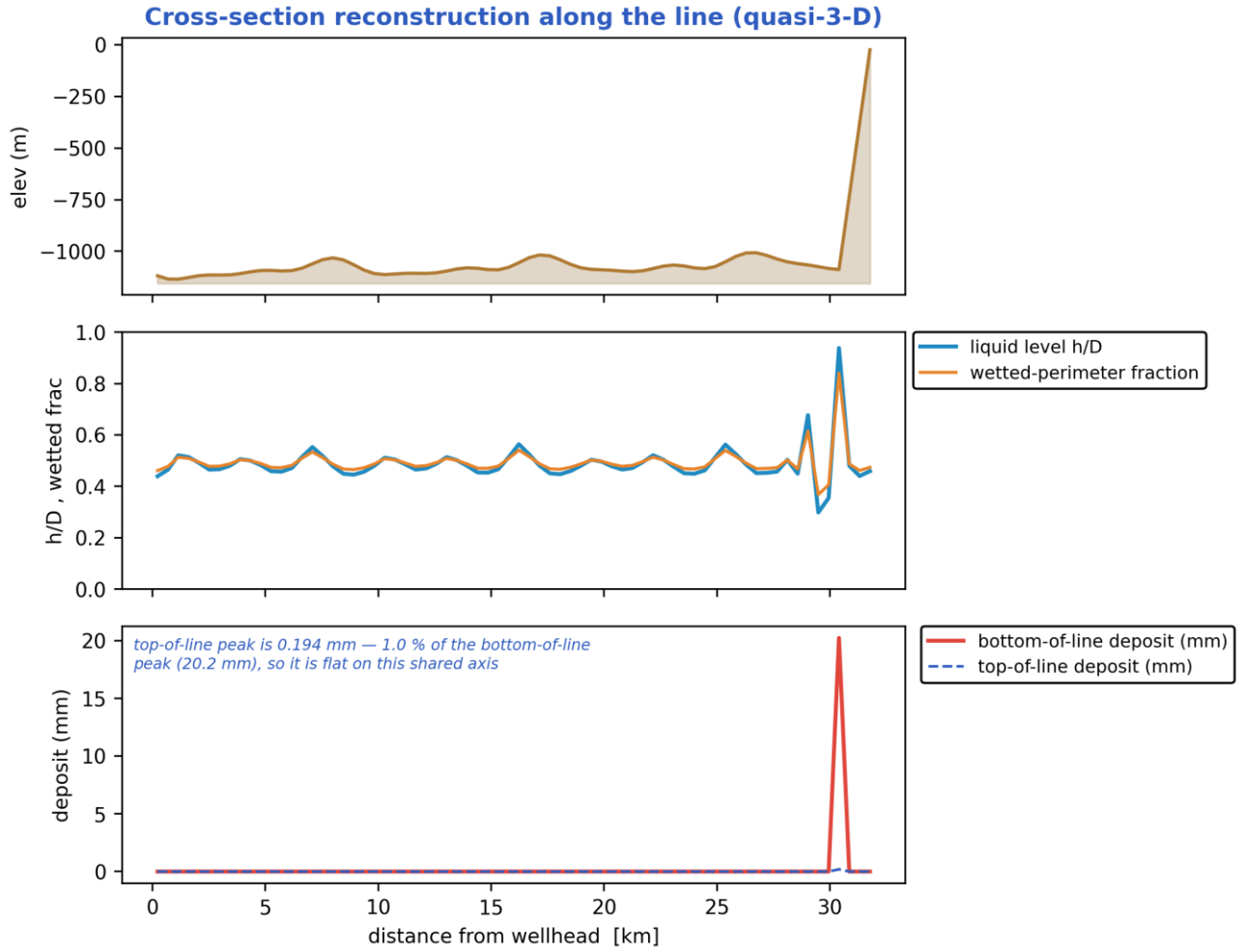


Probabilistic time-to-plug CDF and the max- Φ_{SH} P10–P90 uncertainty band (Monte-Carlo ensemble). [as-operated]

Output — solver diagnostics & balances

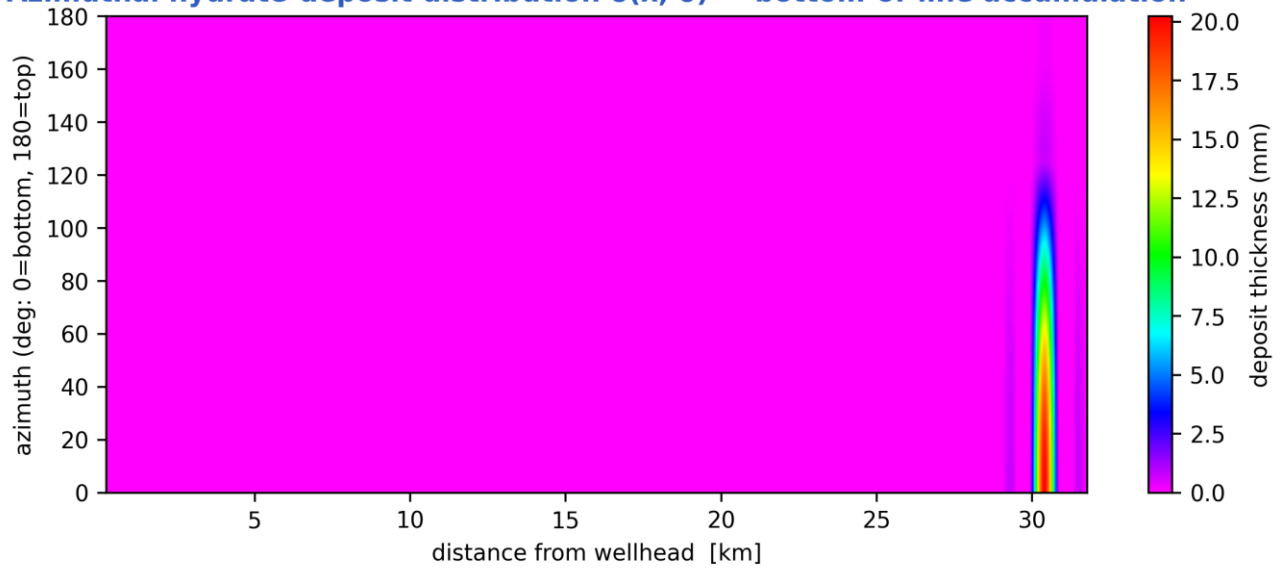


Solver diagnostics: liquid & gas mass balances, clip activity, slug length and numerical consistency. [as-operated]



Quasi-3-D cross-section geometry reconstructed along the line (curves). [as-operated]

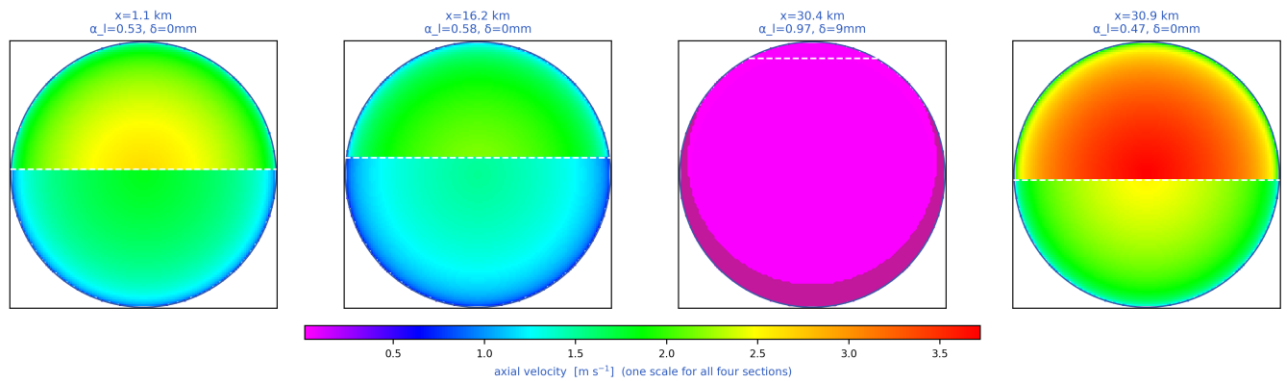
Azimuthal hydrate-deposit distribution $\delta(x, \theta)$ — bottom-of-line accumulation



bottom-of-line thickness: the colour scale is its own range, 0–20.2 mm, against a 127 mm bore radius (15.9 % closed at its thickest). Its circumferential MEAN is the area-mean peak_deposit_mm reported elsewhere, which is smaller by the azimuthal skew.

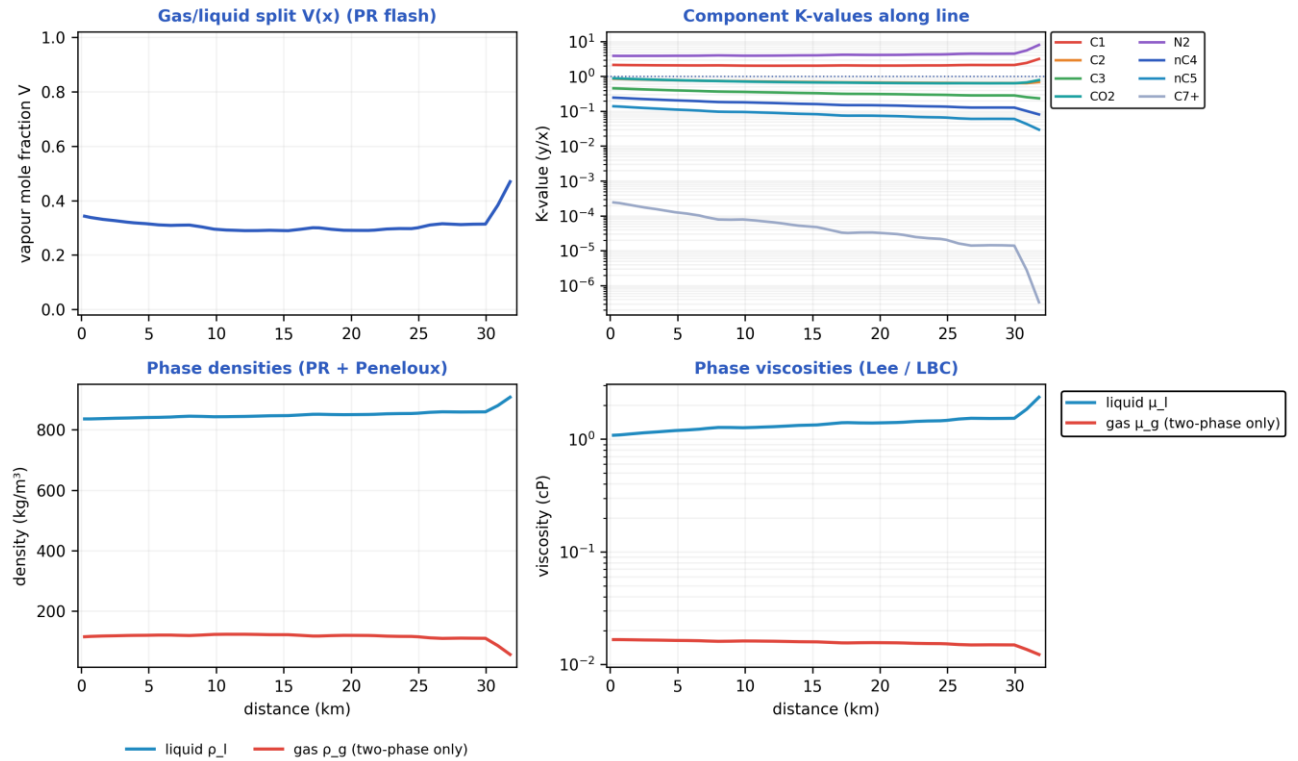
Azimuthal (bottom-of-line) wall-deposit map around the pipe circumference along the route (contour map). [as-operated]

2-D cross-section reconstruction — velocity field, gas/liquid interface (white dashed), wall deposit (magenta)



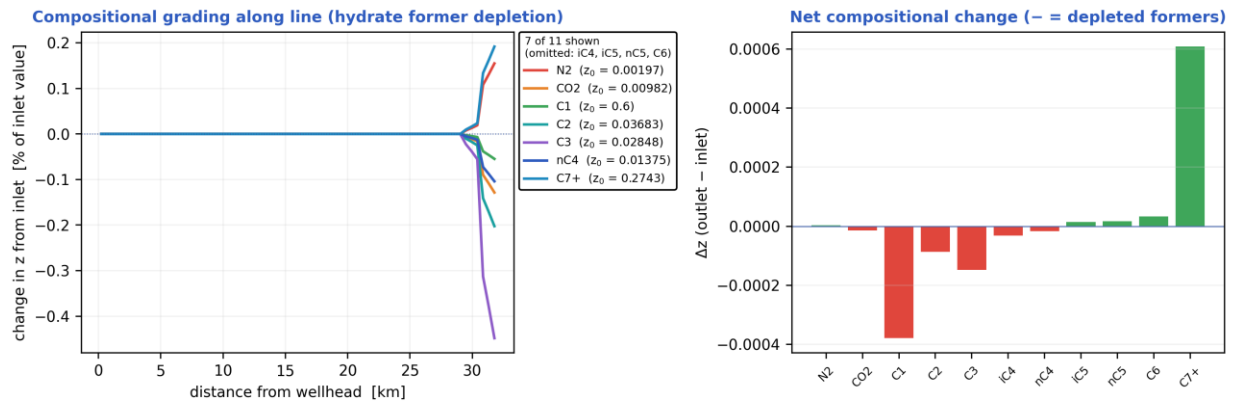
2-D pipe cross-section reconstructions at selected stations (holdup level + deposit ring). [as-operated]

Compositional / PVT tracking along the line (Peng-Robinson EOS)



EOS on the hydrocarbon composition alone: two-phase at 40 of 40 stations, liquid 849 kg/m³ and 1.36 cP. That is within 1.1 % of the flow model's oil (858 kg/m³); its liquid reads 975 kg/m³ only because it carries the 70 % water cut. The PHASE SPLIT agrees in character too: the EOS flashes two-phase at 40 of 40 stations against the flow model's 52 % gas, so this panel and the flow figures describe one fluid.

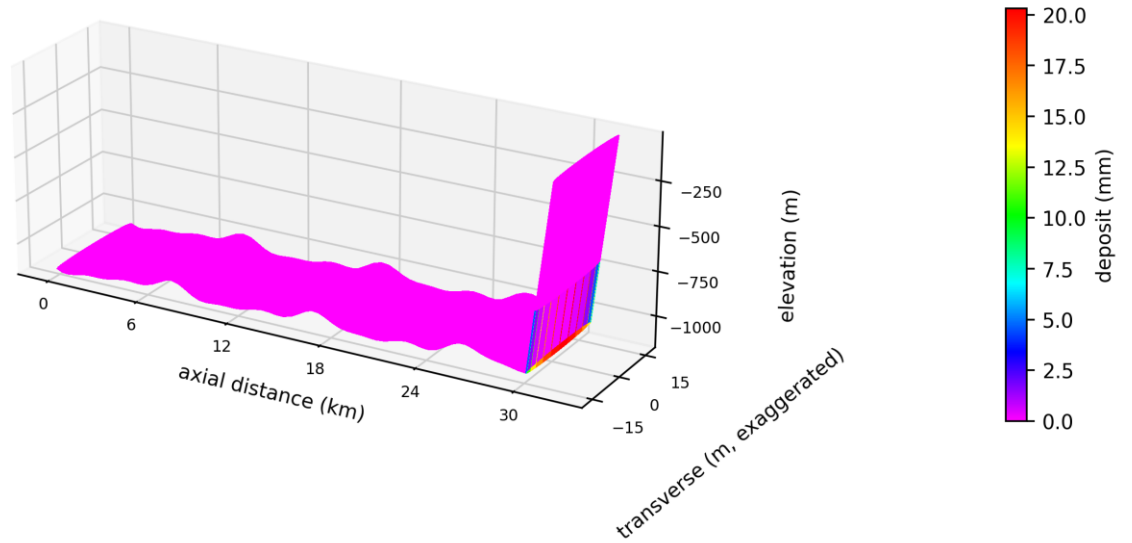
Compositional PVT tracking along the line (Peng-Robinson flash): gas/liquid split, K-values, densities. [as-operated]



Δz is a change in mole FRACTION: a component can rise here while still being consumed, because removing the others renormalises what is left. Total formers consumed: 0.22 % of feed moles.

Compositional grading along the route — preferential depletion of the hydrate-forming light ends. [as-operated]

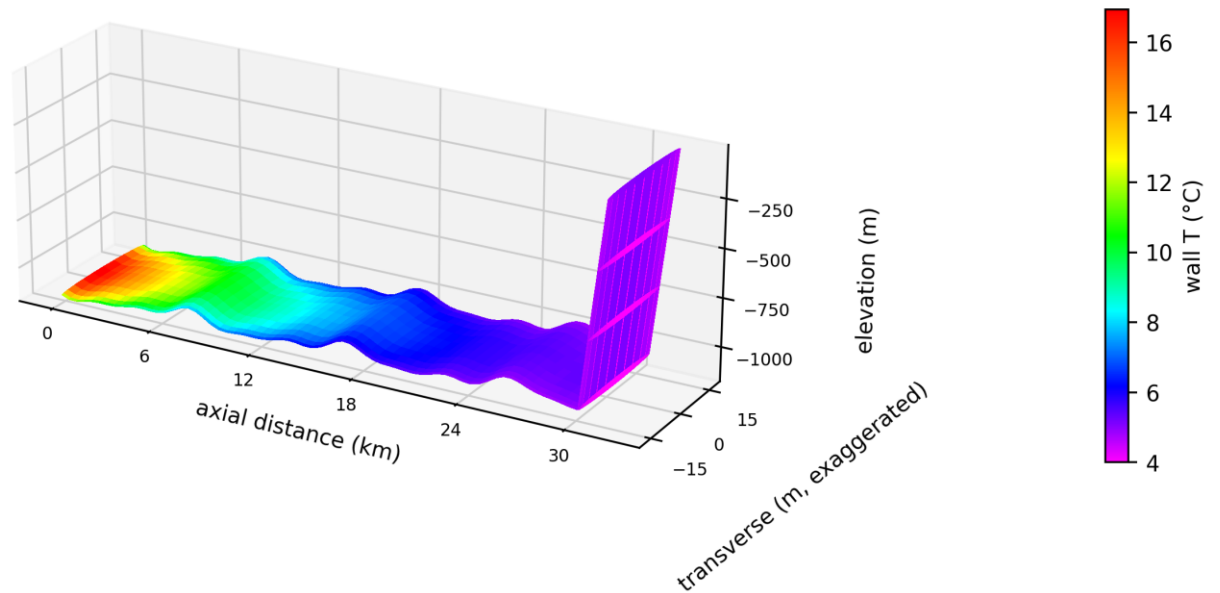
3-D reconstructed pipe — hydrate wall-deposit distribution



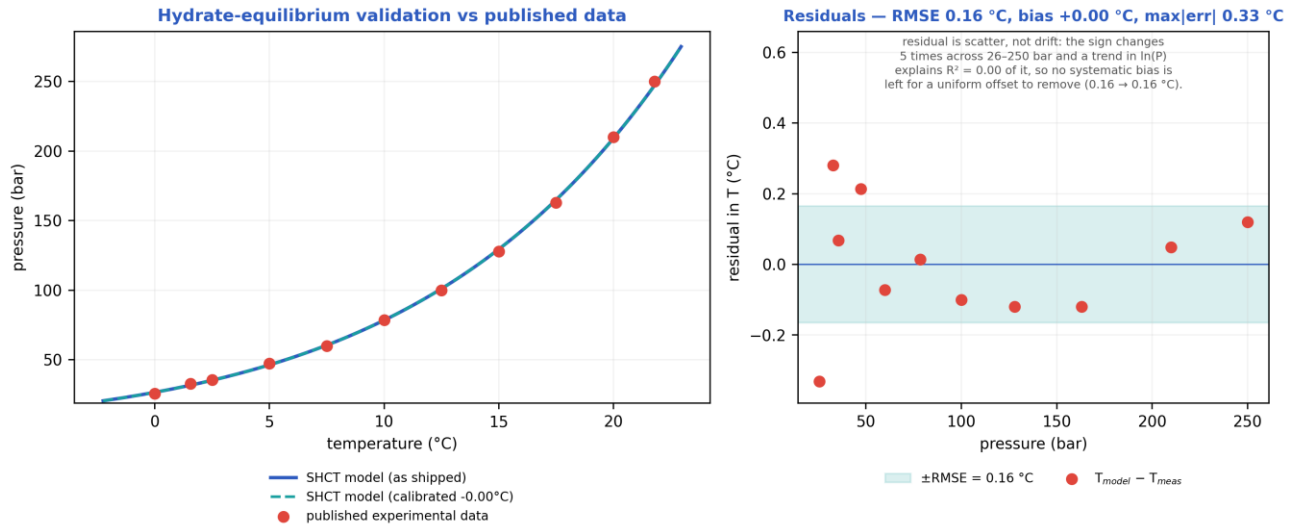
deposit is non-zero only over 30.4–30.4 km (1 % of the route); it peaks at 20.3 mm at 30.4 km. The rest of the tube is at zero, not unmeasured.

3-D reconstructed pipe coloured by wall-deposit thickness. [as-operated]

3-D reconstructed pipe — wall temperature distribution

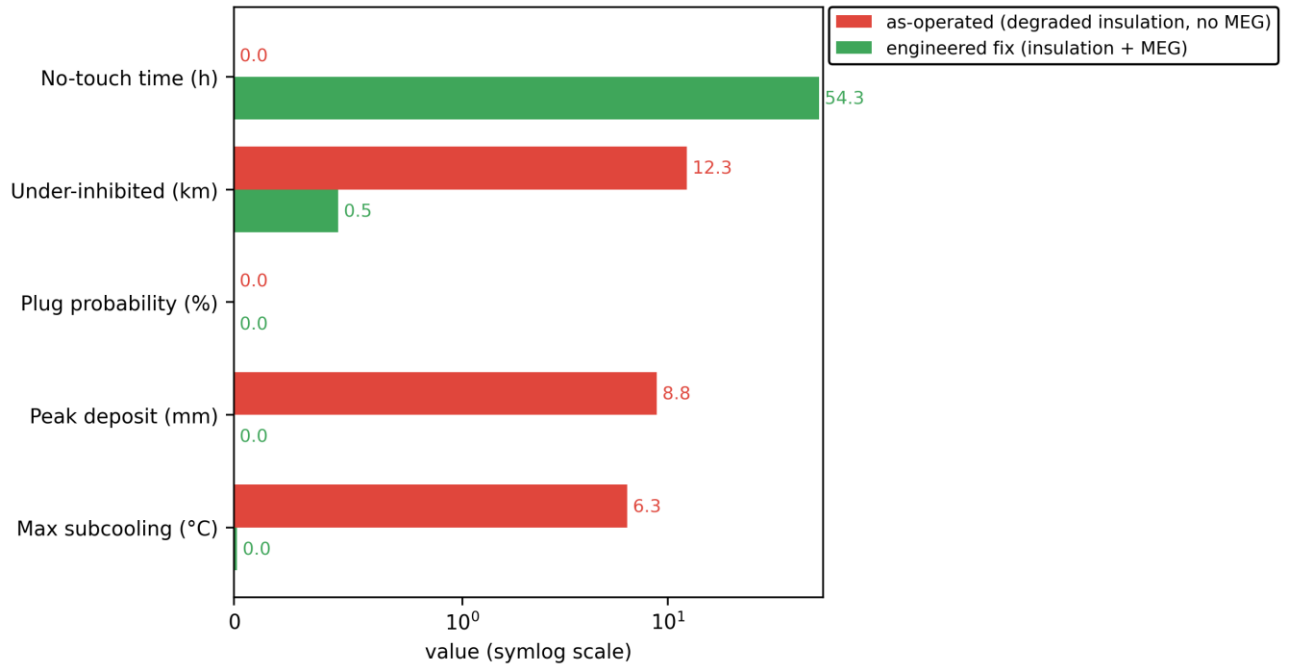


3-D reconstructed pipe coloured by wall temperature. [as-operated]



Solver hydrate-equilibrium closure vs published experimental data (validation curve). [as-operated]

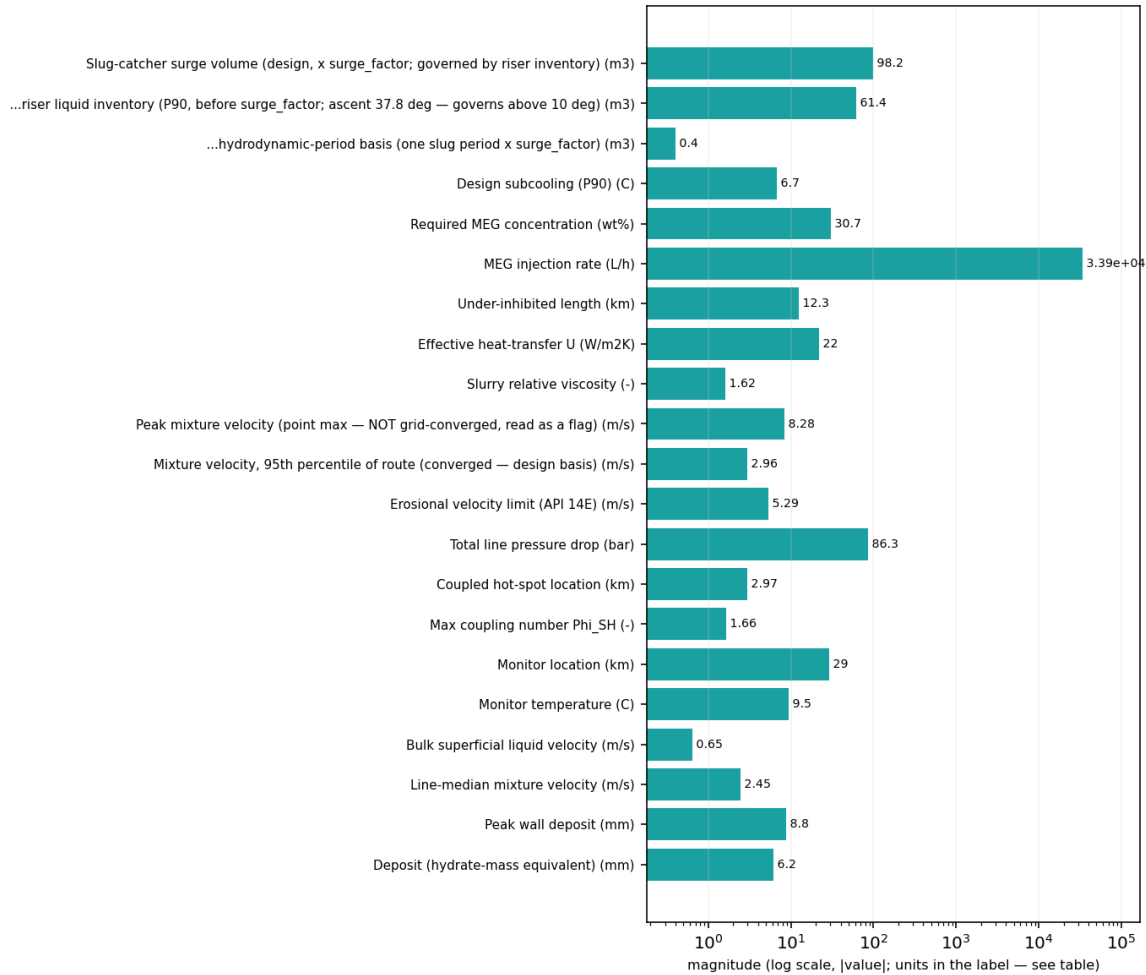
Mitigation comparison — model used as a flow-assurance design tool



Mitigation comparison — as-operated vs the engineered fix across the key flow-assurance metrics (bar chart). [as-operated]

11.1.3 Engineering deliverables (engineering_deliverables.csv)

engineering_deliverables.csv — numeric deliverables [as-operated]
(bar chart: these are unrelated named quantities, not a curve)



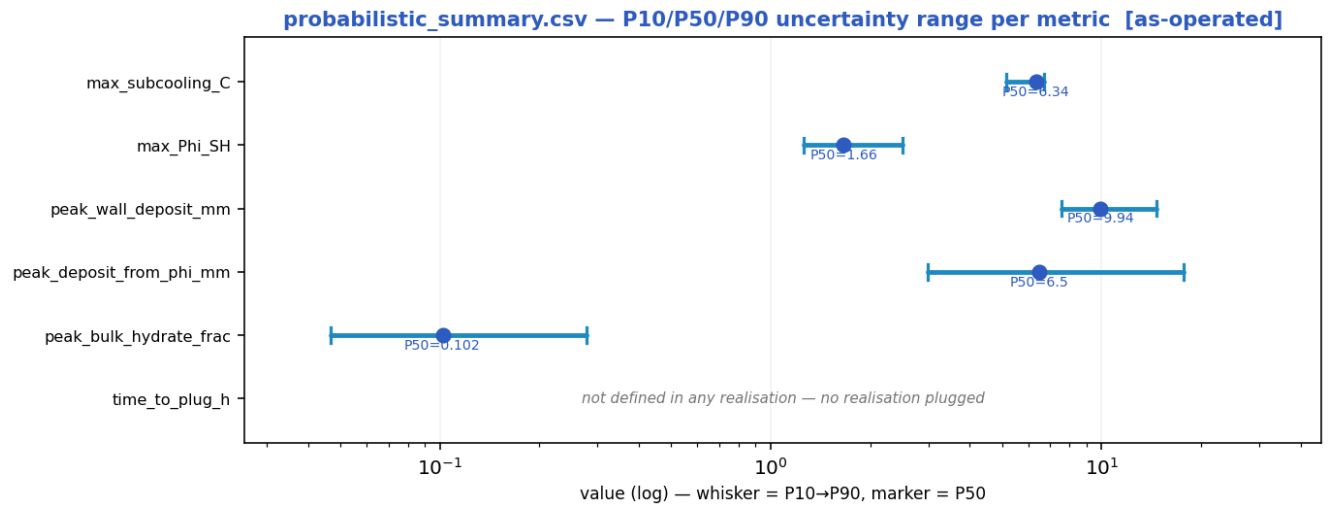
exactly zero, so not drawable on a log axis: MEG injected at inlet (wt%); Cooldown to hydrate (no-touch time) (h)
Probability of plugging (run) (%); Mass-conservation error (%)
Liquid discarded at the bore-closure bound (% of liquid in)

Numeric engineering deliverables as a magnitude chart [as-operated].

deliverable	value	units
Slug-catcher surge volume (design, x surge_factor; governed by riser inventory)	98.19	m3
...riser liquid inventory (P90, before surge_factor; ascent 37.8 deg — governs above 10 deg)	61.37	m3
...hydrodynamic-period basis (one slug period x surge_factor)	0.40	m3
Design subcooling (P90)	6.70	C
Required MEG concentration	30.7	wt%
MEG injection rate	33886.3	L/h
MEG injected at inlet	0.0	wt%
Under-inhibited length	12.34	km
Effective heat-transfer U	22.00	W/m2K
Cooldown to hydrate (no-touch time)	0.00	h

Slurry relative viscosity	1.62	-
Slurry transportable?	True	-
Peak mixture velocity (point max — NOT grid-converged, read as a flag)	8.28	m/s
Mixture velocity, 95th percentile of route (converged — design basis)	2.96	m/s
Slug-unit length (over the slugging reach)	26.7 mean / 80.6 max	m — over 100 % of the route
Erosional velocity limit (API 14E)	5.29	m/s
Total line pressure drop	86.3	bar
Probability of plugging (run)	0	%
Time-to-plug (P50)	n/a — nothing plugged	h
Coupled hot-spot location	2.97	km
Max coupling number Phi_SH	1.659	-
Phi_SH reported at plot cap (saturated)?	False	-
Phi_SH shear-limited (near-zero velocity, magnitude not resolved)?	False	-
Monitor location	29.03	km
Monitor temperature	9.5	C
Bulk superficial liquid velocity	0.65	m/s
Line-median mixture velocity	2.45	m/s
Peak wall deposit	8.8	mm
Wall deposit at full bore?	False	-
Deposit (hydrate-mass equivalent)	6.2	mm
No-touch time source	already-subcooled	-
Mass-conservation error	0.00	%
Liquid discarded at the bore-closure bound	0.00	% of liquid in — a 1-D model limit once the bore shuts; not a solver error
Mass-balance reliability warning	False	-

11.1.4 Probabilistic summary (probabilistic_summary.csv)



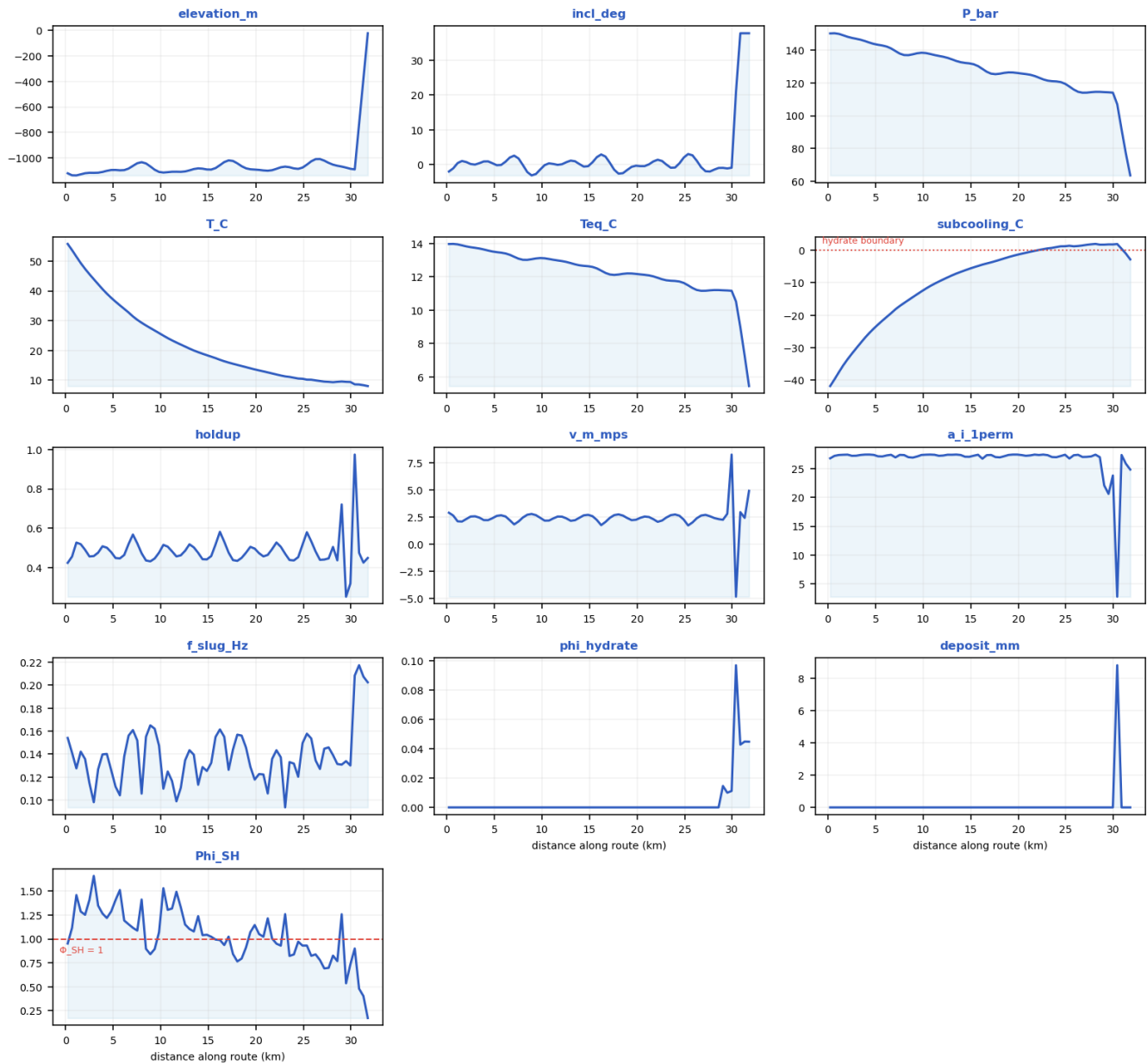
peak_wall_deposit_mm is what the transient actually deposits on the wall, capped at the full-bore radius once the bore shuts.
peak_deposit_from_phi_mm is the annulus equivalent to the peak SUSPENDED bulk hydrate fraction ϕ at one instant, $t = \phi \cdot D/4$.
These are different quantities, not a result and its bound: ϕ is instantaneous while wall deposit accumulates, so on a plugging duty the wall value exceeds the ϕ -equivalent (shut-in: 117 mm against 35 mm) and on a sub-critical one it falls well below it.

P10 → P90 uncertainty range with the P50 marker per metric [as-operated].

metric	P10	P50	P90
max_subcooling_C	5.16	6.34	6.7
max_Phi_SH	1.26	1.66	2.5
peak_wall_deposit_mm	7.55	9.94	14.7
peak_deposit_from_phi_mm	2.98	6.5	17.7
peak_bulk_hydrate_frac	0.0468	0.102	0.278
time_to_plug_h	n/a	n/a	n/a

11.1.5 fields_profile.csv — steady-state spatial profile of every field along the route

fields_profile.csv — every column as a curve vs distance along route (km) [as-operated]



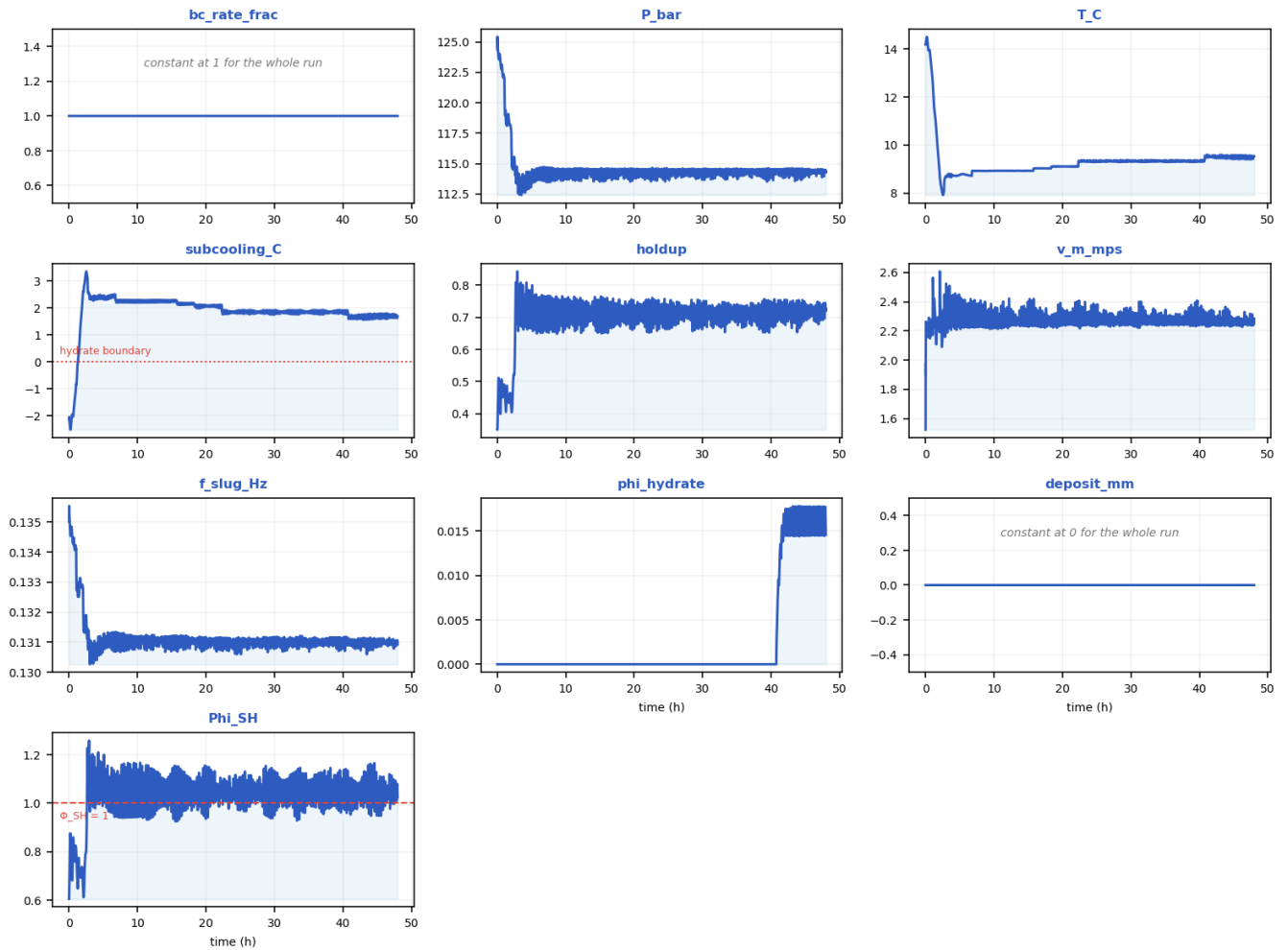
Every column of fields_profile.csv drawn as a curve [as-operated].

x_km	elevation_m	incl_deg	P_bar	T_C	Teq_C	subcooling_C	holdup	v_m_mps	regime	a_i_1perm	f_slug_Hz	phi_hydrate	deposit_mm	Phi_SH
0.22857	-1118.4	-1.9608	150	55.804	13.963	-41.843	0.42292	2.927	slug	26.851	0.15413	0	0	0.95149
2.5143	-1114.5	0.18865	147.26	45.626	13.772	-31.855	0.45553	2.5765	slug	27.287	0.11484	0	0	1.4085
4.8	-1092.7	0.42362	143.64	37.68	13.515	-24.164	0.47775	2.42	slug	27.45	0.1262	0	0	1.2871

7.0857	-1060.5	2.601	139.28	31.416	13.198	-18.216	0.56717	1.849	slug	27.008	0.16096	0	0	1.1161
9.8286	-1107.7	-1.3732	138.25	25.82	13.122	-12.697	0.47485	2.4547	slug	27.435	0.14744	0	0	1.0696
12.114	-1107	0.13701	135.81	21.957	12.939	-9.0155	0.46129	2.5633	slug	27.34	0.11076	0	0	1.3319
14.4	-1082.3	-0.54271	132.07	18.87	12.654	-6.2155	0.44157	2.6844	slug	27.129	0.12883	0	0	1.0395
17.143	-1017.6	0.54352	125.54	15.896	12.136	-3.7608	0.47519	2.4542	slug	27.437	0.12641	0	0	1.0239
19.429	-1087.1	-0.61307	126.26	13.965	12.195	-1.772	0.50462	2.2433	slug	27.503	0.12911	0	0	1.0711
21.714	-1094.2	0.93887	124.13	12.238	12.022	-0.22189	0.49303	2.3308	slug	27.499	0.13586	0	0	1.004
24.457	-1083.8	0.40028	120.42	10.529	11.713	1.1779	0.45188	2.6487	slug	27.25	0.12037	0	0	0.9713
26.743	-1007	-0.74006	113.96	9.6944	11.156	1.4334	0.43833	2.668	slug	27.086	0.12716	0	0	0.77884
29.029	-1066	-0.91009	114.31	9.5279	11.186	1.6875	0.72058	2.2771	slug	22.129	0.13092	0.014589	0	1.2583
31.771	-25	37.781	63.691	7.9865	5.4411	-2.8584	0.44796	4.9421	slug	24.886	0.20239	0.044711	0	0.17269

11.1.6 timeseries_monitor.csv — transient response at the monitored station

timeseries_monitor.csv — every column as a curve vs time (h) [as-operated]



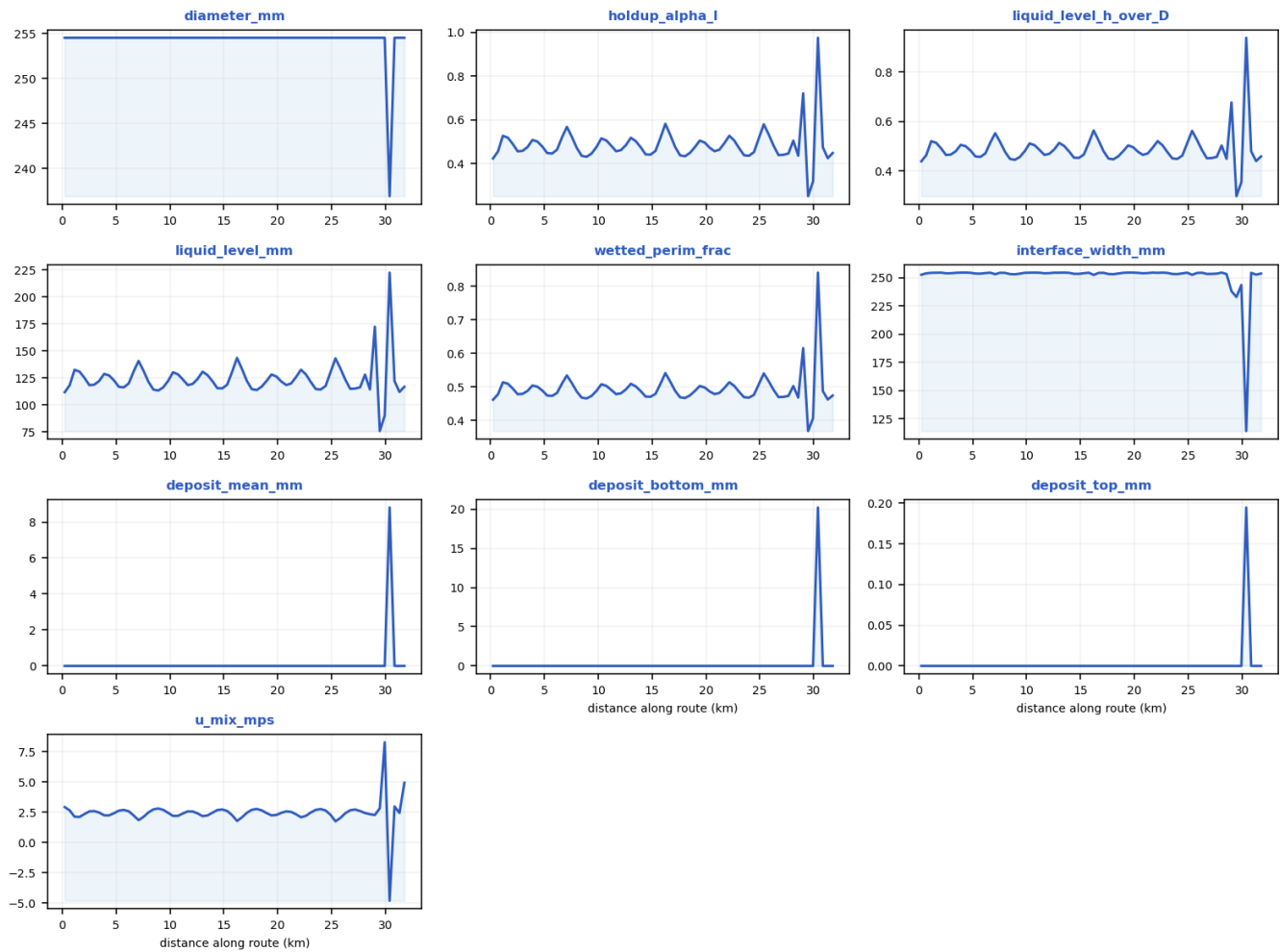
Every column of timeseries_monitor.csv drawn as a curve [as-operated].

time_h	bc_rate_frac	P_bar	T_C	subcooling_C	holdup	v_m_mps	f_slug_Hz	phi_hydrate	deposit_mm	Phi_SH
0	1	124.37	14.165	-2.1326	0.35096	1.5238	0.13502	0	0	0.60534
2.6498	1	114.2	8.0241	3.1537	0.7489	2.2539	0.13121	0	0	1.1205
6.4965	1	114.66	8.729	2.4899	0.76395	2.243	0.13128	0	0	1.1326
10.31	1	114.53	8.9257	2.2805	0.74573	2.2469	0.13112	0	0	1.1156
14.07	1	113.53	8.9259	2.2683	0.67847	2.2695	0.13073	0	0	1.0282
17.868	1	114.33	9.0277	2.1917	0.70805	2.2704	0.13097	0	0	1.0352
21.627	1	114.2	9.1106	2.034	0.70896	2.2604	0.13101	0	0	1.052
25.409	1	114.27	9.3469	1.8233	0.71981	2.3214	0.13099	0	0	1.1114
29.166	1	114.29	9.3236	1.8456	0.71711	2.2468	0.13098	0	0	0.99209
32.931	1	114.06	9.3473	1.8661	0.71334	2.2771	0.13086	0	0	1.0771
36.698	1	114.47	9.3205	1.8519	0.71782	2.265	0.13106	0	0	1.0436
40.468	1	114.47	9.3722	1.8206	0.75495	2.2699	0.13108	0	0	1.0958
44.244	1	114.47	9.4862	1.72	0.7256	2.2573	0.13109	0.017174	0	1.1052

47.995	1	114.31	9.5279	1.6875	0.72058	2.2771	0.13092	0.014589	0	1.0244
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11.1.7 csv_crossection.csv — cross-section / quasi-3-D geometry along the route

csv_crossection.csv — every column as a curve vs distance along route (km) [as-operated]



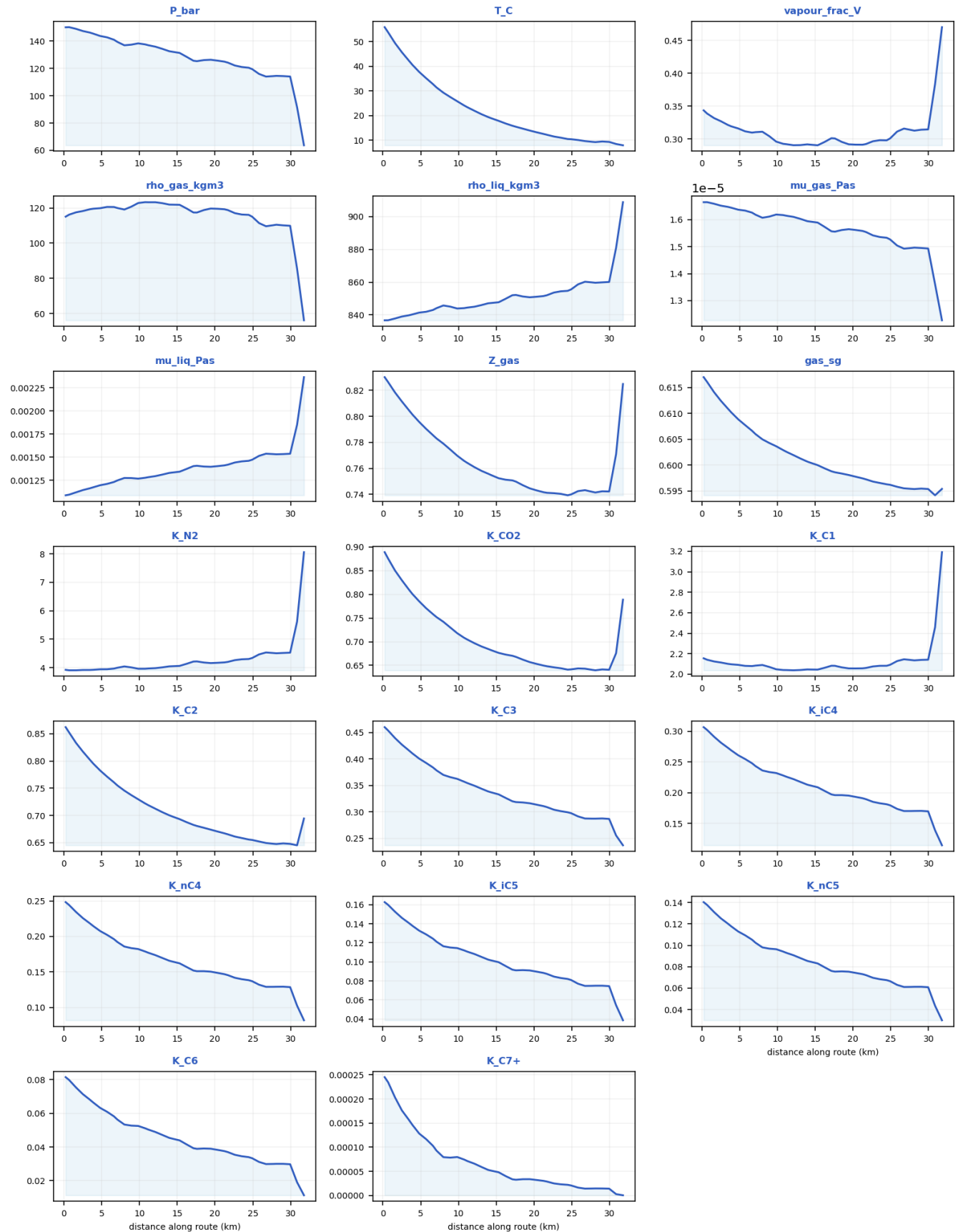
Every column of csv_crossection.csv drawn as a curve [as-operated].

x_k m	diameter _mm	holdup_a lpha_l	liquid_level_h _over_D	liquid_lev el_mm	wetted_peri m_frac	interface_wi dth_mm	deposit_me an_mm	deposit_bott om_mm	deposit_to p_mm	u_mix mps
0.22 857	254.5	0.42292	0.43931	111.8	0.46127	252.62	0	0	0	2.927
2.51 43	254.5	0.45553	0.46504	118.35	0.47773	253.88	0	0	0	2.5765
4.8	254.5	0.47775	0.48252	122.8	0.48887	254.34	0	0	0	2.42
7.08 57	254.5	0.56717	0.55286	140.7	0.53371	253.07	0	0	0	1.849
9.82 86	254.5	0.47485	0.48024	122.22	0.48742	254.3	0	0	0	2.4547
12.1 14	254.5	0.46129	0.46958	119.51	0.48062	254.03	0	0	0	2.5633

14.4	254.5	0.44157	0.45404	115.55	0.4707	253.42	0	0	0	2.6844
17.1 43	254.5	0.47519	0.48051	122.29	0.48759	254.31	0	0	0	2.4542
19.4 29	254.5	0.50462	0.50362	128.17	0.50231	254.49	0	0	0	2.2433
21.7 14	254.5	0.49303	0.49452	125.86	0.49651	254.48	0	0	0	2.3308
24.4 57	254.5	0.45188	0.46217	117.62	0.47589	253.77	0	0	0	2.6487
26.7 43	254.5	0.43833	0.45149	114.9	0.46907	253.3	0	0	0	2.668
29.0 29	254.5	0.72058	0.67702	172.3	0.61519	238.02	0	0	0	2.2771
31.7 71	254.5	0.44796	0.45908	116.84	0.47392	253.65	0	0	0	4.9421

11.1.8 csv_compositional.csv — compositional / PVT state along the route (PR EOS)

csv_compositional.csv — every column as a curve vs distance along route (km) [as-operated]



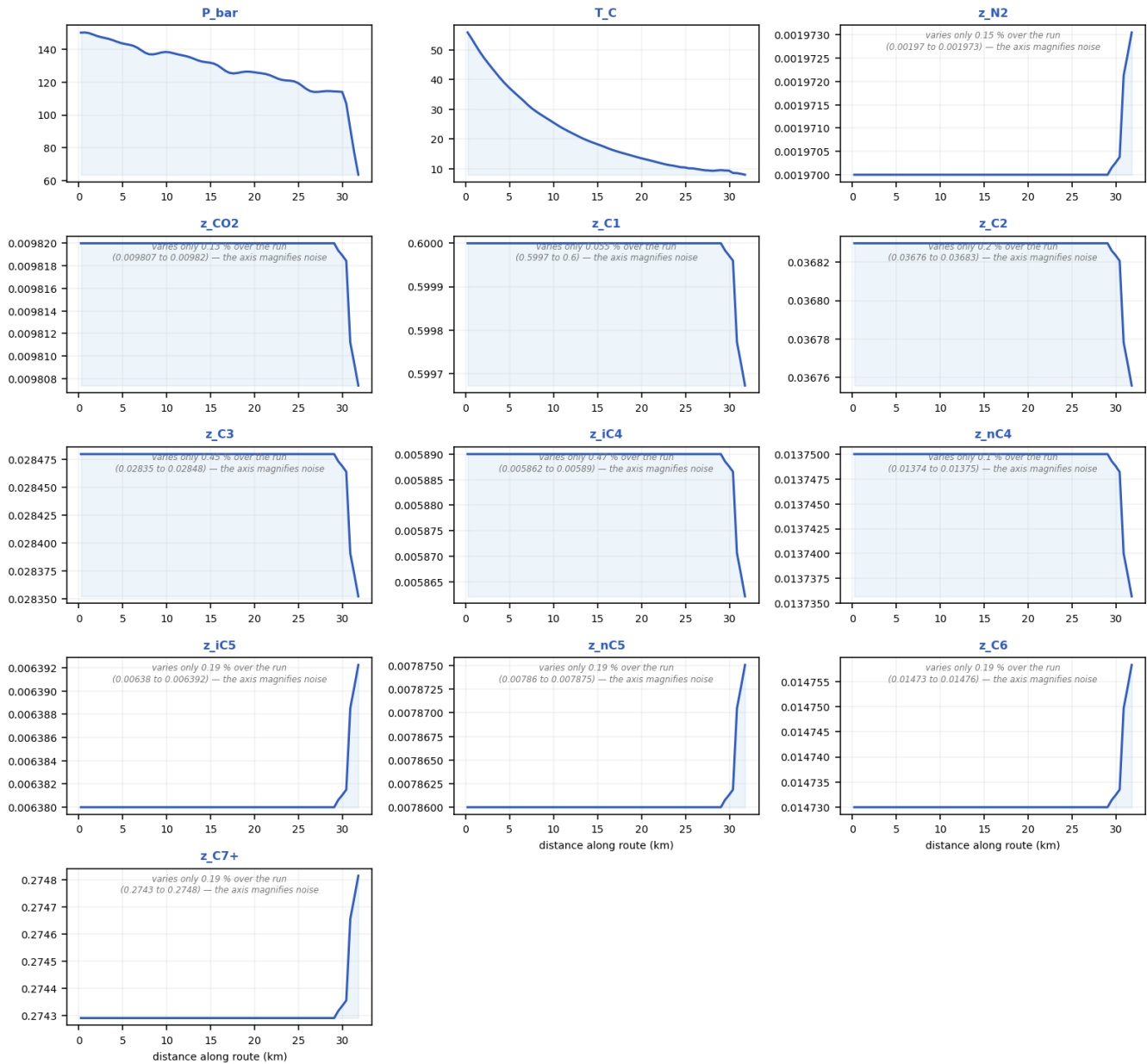
Every column of csv_compositional.csv drawn as a curve [as-operated].

x_k m	P _bar	T_C	pha se_s tate	vapo ur_fr ac_V	rho_ gas_ kgm 3	rho_ liq_k gm3	mu_ gas_ _Pa s	mu_ liq_ _Pa s	Z_ ga s	ga s_ sg	K _N 2	K_ C O2	K _C 1	K_ C2	K_ C3	K_ iC 4	K_ nC 4	K_ i C5	K_ nC 5	K_ C6	K_C 7+
0. 22 85 7	15 0	5 5. 8 0 4	two - pha se	0.34 348	115. 04	836. 74	1.66 43e- 05	0.0 010 864	0. 83 00 6	0. 61 70 3	3. 9 2 0 3	0. 88 89 5 2	2. 1 5 3	0. 86 20 1	0. 46 02 9	0. 30 66 8	0.2 48 2	0.1 62 58	0.1 40 37	0.0 81 40 4	0.0 002 451 5
2. 51 43	14 7. 26	4 5. 6 2 6	two - pha se	0.32 686	118. 24	839. 05	1.65 14e- 05	0.0 011 431	0. 81 13 3	0. 61 24 4	3. 9 1 5 6	0. 82 94 4	2. 1 1 4 2	0. 81 70 5	0. 42 71 8	0. 28 14 2	0.2 25 66	0.1 46 06	0.1 25 18	0.0 71 28 2	0.0 001 761 2
4. 8	14 3. 64	3 7. 6 8	two - pha se	0.31 587	119. 84	841. 43	1.63 66e- 05	0.0 011 955	0. 79 56 7	0. 60 88 7	3. 9 4	0. 78 51 7	2. 0 9 1 2	0. 78 23 8	0. 40 07 5	0. 26 09 4	0.2 07 49	0.1 32 67	0.1 12 93	0.0 63 15 6	0.0 001 283
7. 08 57	13 9. 28	3 1. 4 1 6	two - pha se	0.31 026	119. 92	844. 14	1.61 81e- 05	0.0 012 478	0. 78 30 1	0. 60 60 1	3. 9 9 4 8	0. 75 20 3	2. 0 8 3 9	0. 75 49 2	0. 37 85 8	0. 24 33 6	0.1 91 98	0.1 21 14	0.1 02 44	0.0 56 21	9.2 971 e- 05
9. 82 86	13 8. 25	2 5. 8 2	two - pha se	0.29 583	122. 77	843. 93	1.61 86e- 05	0.0 012 669	0. 76 97 7	0. 60 36 5	3. 9 7 8 8	0. 71 8	2. 0 4 8 1	0. 72 96 5	0. 36 24 7	0. 23 20 2	0.1 82 14	0.1 14 41	0.0 96 36 4	0.0 52 46 4	7.9 656 e- 05
12 .1 14	13 5. 81	2 1. 9 5 7	two - pha se	0.29 033	123. 22	845. 03	1.61 e-05	0.0 012 944	0. 76 09 3	0. 60 19 3	3. 9 7 9 6	0. 69 68	2. 0 3 8 1	0. 71 22 5	0. 34 94 8	0. 22 20 9	0.1 73 5	0.1 08 16	0.0 90 73 3	0.0 48 85 7	6.5 826 e- 05
14 .4	13 2. 07	1 8. 8 7	two - pha se	0.29 114	121. 84	847. 33	1.59 23e- 05	0.0 013 335	0. 75 46 5	0. 60 04 4	4. 0 4 8	0. 68 17 2	2. 0 4 5 9	0. 69 77 9	0. 33 66 9	0. 21 16 3	0.1 64 41	0.1 01 34	0.0 84 61 6	0.0 44 86 8	5.1 169 e- 05
17 .1 43	12 5. 54	1 5. 8 9 6	two - pha se	0.30 108	117. 4	852. 06	1.55 66e- 05	0.0 014 029	0. 75 07 5	0. 59 88	4. 2 1 2 2	0. 67 02 5	2. 0 2 7	0. 68 26 3	0. 32 00 5	0. 19 71 6	0.1 51 88	0.0 91 72 2	0.0 76 03 1	0.0 39 24 3	3.3 431 e- 05
19 .4 29	12 6. 26	1 3. 9 6 5	two - pha se	0.29 175	119. 64	850. 78	1.56 44e- 05	0.0 013 955	0. 74 46 6	0. 59 80 8	4. 1 5 8	0. 65 39 8	2. 0 6 7	0. 67 31 3	0. 31 63 3	0. 19 52 6	0.1 50 31	0.0 90 94 7	0.0 75 38 9	0.0 38 95 2	3.3 739 e- 05
21 .7 14	12 4. 13	1 2. 2 3 8	two - pha se	0.29 216	118. 73	852. 05	1.55 39e- 05	0.0 014 186	0. 74 12 3	0. 59 72 4	4. 2 0 1 2	0. 64 81 9	2. 0 6 1 1	0. 66 52 4	0. 30 89 9	0. 18 94	0.1 45 3	0.0 87 29 1	0.0 72 10 8	0.0 36 91 7	2.8 781 e- 05
24 .4 57	12 0. 42	1 0. 5 2 9	two - pha se	0.29 778	116. 06	854. 72	1.53 27e- 05	0.0 014 602	0. 73 93 1	0. 59 62 8	4. 3 0 1 7	0. 64 10 8	2. 0 8 2 3	0. 65 57 9	0. 29 92 3	0. 18 11 4	0.1 38 26	0.0 82 00 5	0.0 67 43 9	0.0 33 95 3	2.2 083 e- 05
26 .7 43	11 3. 96	9. 6 9 4 4	two - pha se	0.31 573	109. 59	860. 26	1.49 24e- 05	0.0 015 373	0. 74 32 6	0. 59 55	4. 5 3 1 5	0. 64 31 2	2. 1 4 5 6	0. 64 94 3	0. 28 75	0. 17 01 5	0.1 28 89	0.0 74 72 2	0.0 61 03 4	0.0 29 84 9	1.4 178 e- 05
29 .0 29	11 4. 31	9. 5 2 7	two - pha se	0.31 399	110. 1	859. 89	1.49 48e- 05	0.0 015 33	0. 74 23 9	0. 59 54 5	4. 5 1 4	0. 64 15 4	2. 1 3 9	0. 64 75 9	0. 28 75 9	0. 17 04	0.1 29 11	0.0 74 93 7	0.0 61 22 4	0.0 29 98 4	1.4 46e -05

		9								6		7								
31	63	7.	two	0.47	56.2	908.	1.22	0.0	0.	0.	8.	0.	3.	0.	0.	0.	0.0	0.0	0.0	0.0
.7	.6	9	-	027	24	8	73e-	023	82	59	0	78	1	69	23	11	81	38	29	11
71	91	6	pha				05	666	1	9	2	86	9	44	67	43	75	63	87	31
		5	se								5	8	1	2	9	8	8	9	8	5
													3							e-
																				07

11.1.9 csv_compositional_transport.csv — compositional grading (hydrate-former depletion) along the route

csv_compositional_transport.csv — every column as a curve vs distance along route (km) [as-operated]



Every column of csv_compositional_transport.csv drawn as a curve [as-operated].

x_km	P_bar	T_C	z_N2	z_CO2	z_C1	z_C2	z_C3	z_iC4	z_nC4	z_iC5	z_nC5	z_C6	z_C7+
0.228571	150	55.8042	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
2.51429	147.258	45.6261	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
4.8	143.637	37.6796	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
7.08571	139.278	31.4159	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
9.82857	138.252	25.8204	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
12.1143	135.807	21.9569	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
14.4	132.075	18.8704	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
17.1429	125.537	15.896	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
19.4286	126.265	13.9645	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
21.7143	124.135	12.2384	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
24.4571	120.42	10.5289	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
26.7429	113.96	9.69443	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
29.0286	114.307	9.52788	0.00197	0.00982	0.6	0.03683	0.02848	0.00589	0.01375	0.00638	0.00786	0.01473	0.27429
31.7714	63.6912	7.98655	0.00197305	0.00980738	0.599672	0.0367555	0.0283523	0.00586214	0.0137357	0.00639224	0.00787508	0.0147583	0.274816

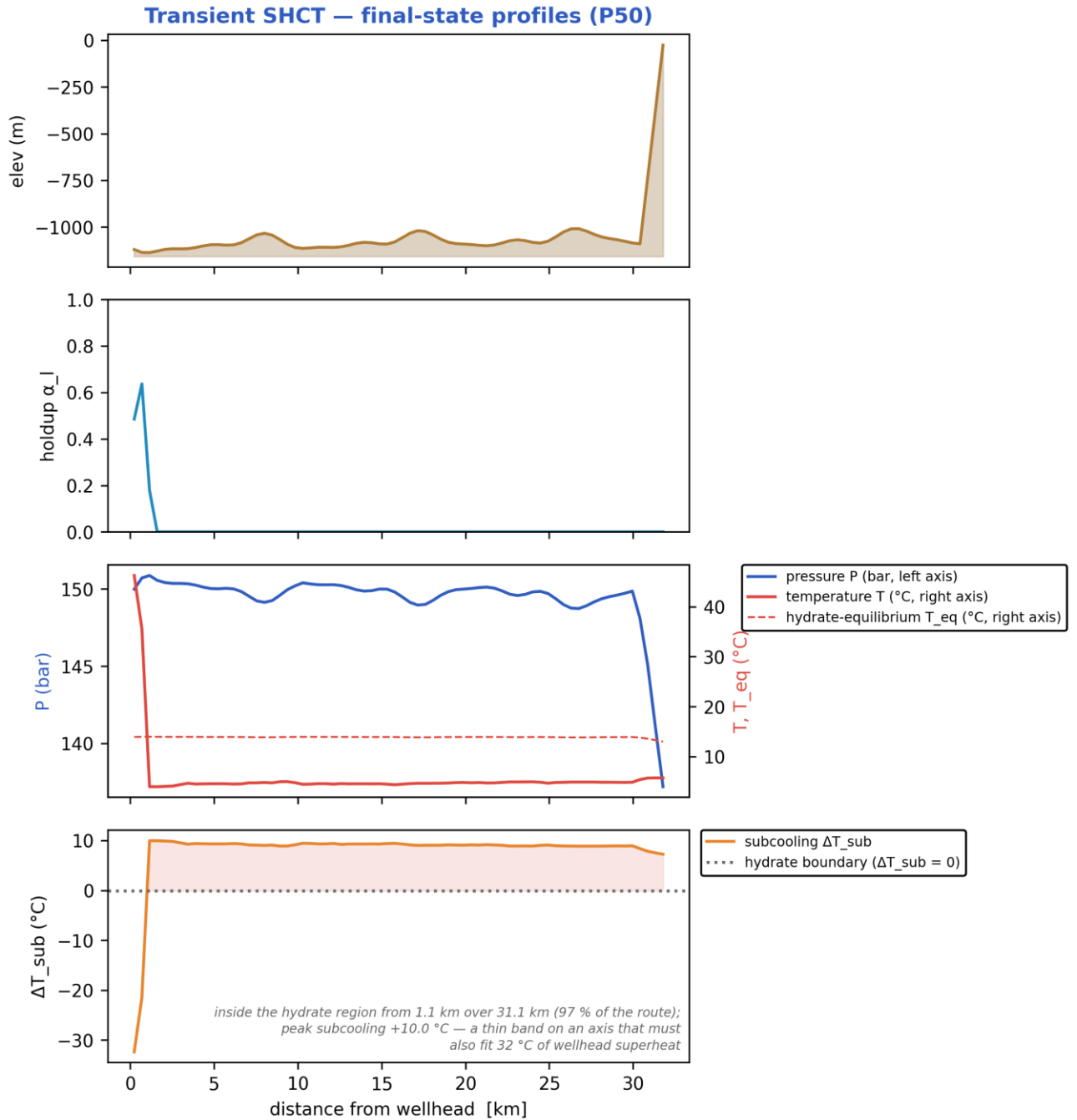
11.2 Unplanned shut-in — cooldown / no-touch time

11.2.1 Headline prediction metrics

Metric	Value	Units
Total ΔP	12.79	bar
Peak velocity	5.301	m/s
Erosional limit (API 14E)	9.389	m/s
Max slug length	14.22	m
Mean slug length	5.652	m
Slug/intermittent fraction	0.04749	—
Slug-catcher surge (design)	1.238	m ³
Max subcooling	10.01	°C
Design subcooling (P90)	10.02	°C
Sustained Φ_{SH} (coupling)	3086	—
Sustained coupling hot-spot	1.143	km
Length above $\Phi_{SH} = 1$	30.17	km
Φ_{SH} running max (startup)	4886	—
...attained at	16.5	h
Plug probability	1	frac
Time-to-plug P50	9.767	h
Time-to-plug P10	8.898	h
Time-to-plug P90	11.05	h
Peak wall deposit	117.1	mm
MEG required	38.5	wt%
MEG rate	4.795e+04	L/h
Under-inhibited length	32	km
No-touch time	0	h
Effective U	22	W/m ² K
Slurry rel. viscosity	3.162e+07	—
Hydrate mass formed	9.399e+05	kg
Min mixture sound speed	264.4	m/s
Wax onset	1.143	km
Max scaling index	0.261	—
Liquid mass error	2.957e-14	frac
Gas mass error	2.445e-15	frac
Solver fallbacks	0	#

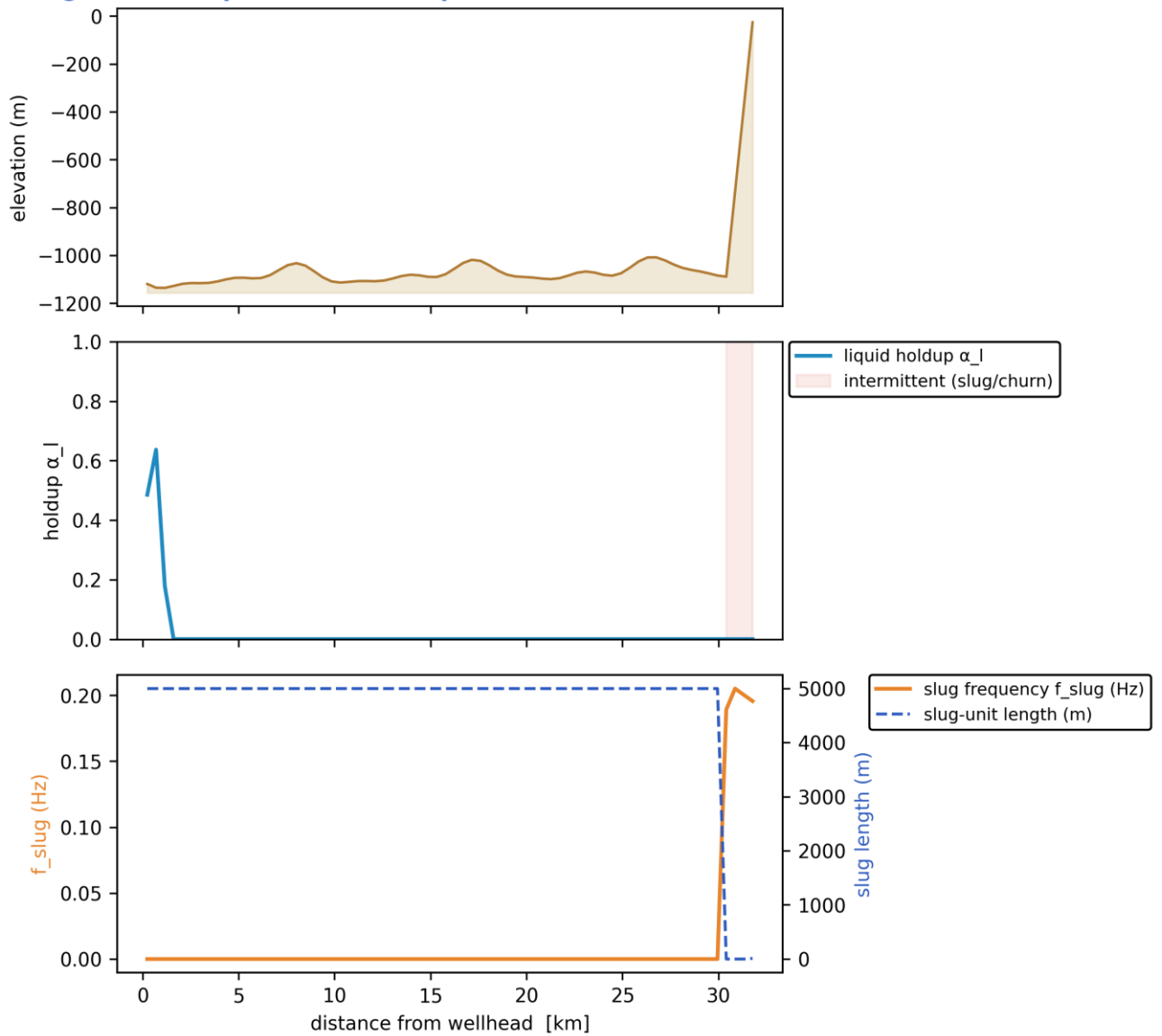
11.2.2 Charts, curves, contours and maps (23 figures)

Note on the resolved-slug figures. The transport grid is $dx \sim 457$ m while a slug unit is of order 10-40 m, so individual slugs are a SUB-GRID quantity: the solver carries them statistically through the slug frequency f_{slug} , the slug unit length $L_u = V_t / f_{\text{slug}}$ and the slug-body holdup α_{ls} . The three figures marked 'sub-grid reconstruction' therefore render a kinematic reconstruction built entirely from those solver outputs - at every station the reconstructed square wave carries the solver's local slug frequency and translational celerity, and the body/film split is chosen so that the unit-averaged holdup reproduces the solver's cell-average α_l exactly. Period, celerity, length and holdup are all run outputs; nothing in them is assumed. Every other figure in this gallery is plotted directly from the solver's space-time history.

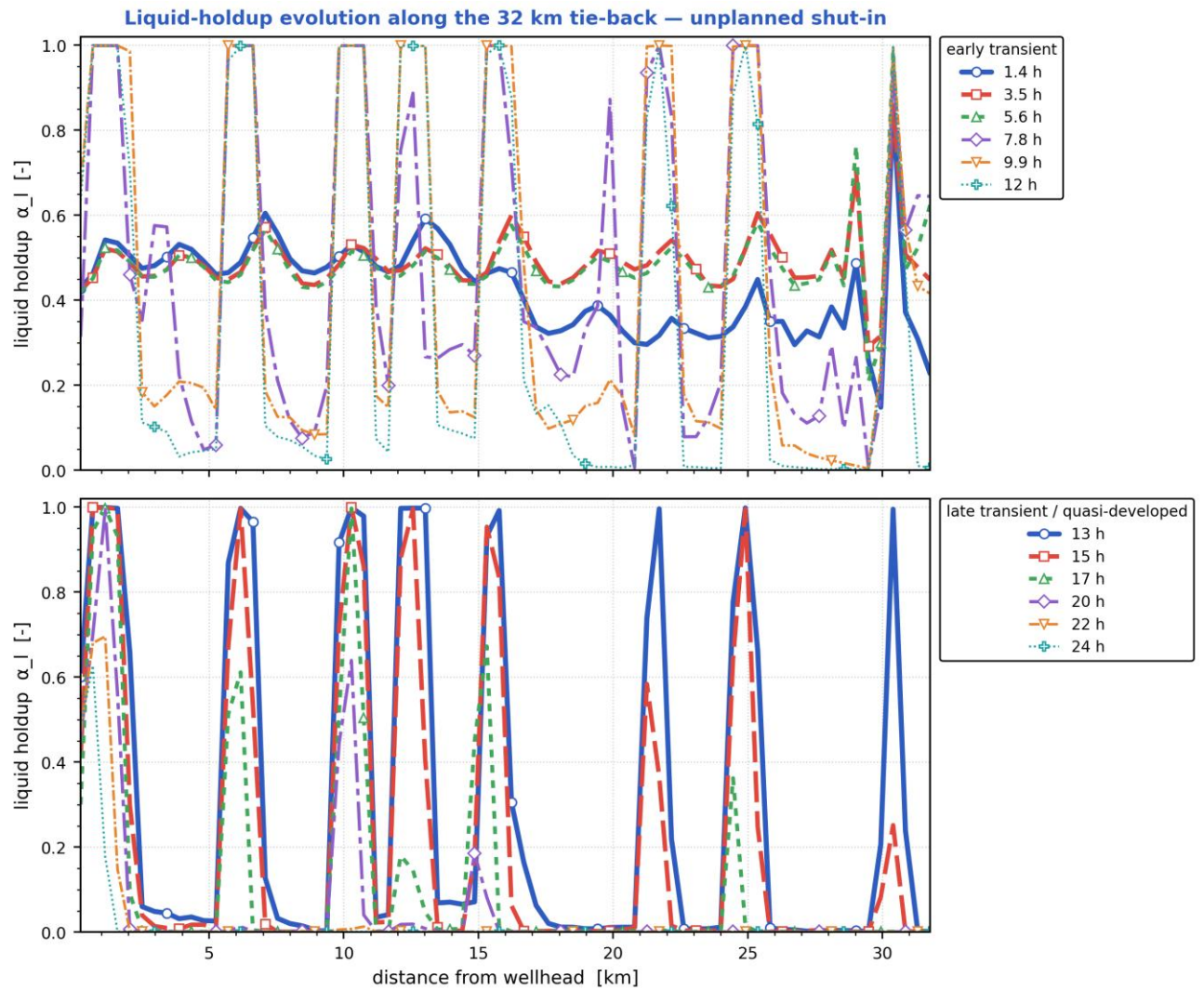


Final-state P50 profiles: seabed elevation, liquid holdup, pressure & temperature vs hydrate T_{eq} , and subcooling along the whole route. [shut-in]

Slug-formation prediction — deepwater medium-crude-oil tie-back

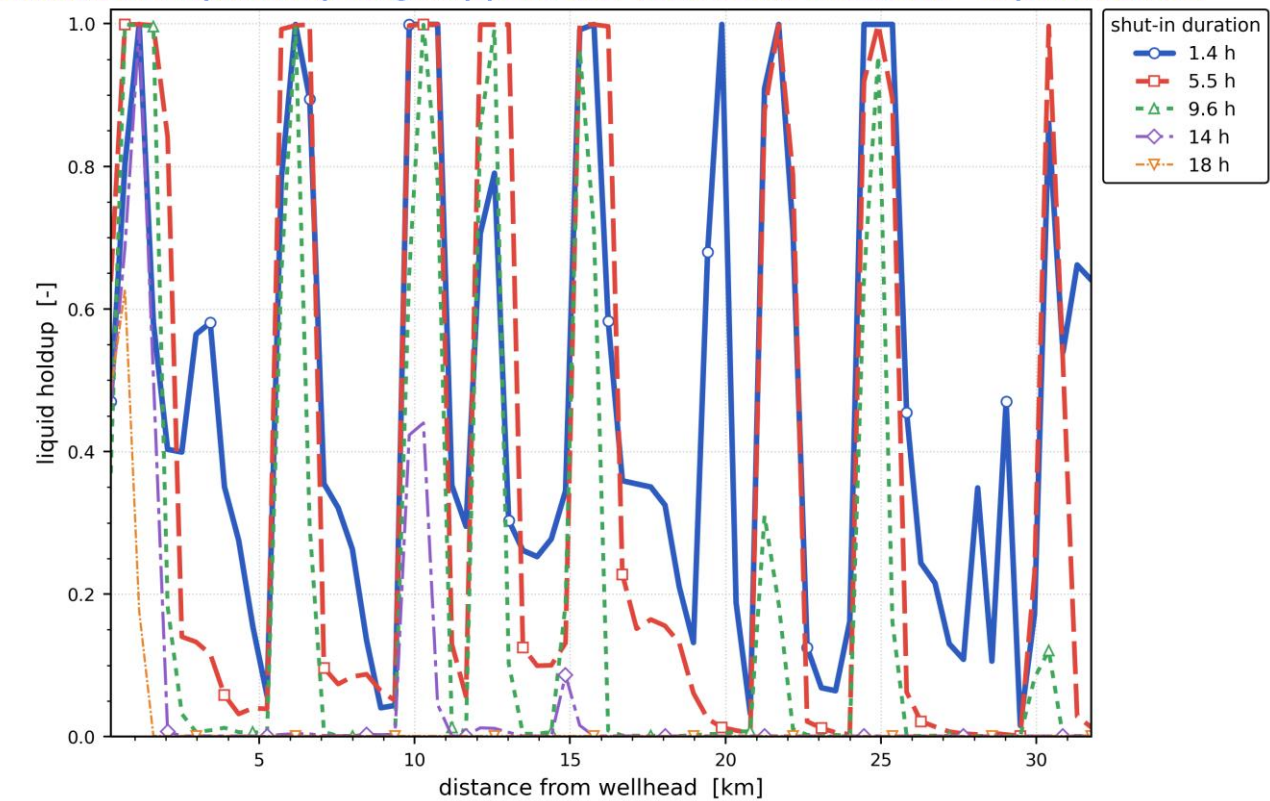


Slug-formation prediction: terrain, intermittent (slug/churn) bands, slug frequency and slug-unit length, and liquid holdup. [shut-in]



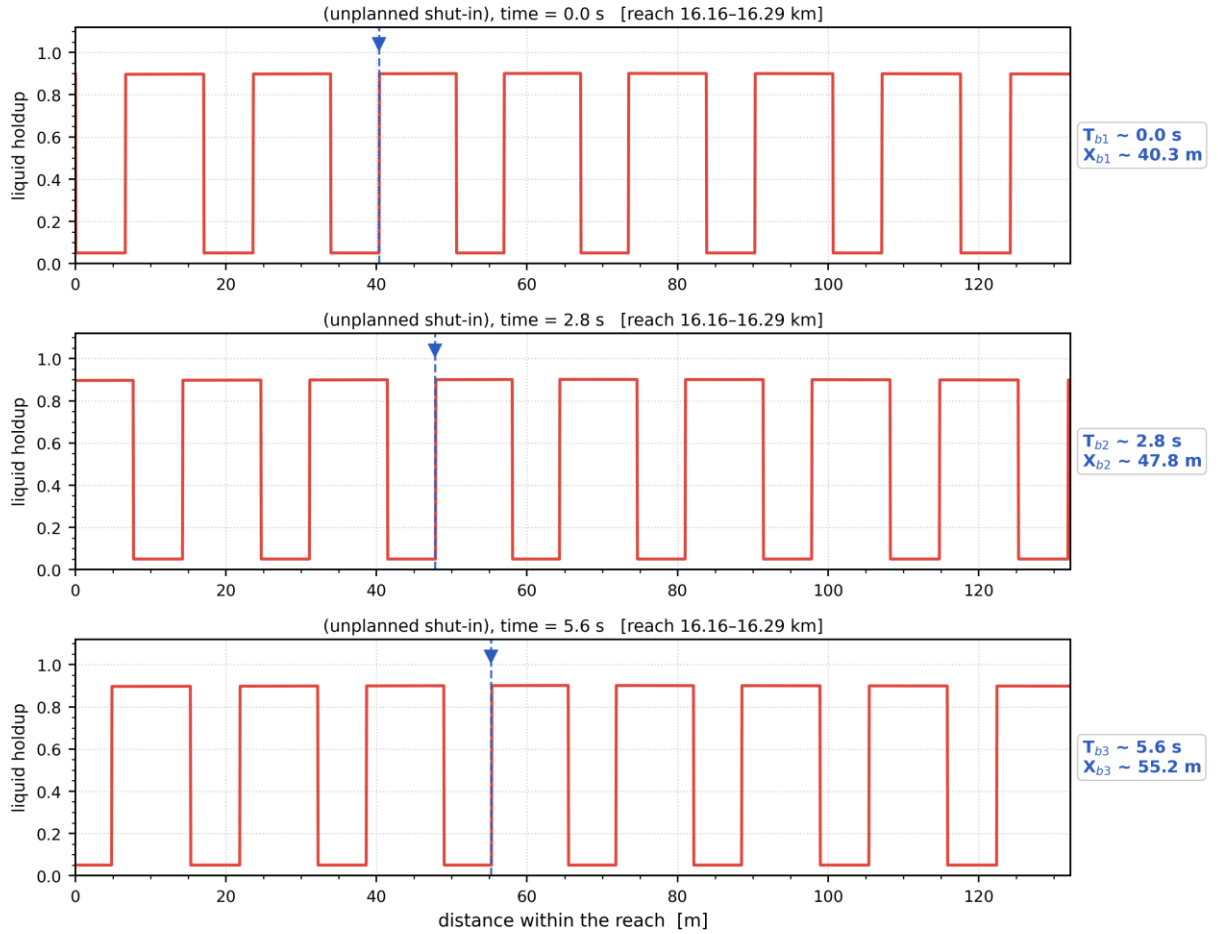
*Liquid holdup along the whole route at successive times — the early transient (upper) and the late, quasi-developed state (lower).
The travelling and terrain-locked holdup structure is the slug activity resolved in time. [shut-in]*

Distribution of liquid holdup along the pipeline after different shut-in durations — unplanned shut-in



Distribution of liquid holdup along the pipeline after successive elapsed durations (shut-in duration for the shut-in scenario, production time otherwise) — one curve per duration. [shut-in]

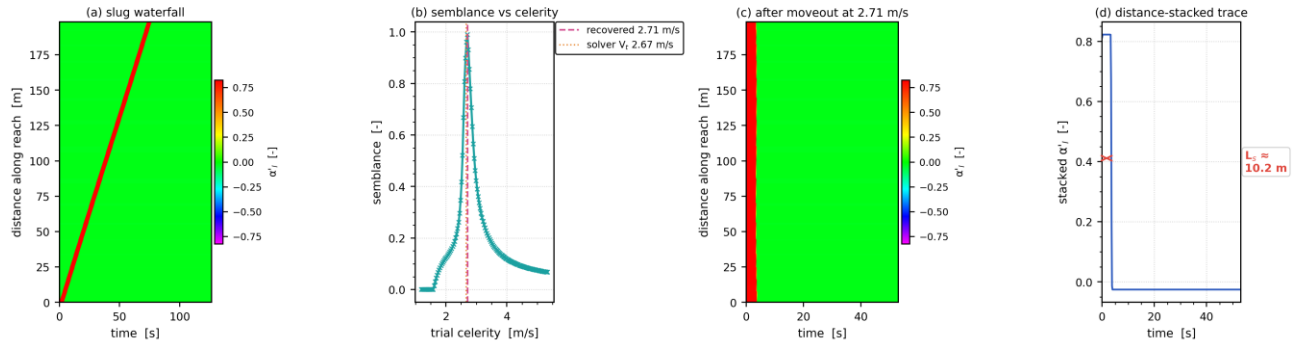
Slug propagation and front tracking — resolved slug units at $t = 5.9$ h ($V_t = 2.67$ m/s, $f_{slug} = 0.16$ Hz, $L_u = 16.5$ m)



Mass-consistent sub-grid reconstruction: period, celerity, unit length and unit-average holdup are the solver's own fields.

Slug propagation and front tracking: three successive snapshots of the resolved slug units over a short reach, with one front followed across the panels (its arrival time T_b and position X_b are annotated). Sub-grid reconstruction — see §note. [shut-in]

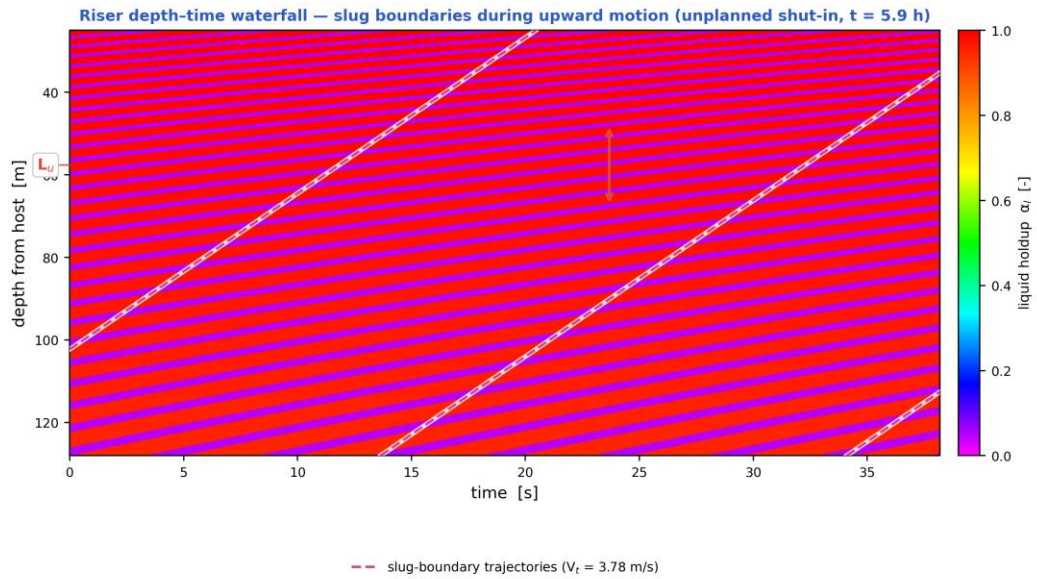
Slug tracking in the space-time plane — unplanned shut-in, reach 16.13–16.33 km at $t = 5.9$ h ($L_u = 16.5$ m, $f_{slug} = 0.16$ Hz)



One slug unit of the mass-consistent sub-grid reconstruction; the moveout scan (b) recovers 2.71 m/s against the solver's $V_t = 2.67$ m/s at the tracked cell (+1.5 %), the difference being the variation of V_t across the 198 m window.

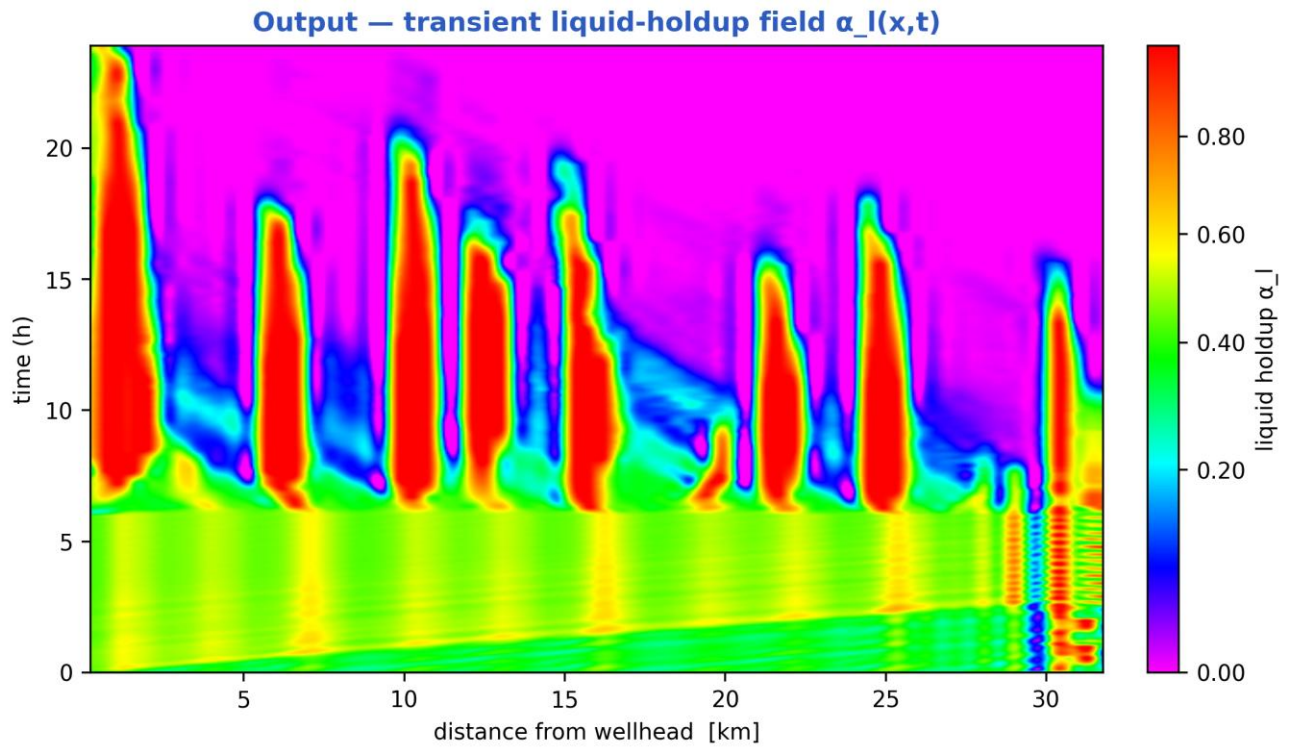
Slug tracking in the space-time plane: (a) the distance-time waterfall of one slug unit, (b) semblance against trial celerity, (c) the waterfall after linear moveout at the recovered celerity, (d) the distance-stacked trace whose width gives the slug body length. The

recovered celerity is reported on the figure against the solver's own translational velocity V_t at the tracked cell. Sub-grid reconstruction — see §note. [shut-in]



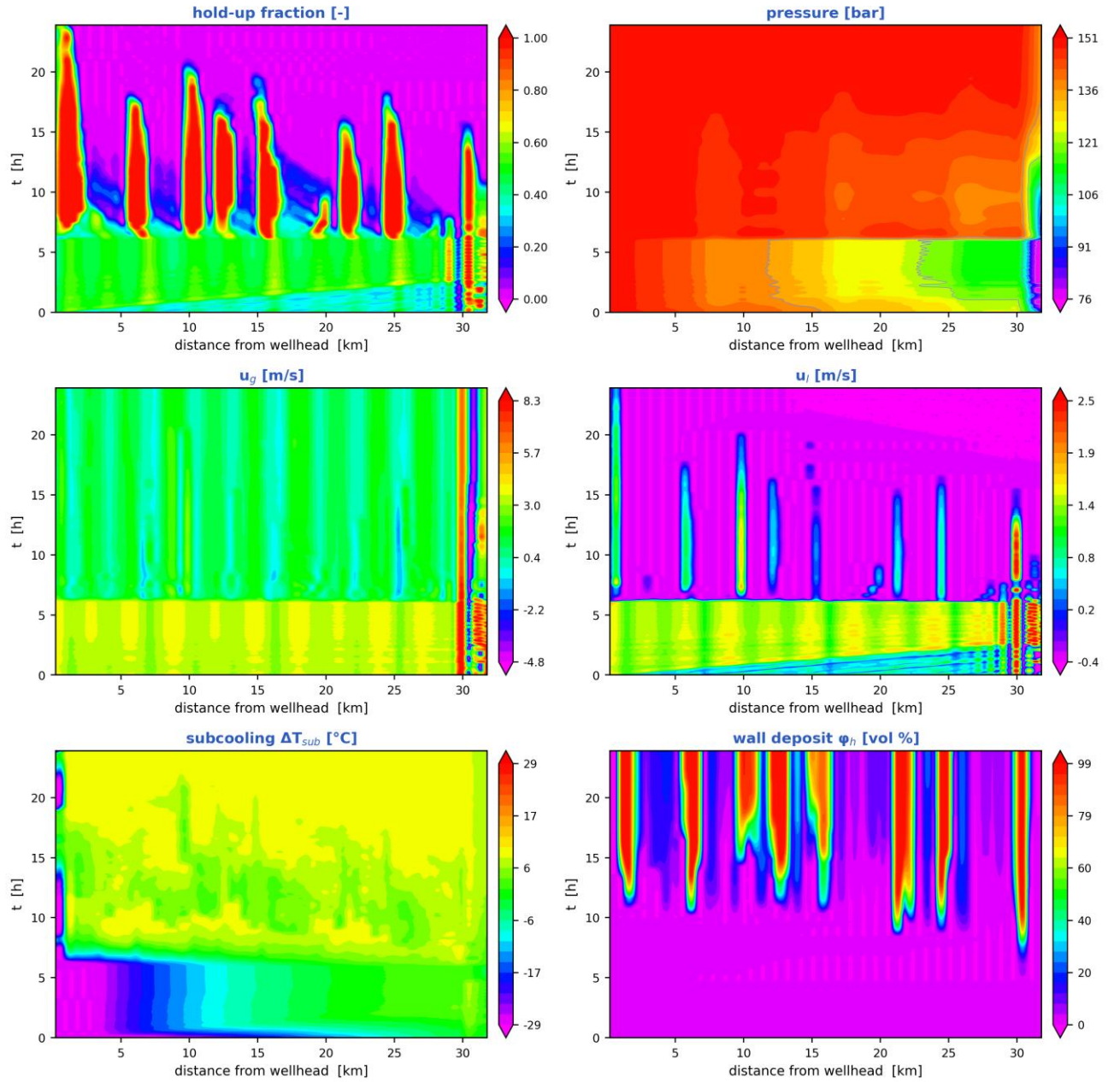
Mass-consistent sub-grid reconstruction over the steel-catenary riser ($L_u = 25.8$ m, slug body 14.7 m, period 6.8 s); celerity, period and length are solver outputs. The arrow marks one slug unit projected onto the depth axis, 20.0 m.

Depth-time waterfall over the steel-catenary riser: slug boundaries during upward motion, their trajectories (slope = translational celerity) and the slug unit length marked on the depth axis. Sub-grid reconstruction — see §note. [shut-in]

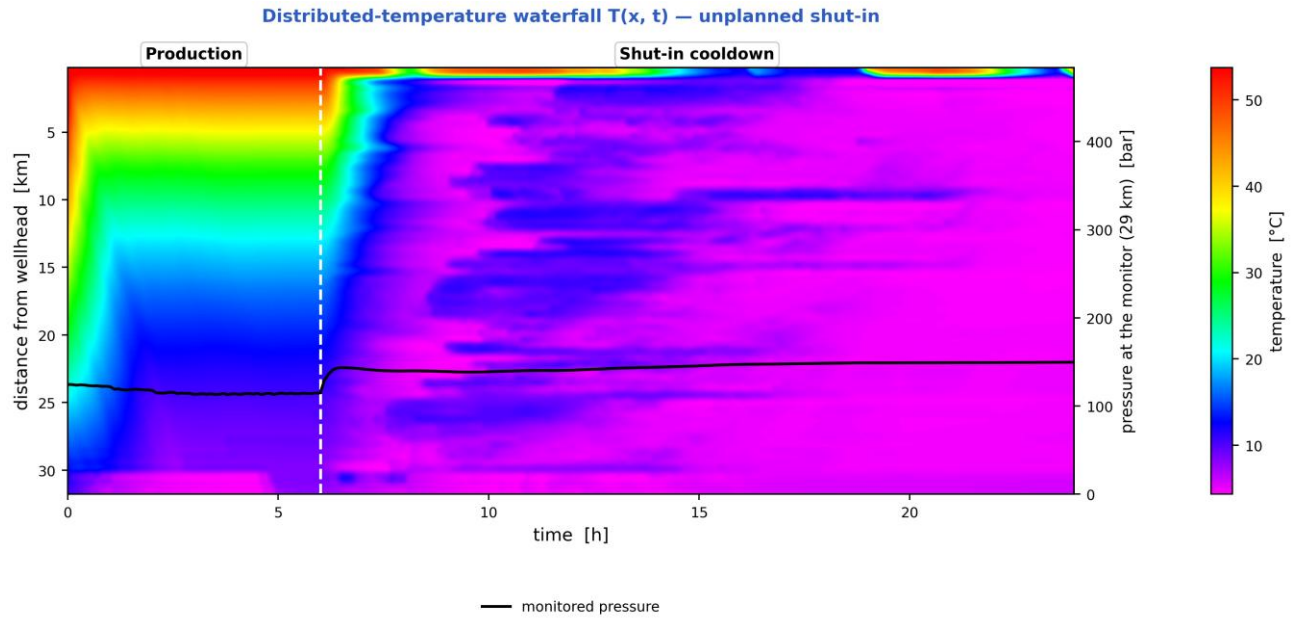


Liquid-holdup field $\alpha_l(x,t)$ — the bands are slug activity migrating along the line in time (space-time contour map). [shut-in]

Space-time solution of the 32 km tie-back — unplanned shut-in (N = 70 cells, 240 snapshots)

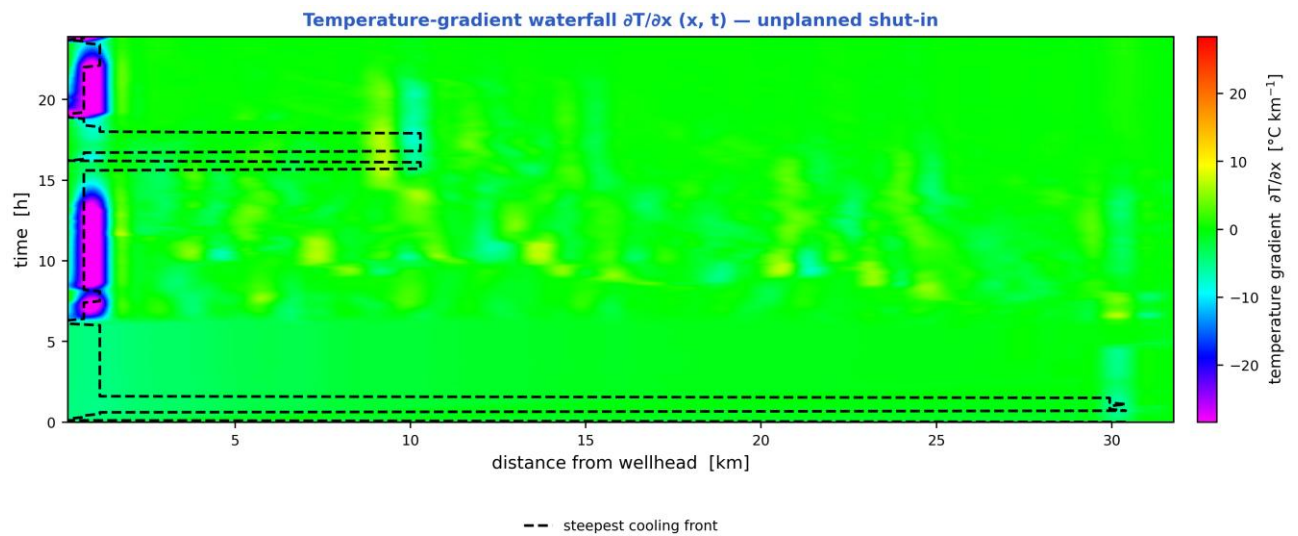


The TRUE space-time solution of the tie-back: liquid holdup, pressure, gas and liquid velocities, subcooling and wall-deposit fraction, each as a filled-contour field over (distance, time). [shut-in]



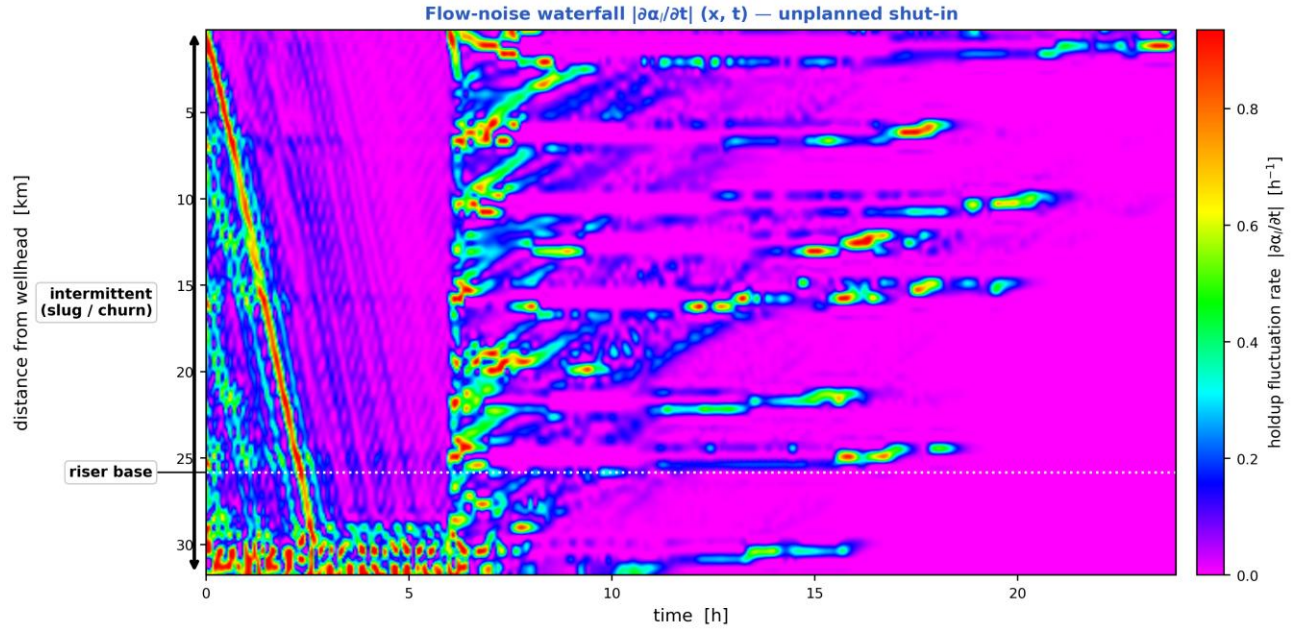
At the final state the whole line is inside the hydrate region (97 % of the route), so there is no onset distance to mark.

Distributed-temperature waterfall $T(x, t)$ — the thermal field over distance and time, with the monitored pressure overlaid, the operating stages marked and the hydrate-onset distance annotated. [shut-in]



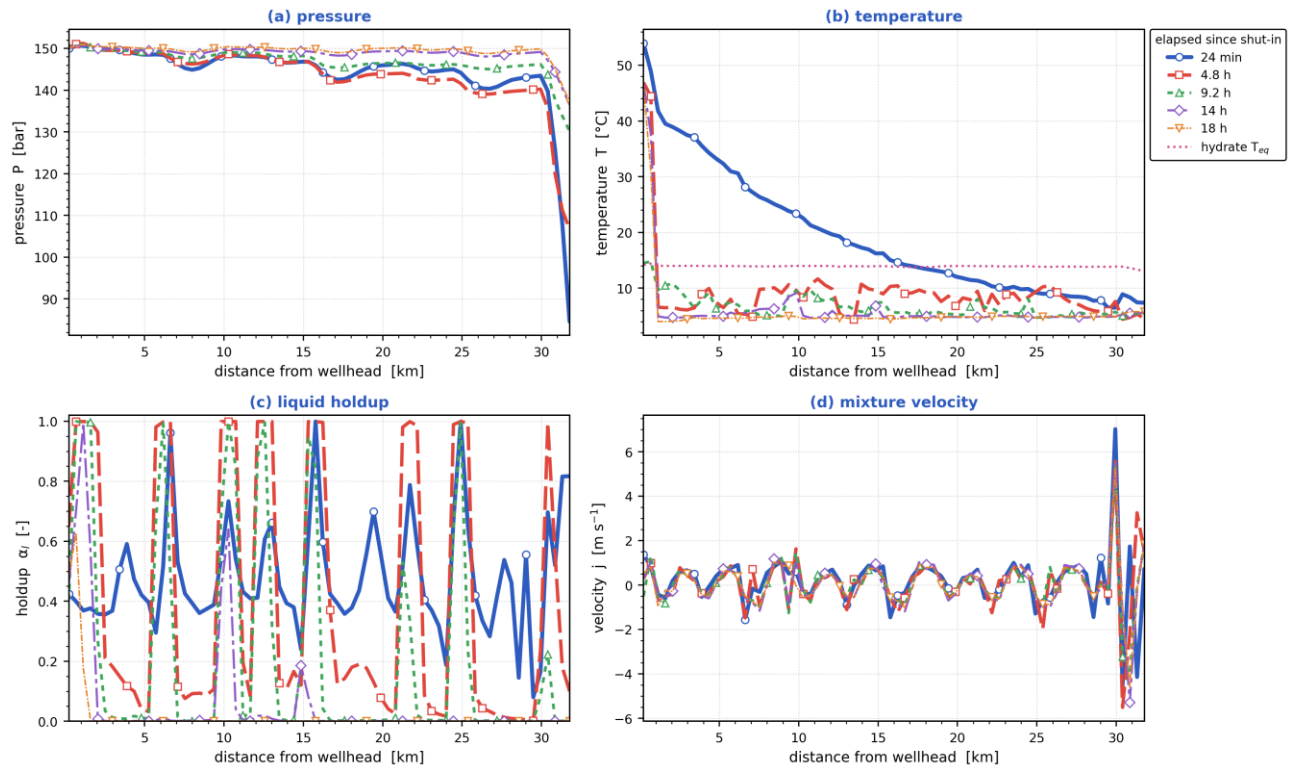
A travelling thermal front is a narrow band of steep gradient; the dashed line tracks the steepest cooling at each instant.

Temperature-gradient waterfall $\partial T / \partial x(x, t)$. A travelling thermal front is a narrow band of steep gradient, so it is localised here even where the temperature map looks smooth; the dashed line tracks the steepest cooling. [shut-in]



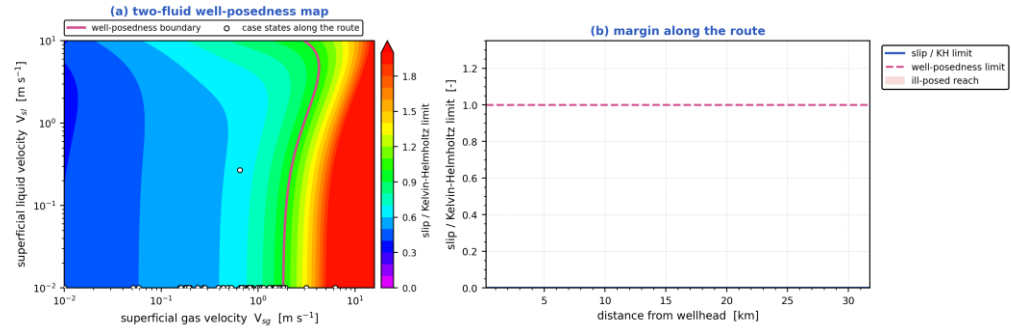
Flow-noise waterfall $|\partial\alpha/\partial t|(x, t)$ — where the liquid holdup changes fastest is where the flow is most unsteady, with the intermittent (slug/churn) reach and the riser base marked. [shut-in]

Parameters along the pipeline at successive times — unplanned shut-in



Pressure, temperature, liquid holdup and mixture velocity along the route, each at the same successive times. [shut-in]

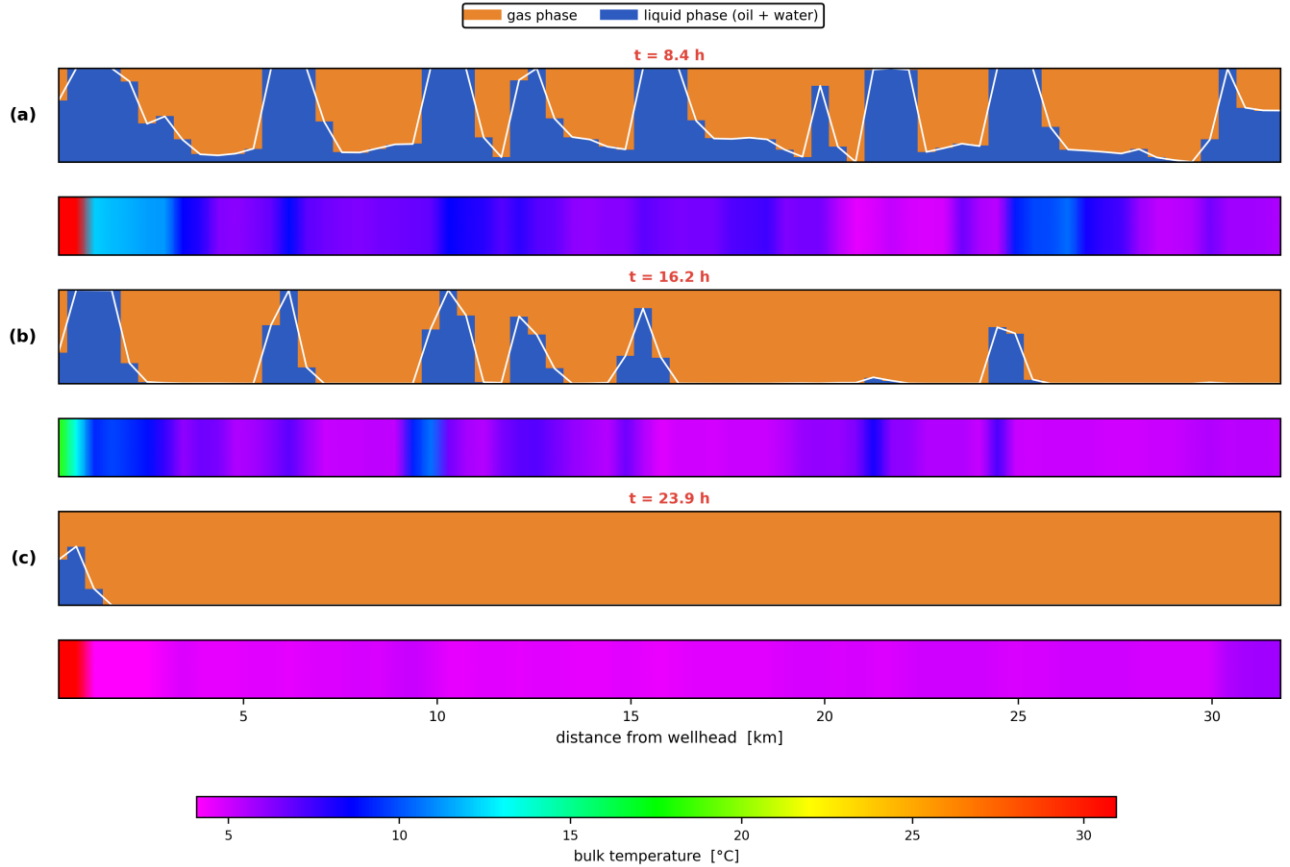
Well-posedness of the two-fluid description — unplanned shut-in



the interfacial-pressure term ($C = 1.20 \pm 1$) makes the two-fluid system unconditionally hyperbolic, so the initial-value problem is well posed over the whole route; against the *INVISCID* criterion ($C = 0$) 0 % of the route would be past the limit, and that is the reference the literature quotes

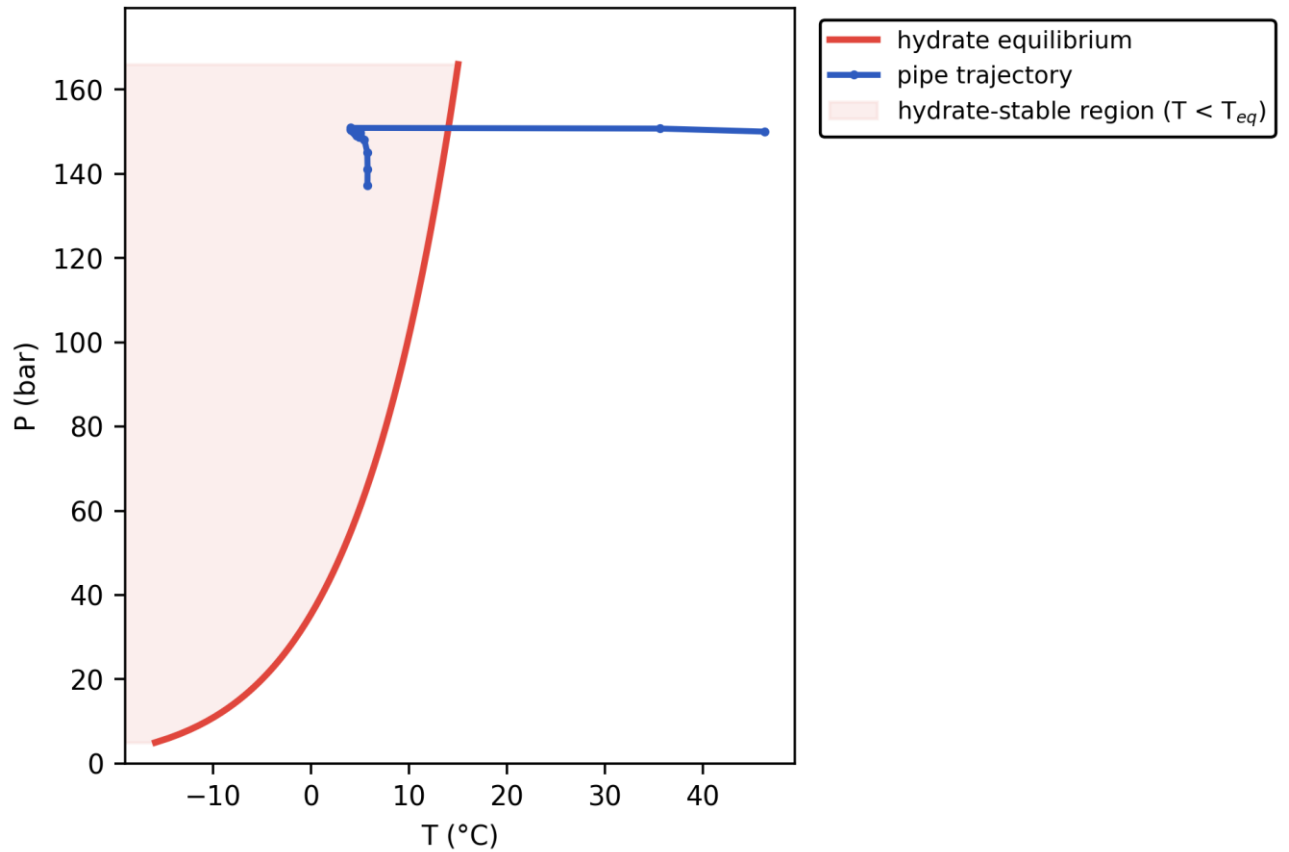
Well-posedness of the two-fluid description: (a) the inviscid Kelvin–Helmholtz boundary over the superficial-velocity plane with the case's own states, (b) the slip/KH margin along the route. Past the boundary the 1-D two-fluid initial-value problem is ill-posed and growth rates are grid-dependent. [shut-in]

Pipeline cloud maps at successive times — phase distribution (upper) and temperature (lower), unplanned shut-in

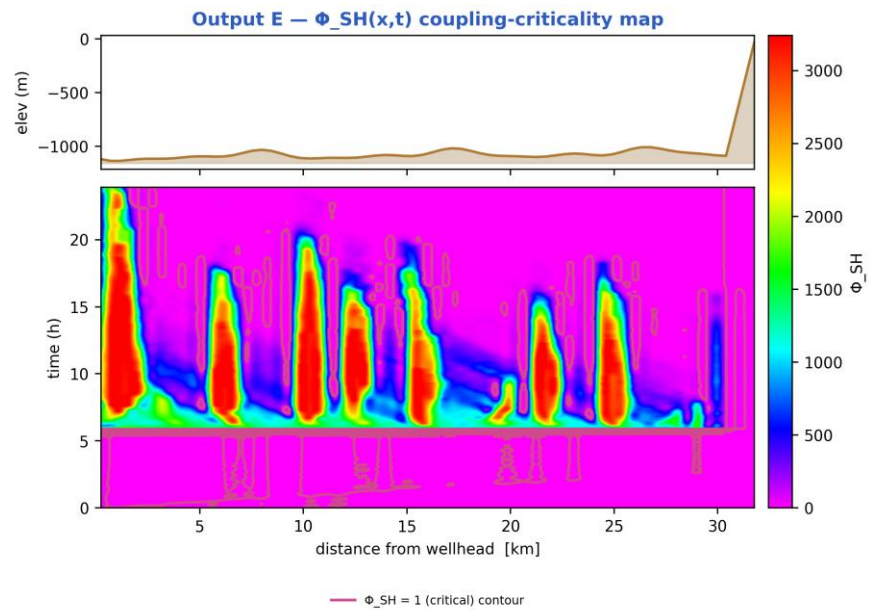


Pipeline cloud maps at successive times: the gas/liquid phase distribution inside the bore (upper strip of each pair) above the bulk-temperature field along the same reach (lower strip), on a shared temperature scale. [shut-in]

Output C — P-T trajectory vs hydrate envelope



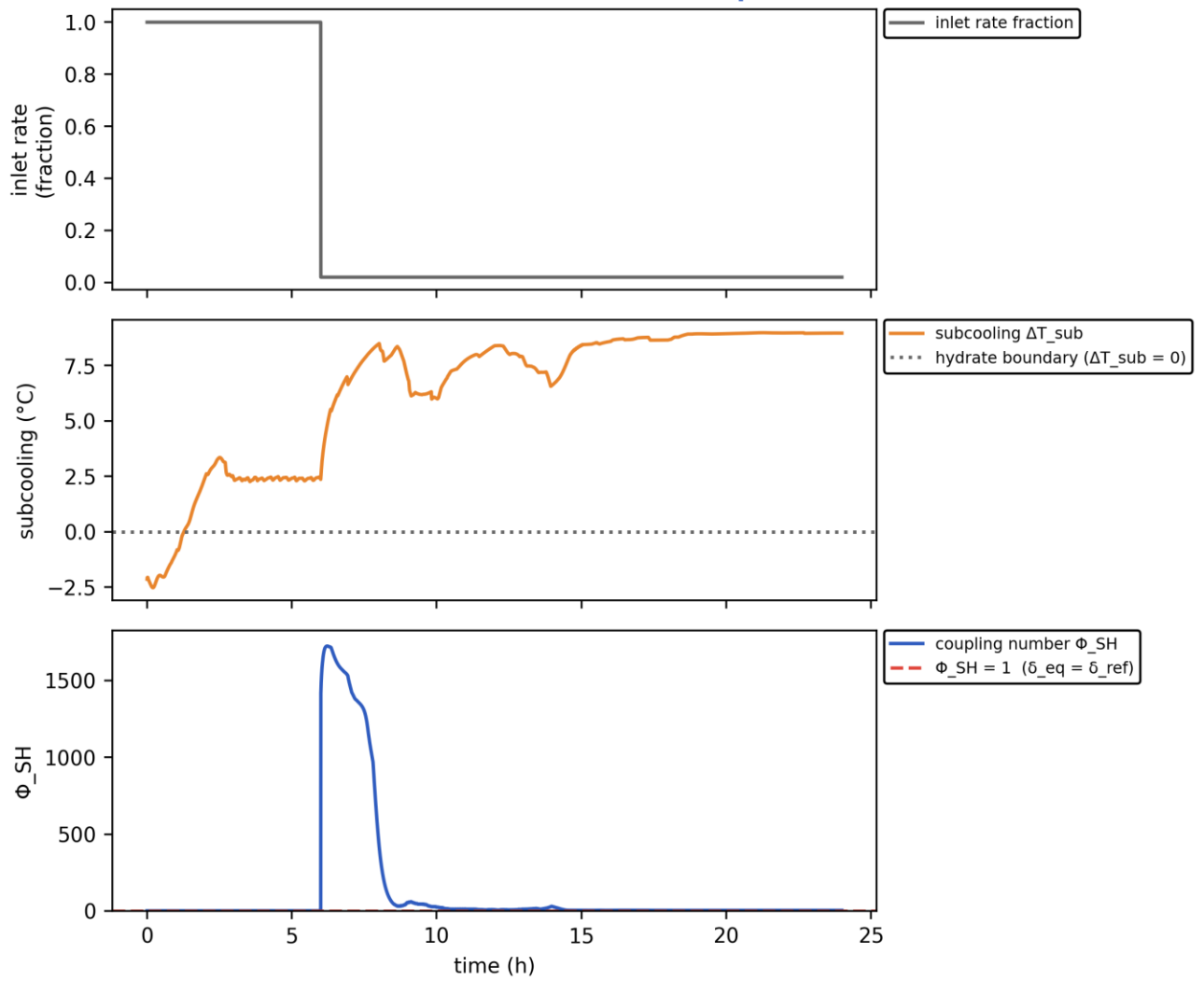
Production P–T trajectory against the hydrate-stability envelope (curve). [shut-in]



the field reaches $\Phi_{SH} = 3.72e+03$, but the line is at rest: the shear-removal term in Φ_{SH} goes to zero, so the MAGNITUDE is set by that, not by resolved physics — read the super-critical EXTENT, not the number

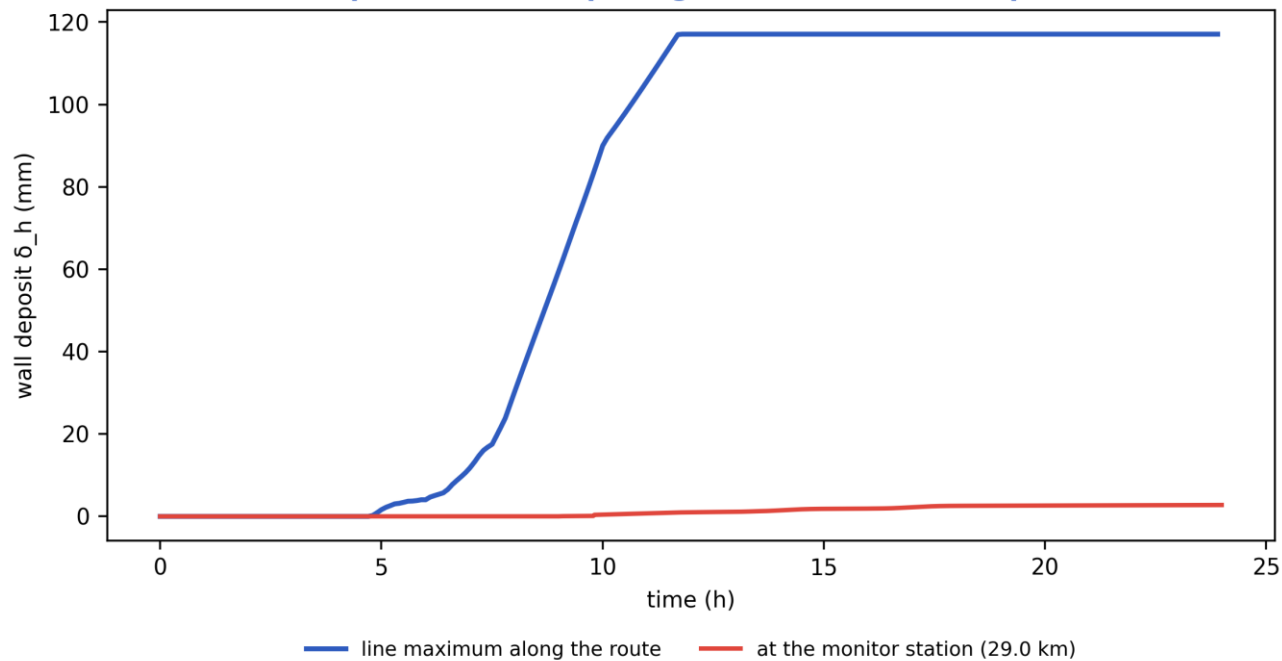
Slug-Hydrate coupling-criticality map $\Phi_{SH}(x,t)$. Φ_{SH} is the equilibrium deposit thickness in units of $\delta_{ref} = 21.2$ mm; the contour drawn is $\Phi_{SH} = 1$, just below the derived runaway threshold $\Phi_{crit} = 1.08$, which separates slug-scoured from plugging-critical (space-time map). [shut-in]

Transient scenario 'shutin' — monitor response



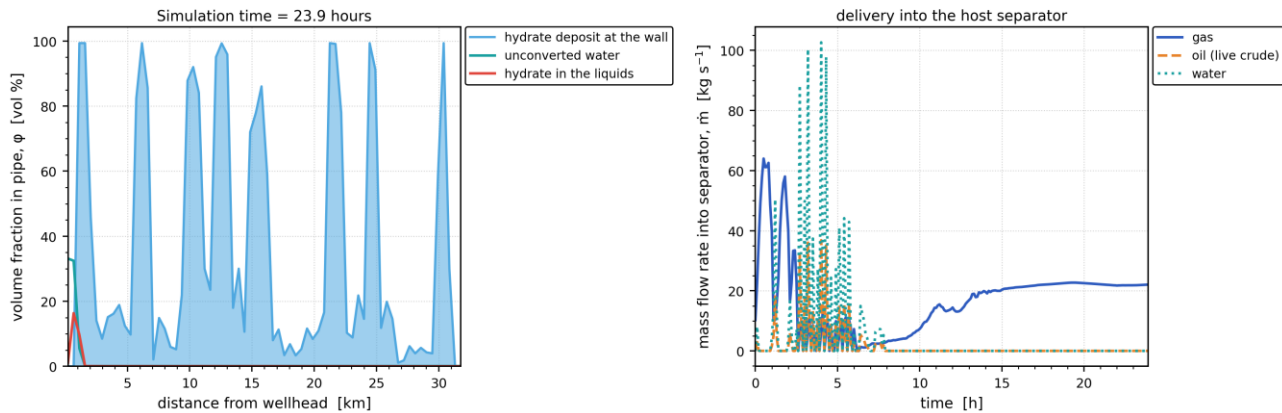
Monitored-station transient response vs time (curves). [shut-in]

Output D — wall-deposit growth (transient, coupled)



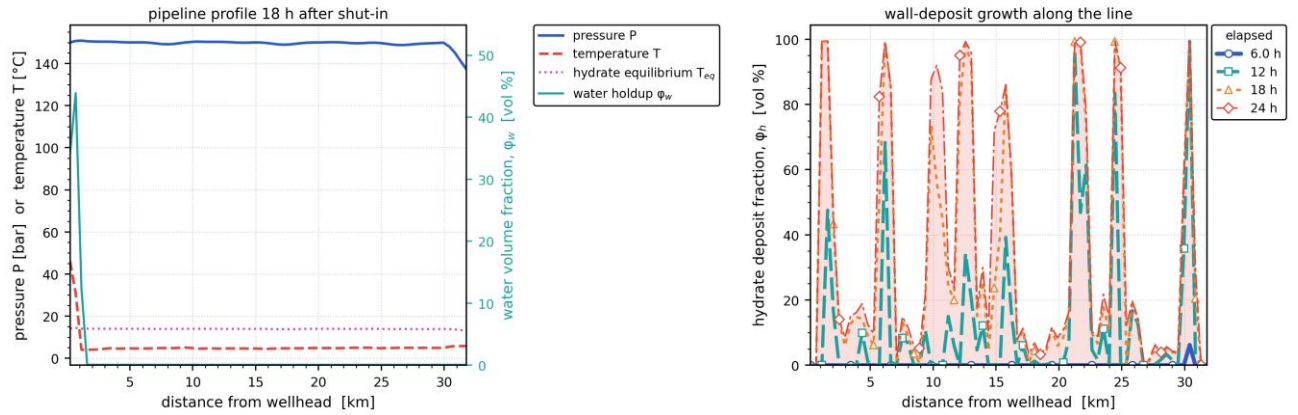
Wall-deposit growth at the monitor station vs time (curve). [shut-in]

In-pipe hydrate/water distribution and host delivery — unplanned shut-in



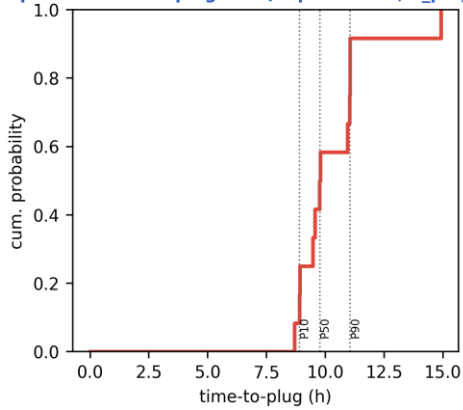
(a) In-pipe volume fractions along the route at the reported time — unconverted water, hydrate carried in the liquids, and the hydrate deposit standing on the wall; (b) the gas, oil and water mass rates delivered into the host separator against time. [shut-in]

Late-time pipeline state and deposit growth — unplanned shut-in

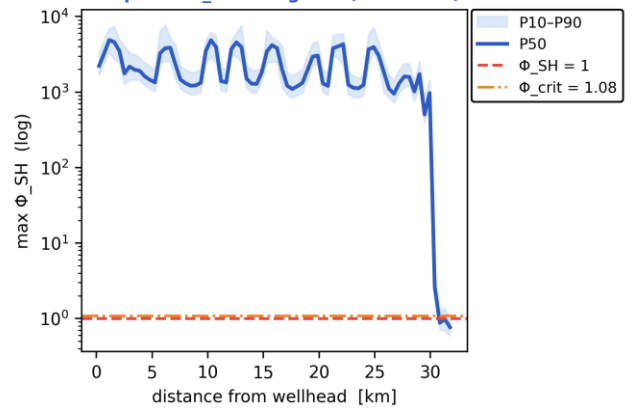


(a) The late-time pipeline profile — pressure, temperature against the hydrate equilibrium temperature, and the water volume fraction; (b) the wall-deposit volume fraction along the line at successive elapsed times. [shut-in]

Output G — time-to-plug CDF (Kaplan-Meier, $P_{plug}=100\%$)



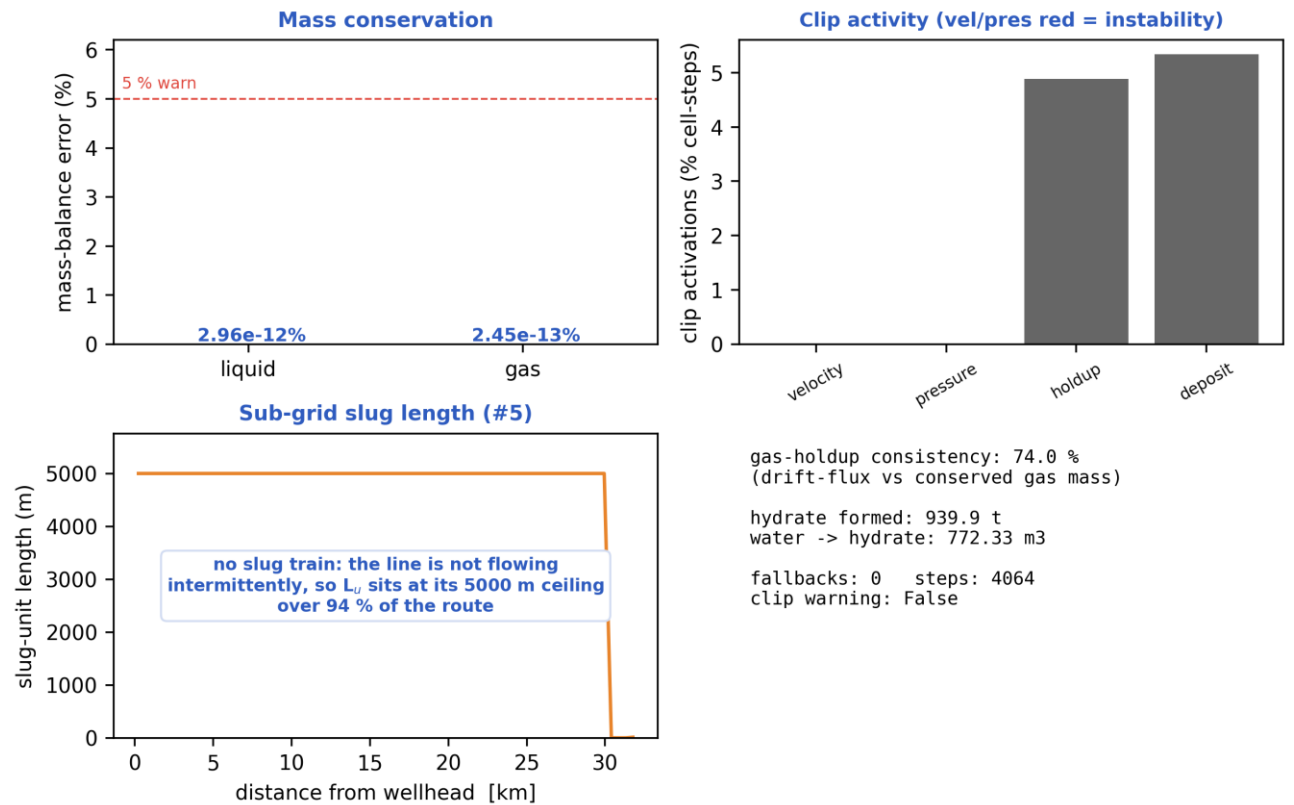
Output — Φ_{SH} along line (ensemble)



log axis: the field reaches $4.89e+03$ while the thresholds it is read against are 1 and 1.08; on a linear scale they coincide with zero.

Probabilistic time-to-plug CDF and the $\max\Phi_{SH}$ P10–P90 uncertainty band (Monte-Carlo ensemble). [shut-in]

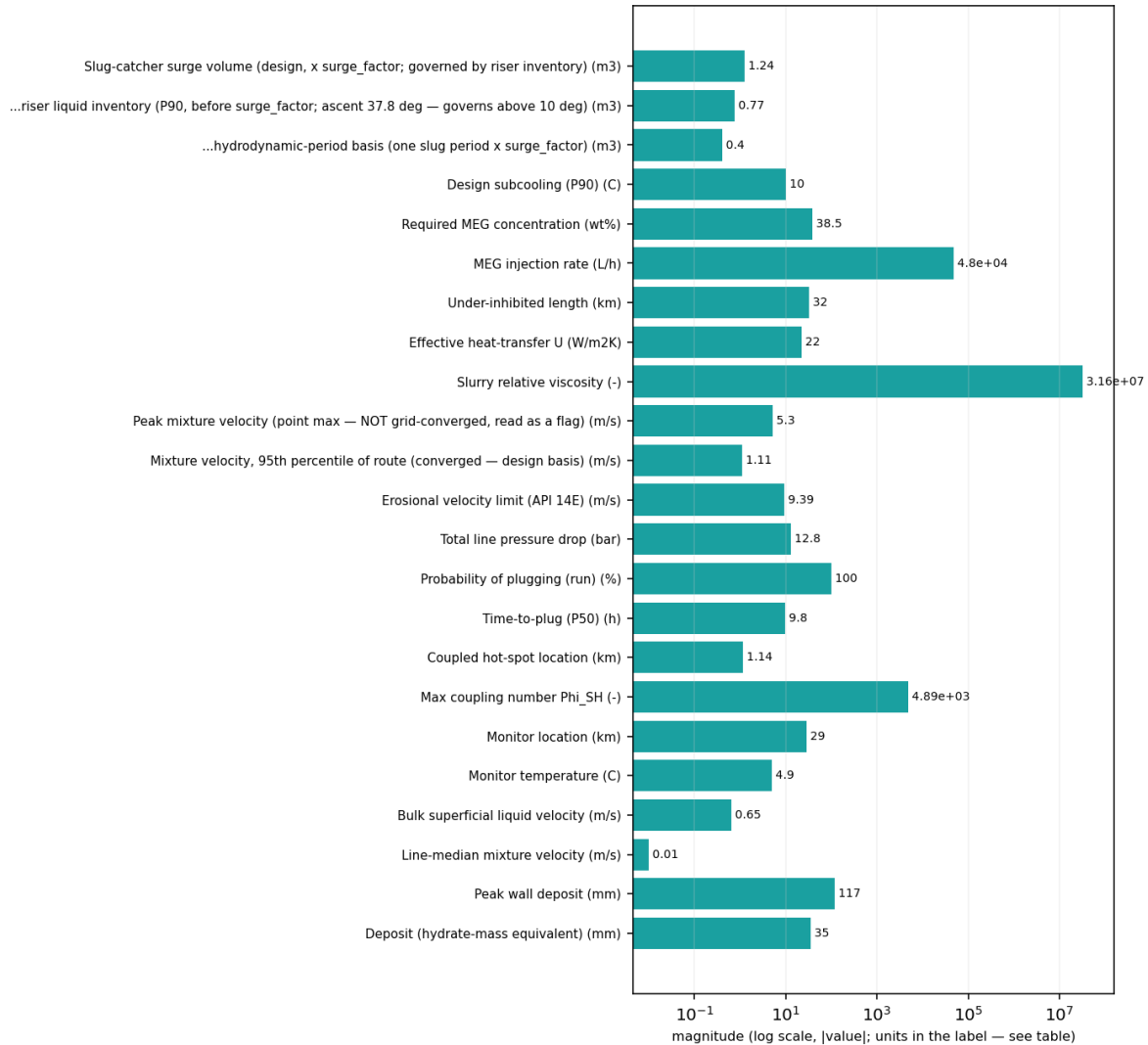
Output — solver diagnostics & balances



Solver diagnostics: liquid & gas mass balances, clip activity, slug length and numerical consistency. [shut-in]

11.2.3 Engineering deliverables (engineering_deliverables.csv)

engineering_deliverables.csv — numeric deliverables [shut-in]
(bar chart: these are unrelated named quantities, not a curve)



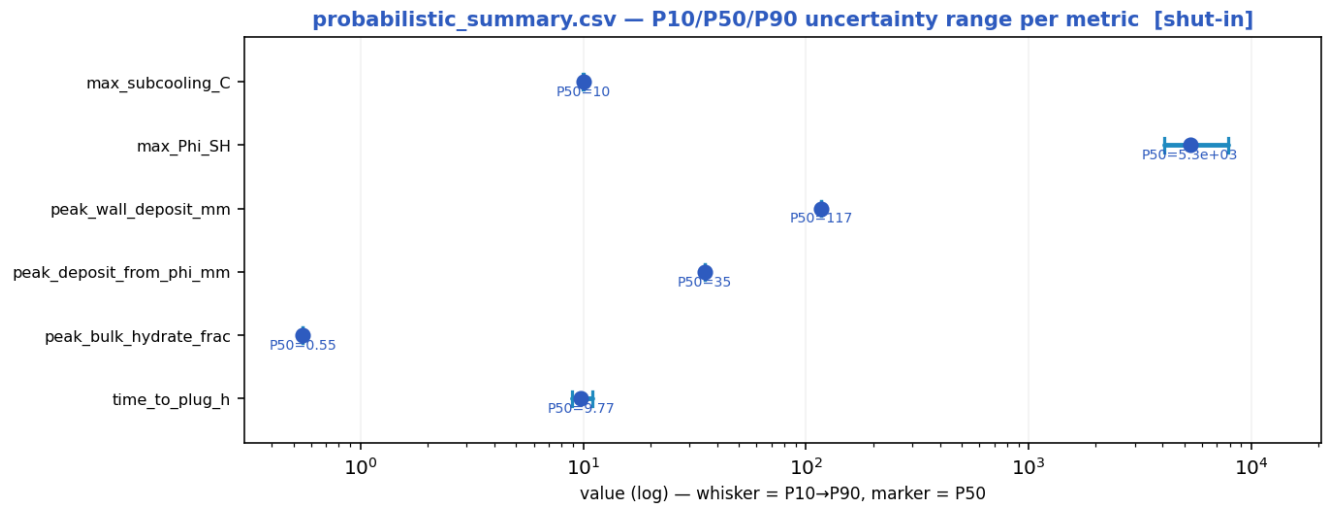
exactly zero, so not drawable on a log axis: MEG injected at inlet (wt%); Cooldown to hydrate (no-touch time) (h)
Mass-conservation error (%); Liquid discarded at the bore-closure bound (% of liquid in)

Numeric engineering deliverables as a magnitude chart [shut-in].

deliverable	value	units
Slug-catcher surge volume (design, x surge_factor; governed by riser inventory)	1.24	m3
...riser liquid inventory (P90, before surge_factor; ascent 37.8 deg — governs above 10 deg)	0.77	m3
...hydrodynamic-period basis (one slug period x surge_factor)	0.40	m3
Design subcooling (P90)	10.02	C
Required MEG concentration	38.5	wt%
MEG injection rate	47953.2	L/h
MEG injected at inlet	0.0	wt%

Under-inhibited length	32.00	km
Effective heat-transfer U	22.00	W/m2K
Cooldown to hydrate (no-touch time)	0.00	h
Slurry relative viscosity	31622776.60	-
Slurry transportable?	False	-
Peak mixture velocity (point max — NOT grid-converged, read as a flag)	5.30	m/s
Mixture velocity, 95th percentile of route (converged — design basis)	1.11	m/s
Slug-unit length (over the slugging reach)	5.7 mean / 14.2 max	m — over 6 % of the route
Erosional velocity limit (API 14E)	9.39	m/s
Total line pressure drop	12.8	bar
Probability of plugging (run)	100	%
Time-to-plug (P50)	9.8	h
Coupled hot-spot location	1.14	km
Max coupling number Phi_SH	4886	-
Phi_SH reported at plot cap (saturated)?	False	-
Phi_SH shear-limited (near-zero velocity, magnitude not resolved)?	True	-
Monitor location	29.03	km
Monitor temperature	4.9	C
Bulk superficial liquid velocity	0.65	m/s
Line-median mixture velocity	0.01	m/s
Peak wall deposit	117.1	mm
Wall deposit at full bore?	True	-
Deposit (hydrate-mass equivalent)	35.0	mm
No-touch time source	already-subcooled	-
Mass-conservation error	0.00	%
Liquid discarded at the bore-closure bound	0.00	% of liquid in — a 1-D model limit once the bore shuts; not a solver error
Mass-balance reliability warning	False	-

11.2.4 Probabilistic summary (probabilistic_summary.csv)



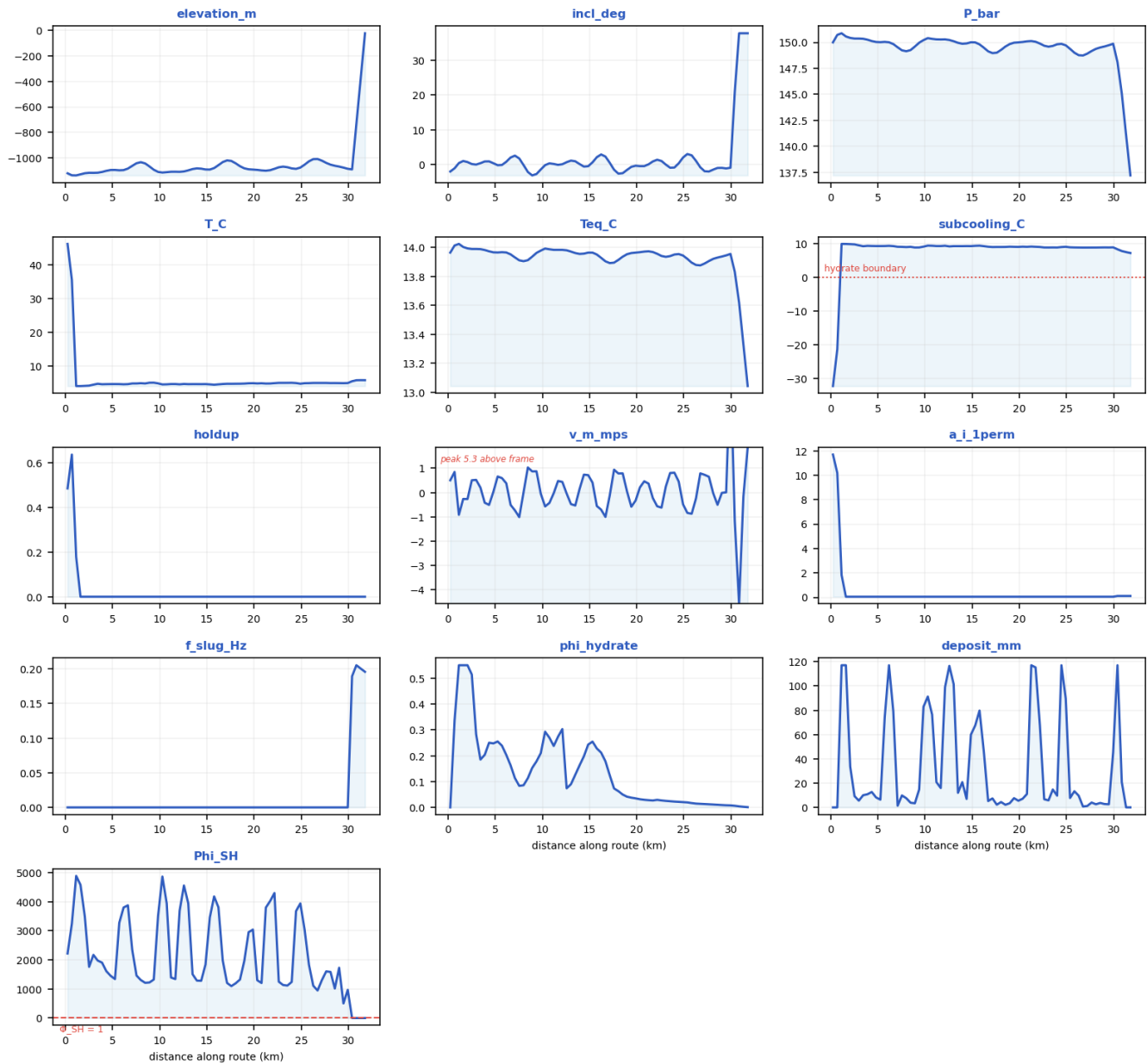
peak_wall_deposit_mm is what the transient actually deposits on the wall, capped at the full-bore radius once the bore shuts.
peak_deposit_from_phi_mm is the annulus equivalent to the peak SUSPENDED bulk hydrate fraction ϕ at one instant, $t = \phi \cdot D/4$.
These are different quantities, not a result and its bound: ϕ is instantaneous while wall deposit accumulates, so on a plugging duty the wall value exceeds the ϕ -equivalent (shut-in: 117 mm against 35 mm) and on a sub-critical one it falls well below it.

P10 → P90 uncertainty range with the P50 marker per metric [shut-in].

metric	P10	P50	P90
max_subcooling_C	9.92	10	10
max_Phi_SH	4.07e+03	5.3e+03	7.89e+03
peak_wall_deposit_mm	117	117	117
peak_deposit_from_phi_mm	35	35	35
peak_bulk_hydrate_frac	0.55	0.55	0.55
time_to_plug_h	8.9	9.77	11

11.2.5 fields_profile.csv — steady-state spatial profile of every field along the route

fields_profile.csv — every column as a curve vs distance along route (km) [shut-in]



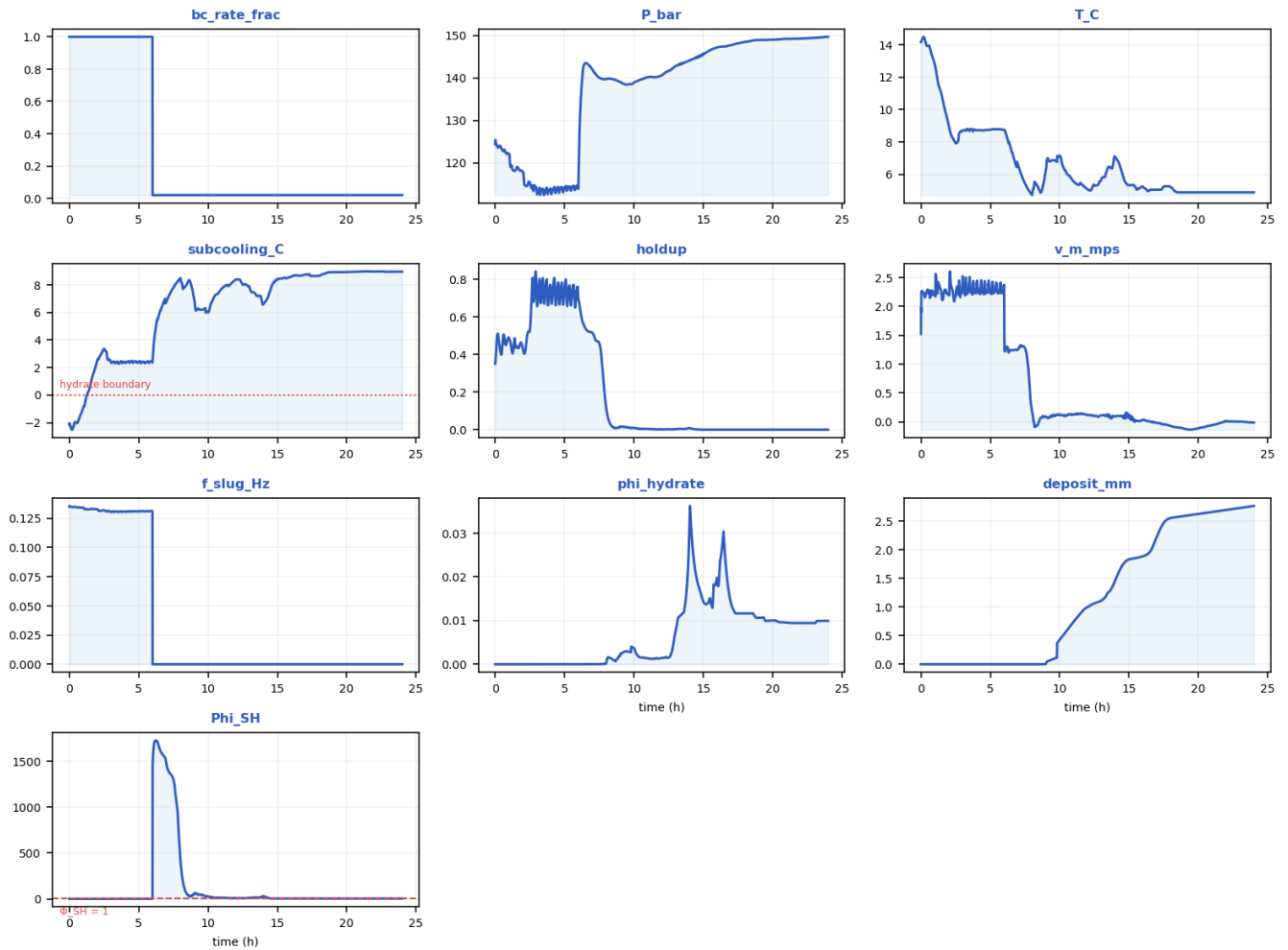
Every column of fields_profile.csv drawn as a curve [shut-in].

x_km	elevation_m	incl_deg	P_bar	T_C	Teq_C	subcooling_C	holdup	v_m_mps	regime	a_i_lperm	f_slug_Hz	phi_hydrate	deposit_mm	Phi_SH
0.22857	-1118.4	-1.9608	150	46.299	13.963	-32.32	0.48612	0.5023	stratified-smooth	11.687	0.0001	0.0008054	0	2218
2.5143	-1114.5	0.18865	150.37	4.1313	13.988	9.8611	0.001	0.50472	stratified-smooth	0.047104	0.0001	0.51414	9.2801	1760.8

									h					
4.8	-1092.7	0.42362	150.04	4.5987	13.965	9.3623	0.001	0.028692	stratified-smooth	0.047104	0.0001	0.24799	8.1977	1451.5
7.0857	-1060.5	2.601	149.53	4.7795	13.93	9.1427	0.001	-0.73329	stratified-smooth	0.047104	0.0001	0.11392	1.3639	2348.6
9.8286	-1107.7	-1.3732	150.22	4.8062	13.977	9.1831	0.001	-0.048655	stratified-smooth	0.047104	0.0001	0.21046	82.883	3520.2
12.114	-1107	0.13701	150.29	4.5256	13.982	9.4526	0.001	0.43654	stratified-smooth	0.047104	0.0001	0.30336	99.046	3689.9
14.4	-1082.3	-0.54271	149.9	4.5795	13.956	9.3561	0.001	0.73569	stratified-smooth	0.047104	0.0001	0.19832	6.9583	1281.5
17.143	-1017.6	0.54352	148.97	4.6901	13.891	9.0704	0.001	-0.11872	stratified-smooth	0.047104	0.0001	0.12483	7.4147	1207.8
19.429	-1087.1	-0.61307	149.97	4.8345	13.96	9.1328	0.001	-0.58533	stratified-smooth	0.047104	0.0001	0.038406	7.6266	2951.3
21.714	-1094.2	0.93887	150.06	4.771	13.967	9.1433	0.001	-0.23033	stratified-smooth	0.047104	0.0001	0.027472	115.27	4023.5
24.457	-1083.8	0.40028	149.86	4.9009	13.953	9.0678	0.001	0.44961	stratified-smooth	0.047104	0.0001	0.022274	117.07	3676.2
26.743	-1007	-0.74006	148.74	4.9385	13.875	8.9057	0.001	0.78833	stratified-smooth	0.047104	0.0001	0.014611	0.70837	948.69
29.029	-1066	-0.91009	149.61	4.8876	13.936	8.956	0.001	-0.014125	stratified-smooth	0.047104	0.0001	0.0099305	2.7657	1730.2
31.771	-25	37.781	137.21	5.757	13.044	7.2883	0.001	1.8593	slug	0.10991	0.19581	0.0017126	0.026989	0.76521

11.2.6 timeseries_monitor.csv — transient response at the monitored station

timeseries_monitor.csv — every column as a curve vs time (h) [shut-in]



Every column of timeseries_monitor.csv drawn as a curve [shut-in].

time_h	bc_rate_frac	P_bar	T_C	subcooling_C	holdup	v_m_mps	f_slug_Hz	phi_hydrate	deposit_mm	Phi_SH
0	1	124.37	14.165	-2.1326	0.35096	1.5238	0.13502	0	0	0.60534
1.1214	1	118.97	12.23	-0.66947	0.45802	2.3931	0.13273	0	0	0.73719
2.4192	1	115.52	8.0406	3.2452	0.52128	2.1143	0.13187	0	0	0.7926
4.2269	1	114.02	8.7273	2.4408	0.79559	2.2377	0.13101	0	0	1.1932
6.0116	0.02	117.46	8.7489	2.6822	0.69205	1.4271	0.0001	0	0	1485.9
8.229	0.02	139.81	5.4942	7.7313	0.042237	-0.078871	0.0001	0.0014869	0	156.28
10.281	0.02	139.27	6.4093	6.7804	0.0059287	0.11397	0.0001	0.0019452	0.51337	15.48
12.171	0.02	140.98	5.0032	8.3962	0.0021902	0.11249	0.0001	0.0014651	1.0231	6.6584
14.062	0.02	144.13	7.0001	6.6929	0.008332	0.092245	0.0001	0.035091	1.5002	25
15.914	0.02	147.05	5.2392	8.5624	0.001	0.02172	0.0001	0.018641	1.8807	3.9085
17.831	0.02	148.34	5.2654	8.642	0.001	-0.061331	0.0001	0.011618	2.541	4.031
20.166	0.02	148.99	4.8864	8.9342	0.001	-0.10227	0.0001	0.010014	2.6323	3.4827
22.16	0.02	149.25	4.8863	8.9646	0.001	0.0061847	0.0001	0.0094257	2.7018	3.4872

23.994	0.02	149.61	4.8876	8.956	0.001	-0.014125	0.0001	0.0099305	2.7657	3.6977
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11.3 Engineered mitigation — restored insulation + continuous MEG

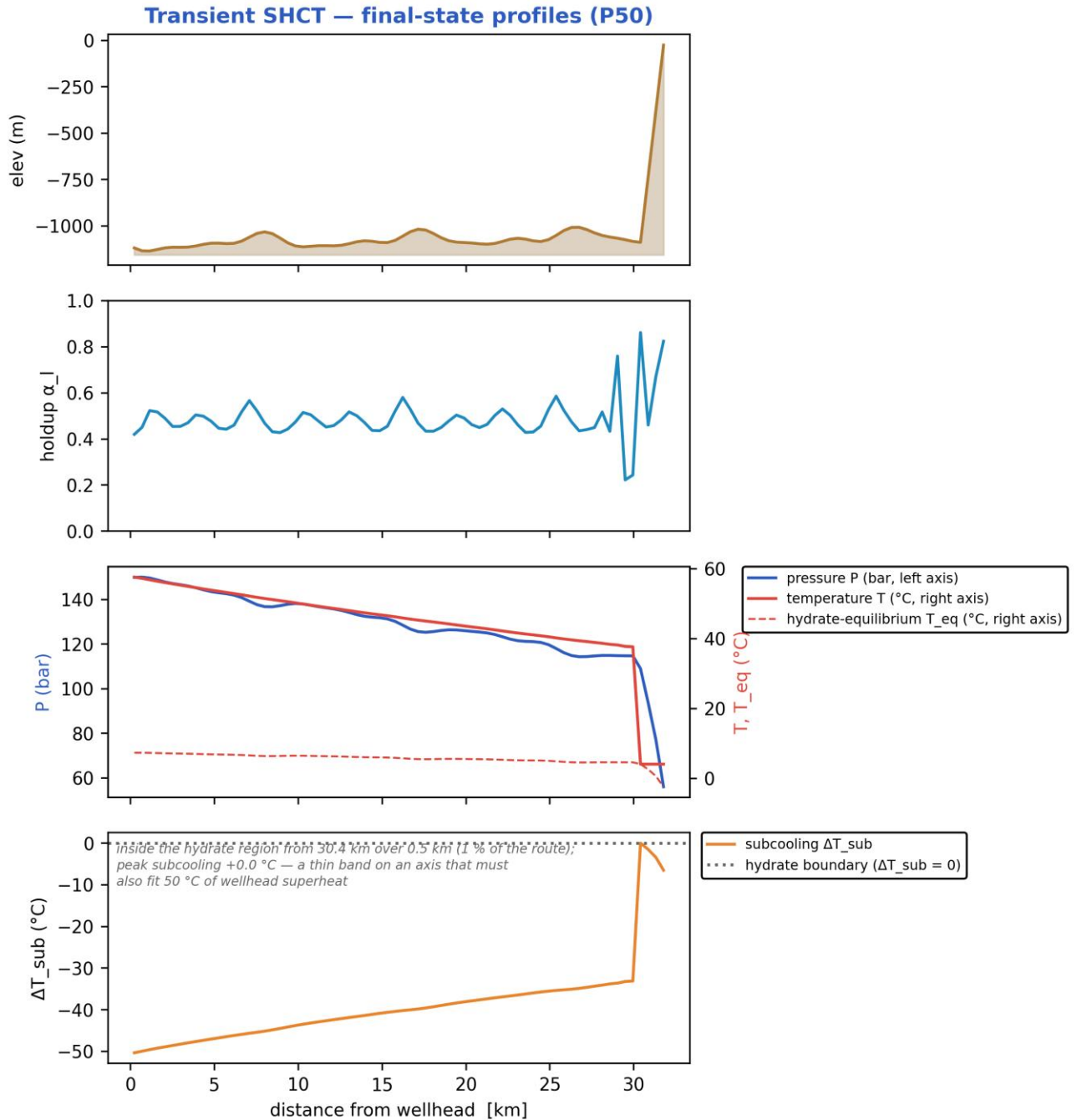
11.3.1 Headline prediction metrics

Metric	Value	Units
Total ΔP	93.98	bar
Peak velocity	8.352	m/s
Erosional limit (API 14E)	5.306	m/s
Max slug length	84.29	m
Mean slug length	27.31	m
Slug/intermittent fraction	0.7241	—
Slug-catcher surge (design)	105.6	m ³
Max subcooling	0.01365	°C
Design subcooling (P90)	0.02784	°C
Sustained Φ_{SH} (coupling)	0	—
Sustained coupling hot-spot	30.4	km
Length above $\Phi_{SH} = 1$	0	km
Φ_{SH} running max (startup)	0.302	—
...attained at	0.4015	h
Plug probability	0	frac
Peak wall deposit	0	mm
MEG required	8.96	wt%
MEG rate	7537	L/h
Under-inhibited length	0.4571	km
No-touch time	54.32	h
Effective U	3.448	W/m ² K
Slurry rel. viscosity	1	—
Hydrate mass formed	0	kg
Min mixture sound speed	210.5	m/s
Wax onset	30.4	km
Max scaling index	0.4879	—
Liquid mass error	4.685e-15	frac
Gas mass error	0	frac
Solver fallbacks	0	#

11.3.2 Charts, curves, contours and maps (23 figures)

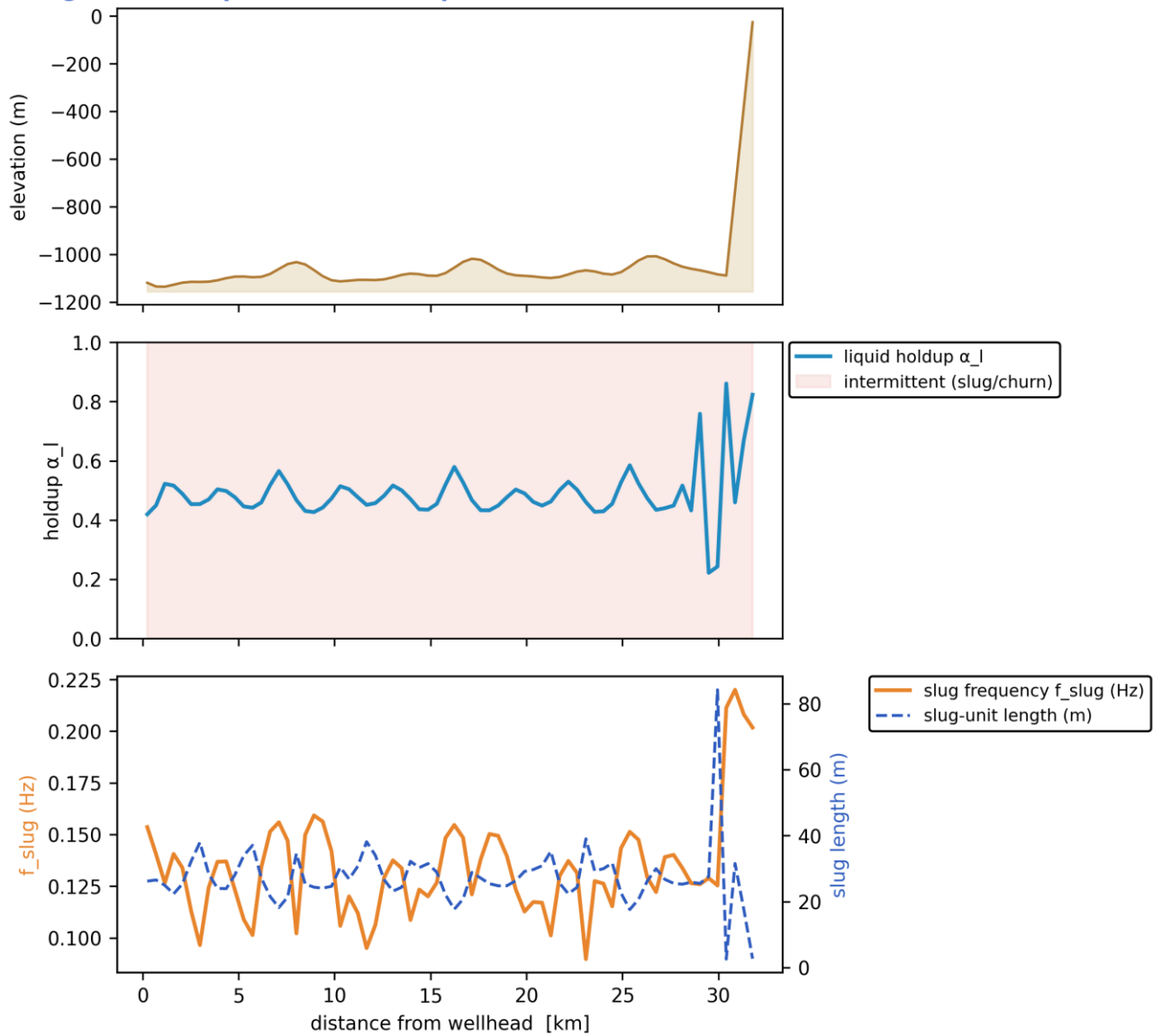
Note on the resolved-slug figures. The transport grid is $dx \sim 457$ m while a slug unit is of order 10-40 m, so individual slugs are a SUB-GRID quantity: the solver carries them statistically through the slug frequency f_{slug} , the slug unit length $L_u = V_t / f_{slug}$ and the slug-body holdup α_{ls} . The three

figures marked 'sub-grid reconstruction' therefore render a kinematic reconstruction built entirely from those solver outputs - at every station the reconstructed square wave carries the solver's local slug frequency and translational celerity, and the body/film split is chosen so that the unit-averaged holdup reproduces the solver's cell-average α_l exactly. Period, celerity, length and holdup are all run outputs; nothing in them is assumed. Every other figure in this gallery is plotted directly from the solver's space-time history.



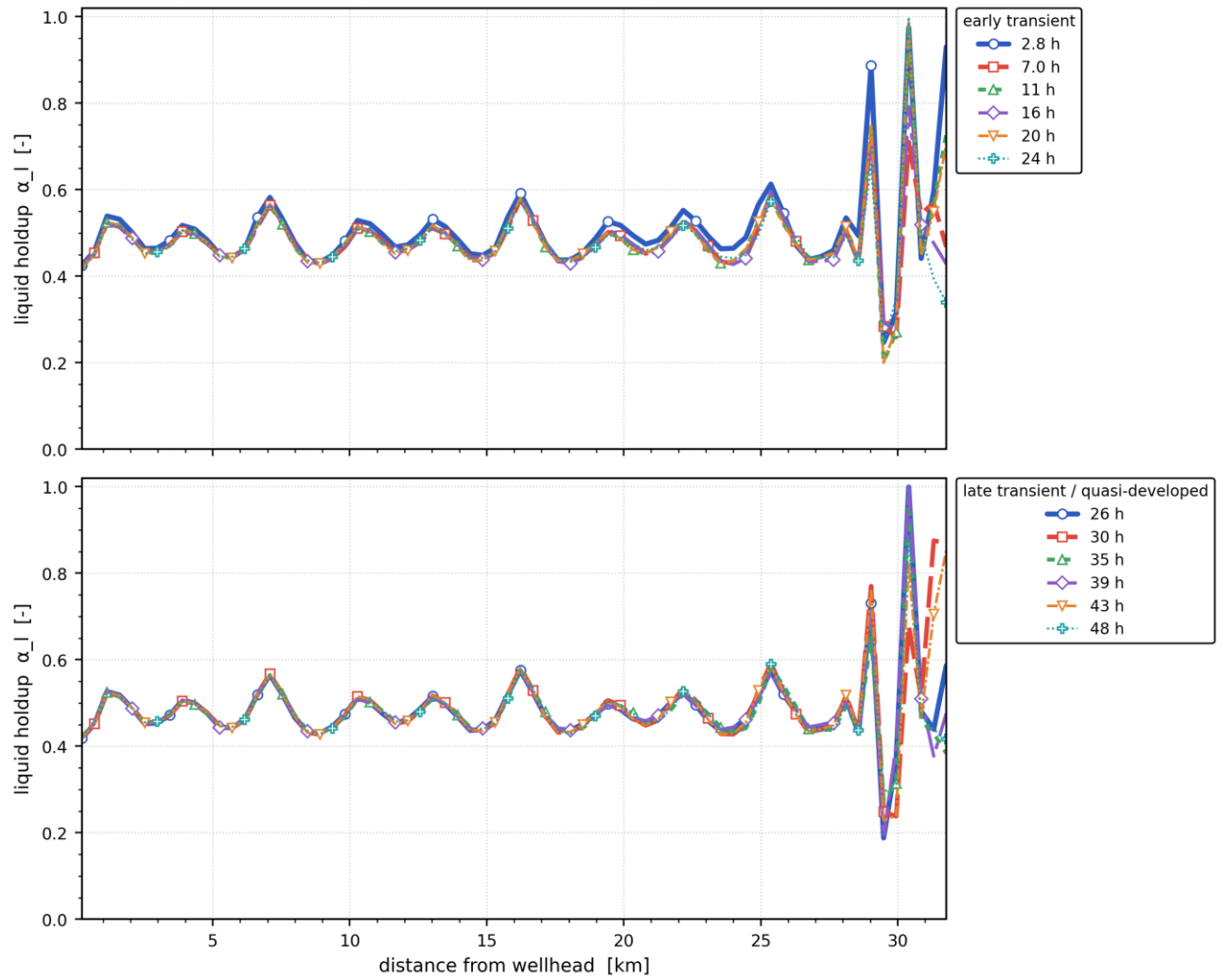
Final-state P50 profiles: seabed elevation, liquid holdup, pressure & temperature vs hydrate T_{eq} , and subcooling along the whole route. [mitigated]

Slug-formation prediction — deepwater medium-crude-oil tie-back



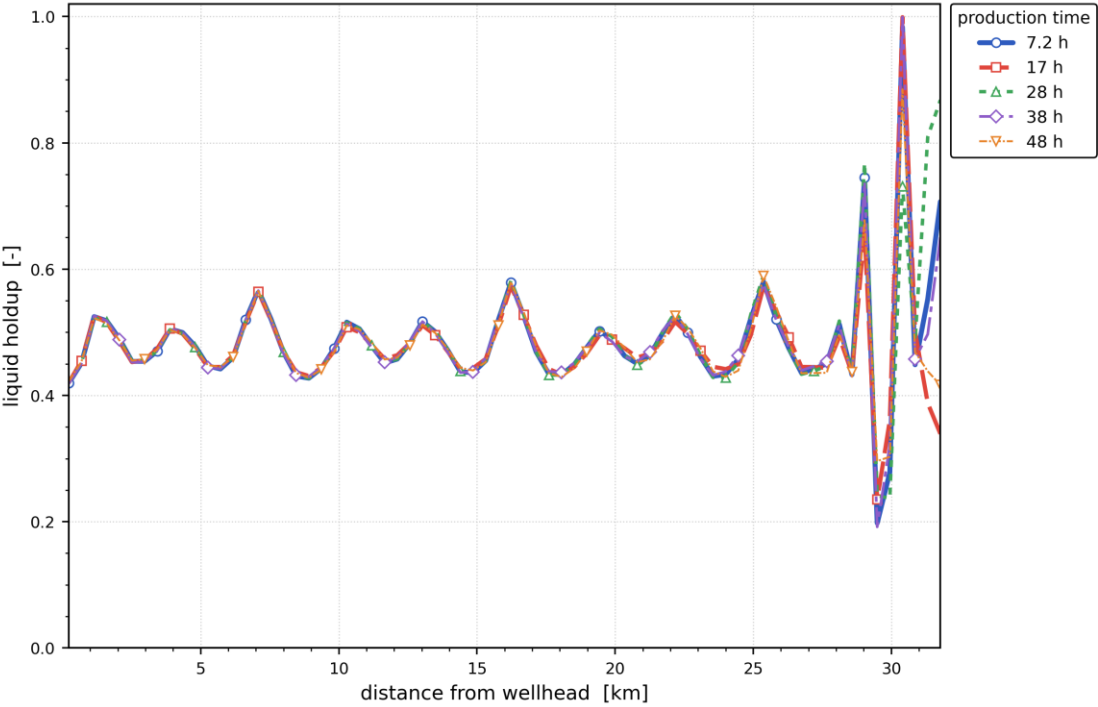
Slug-formation prediction: terrain, intermittent (slug/churn) bands, slug frequency and slug-unit length, and liquid holdup.
[mitigated]

Liquid-holdup evolution along the 32 km tie-back — engineered fix (insulation + MEG)



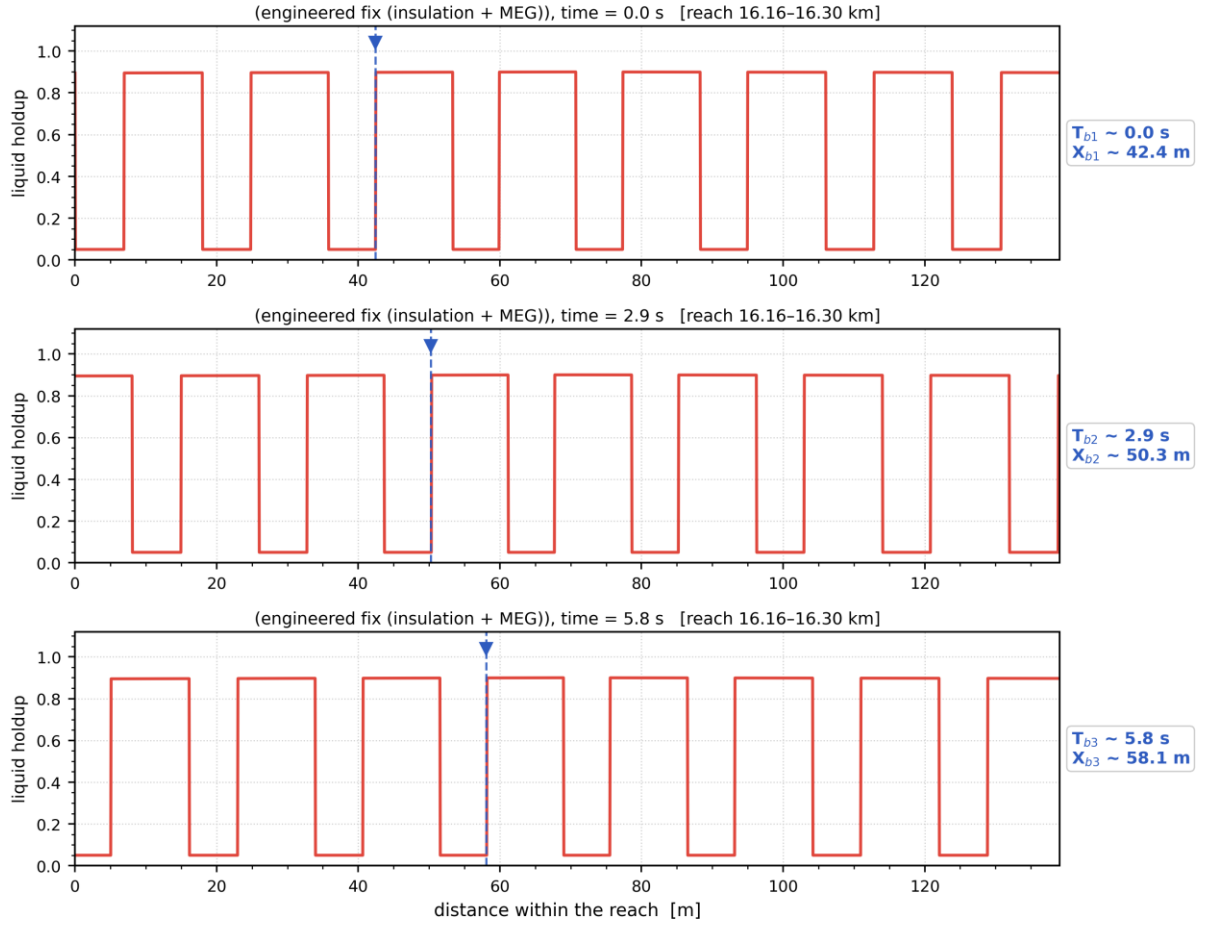
Liquid holdup along the whole route at successive times — the early transient (upper) and the late, quasi-developed state (lower).
The travelling and terrain-locked holdup structure is the slug activity resolved in time. [mitigated]

Distribution of liquid holdup along the pipeline after different production durations — engineered fix (insulation + MEG)



Distribution of liquid holdup along the pipeline after successive elapsed durations (shut-in duration for the shut-in scenario, production time otherwise) — one curve per duration. [mitigated]

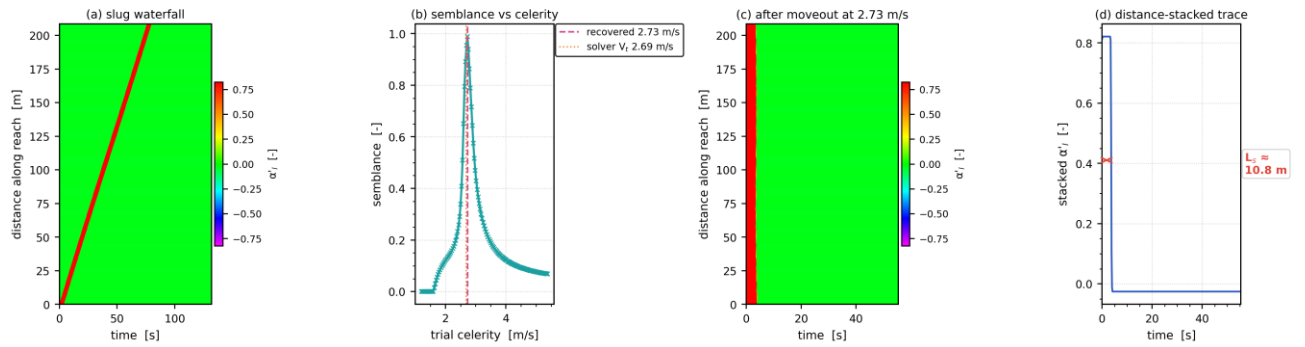
Slug propagation and front tracking — resolved slug units at $t = 47.8$ h ($V_t = 2.69$ m/s, $f_{slug} = 0.15$ Hz, $L_u = 17.4$ m)



Mass-consistent sub-grid reconstruction: period, celerity, unit length and unit-average holdup are the solver's own fields.

Slug propagation and front tracking: three successive snapshots of the resolved slug units over a short reach, with one front followed across the panels (its arrival time T_b and position X_b are annotated). Sub-grid reconstruction — see §note. [mitigated]

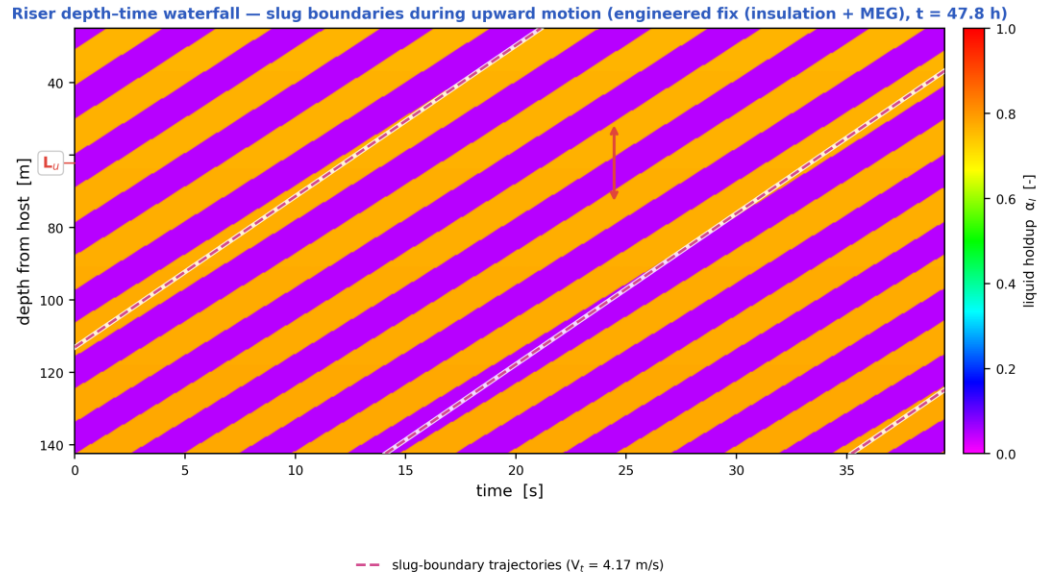
Slug tracking in the space-time plane — engineered fix (insulation + MEG), reach 16.12–16.33 km at $t = 47.8$ h ($L_u = 17.4$ m, $f_{slug} = 0.15$ Hz)



One slug unit of the mass-consistent sub-grid reconstruction; the moveout scan (b) recovers 2.73 m/s against the solver's $V_t = 2.69$ m/s at the tracked cell (+1.5 %), the difference being the variation of V_t across the 209 m window.

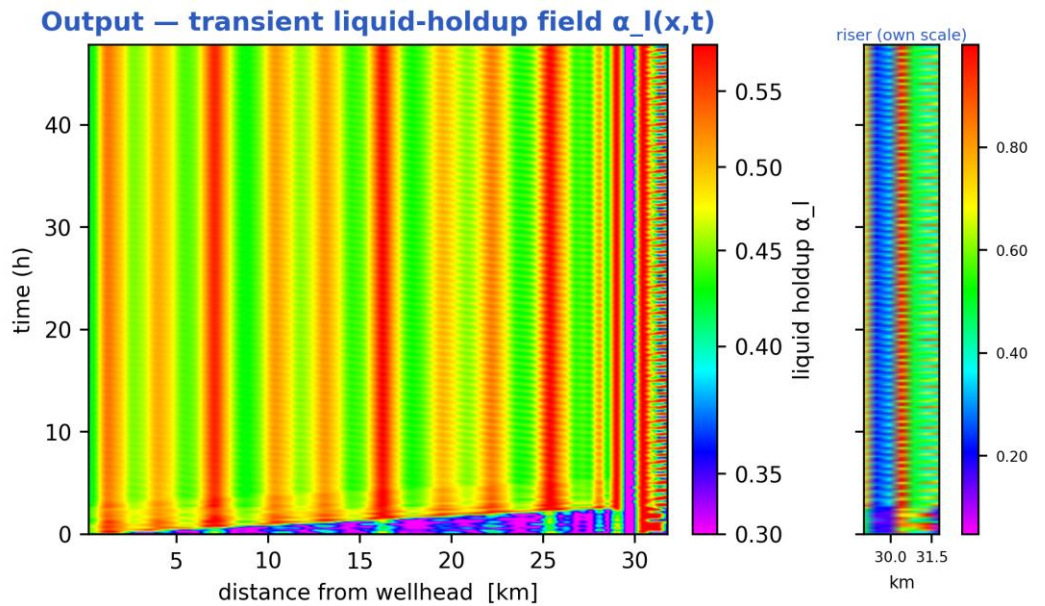
Slug tracking in the space-time plane: (a) the distance-time waterfall of one slug unit, (b) semblance against trial celerity, (c) the waterfall after linear moveout at the recovered celerity, (d) the distance-stacked trace whose width gives the slug body length. The

recovered celerity is reported on the figure against the solver's own translational velocity V_t at the tracked cell. Sub-grid reconstruction — see §note. [mitigated]



Mass-consistent sub-grid reconstruction over the steel-catenary riser ($L_u = 29.4$ m, slug body 16.4 m, period 7.0 s); celerity, period and length are solver outputs. The arrow marks one slug unit projected onto the depth axis, 22.8 m.

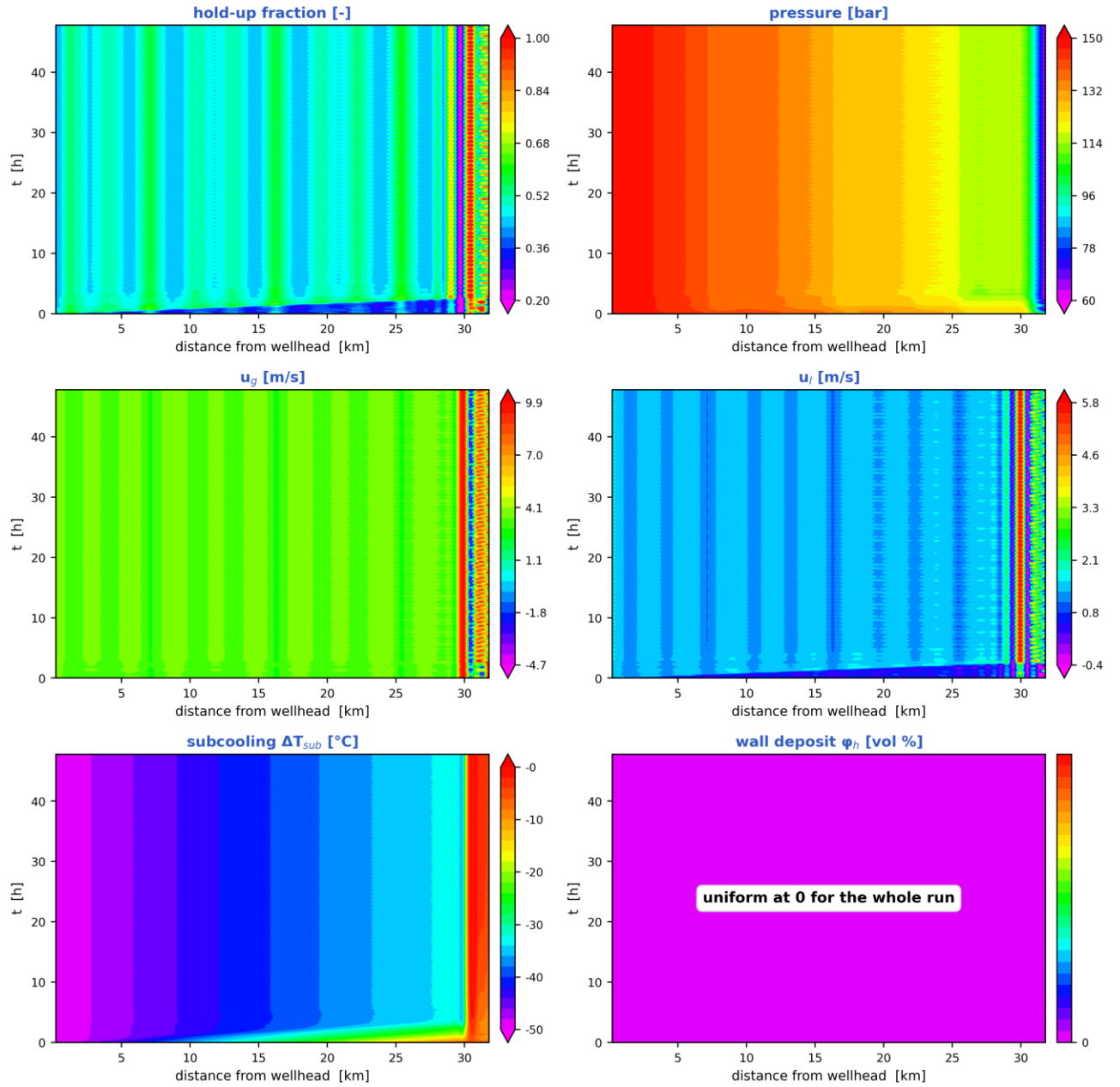
Depth-time waterfall over the steel-catenary riser: slug boundaries during upward motion, their trajectories (slope = translational celerity) and the slug unit length marked on the depth axis. Sub-grid reconstruction — see §note. [mitigated]



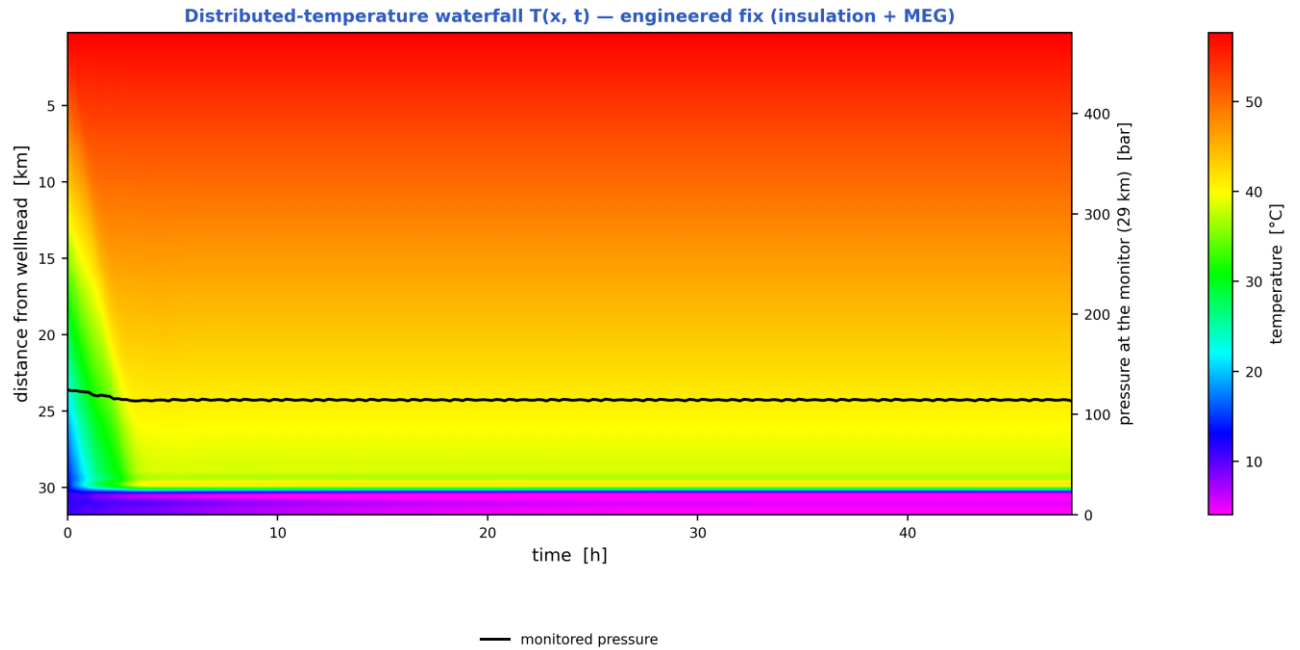
the colour scale is set by the flowline ($\alpha_l = 0.34$ – 0.58); the riser, the last 10 % of the route, runs 0.05–1.00 and saturates at both ends

Liquid-holdup field $\alpha_l(x,t)$ — the bands are slug activity migrating along the line in time (space-time contour map). [mitigated]

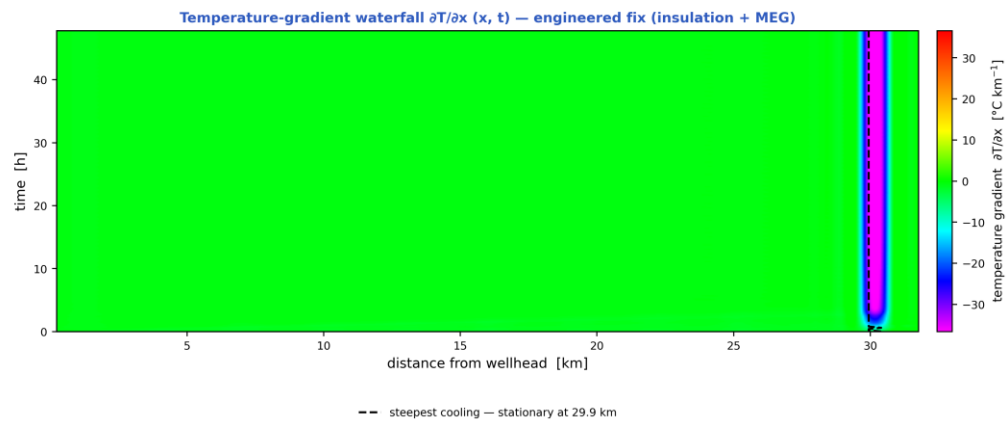
Space-time solution of the 32 km tie-back — engineered fix (insulation + MEG) (N = 70 cells, 240 snapshots)



The TRUE space-time solution of the tie-back: liquid holdup, pressure, gas and liquid velocities, subcooling and wall-deposit fraction, each as a filled-contour field over (distance, time). [mitigated]

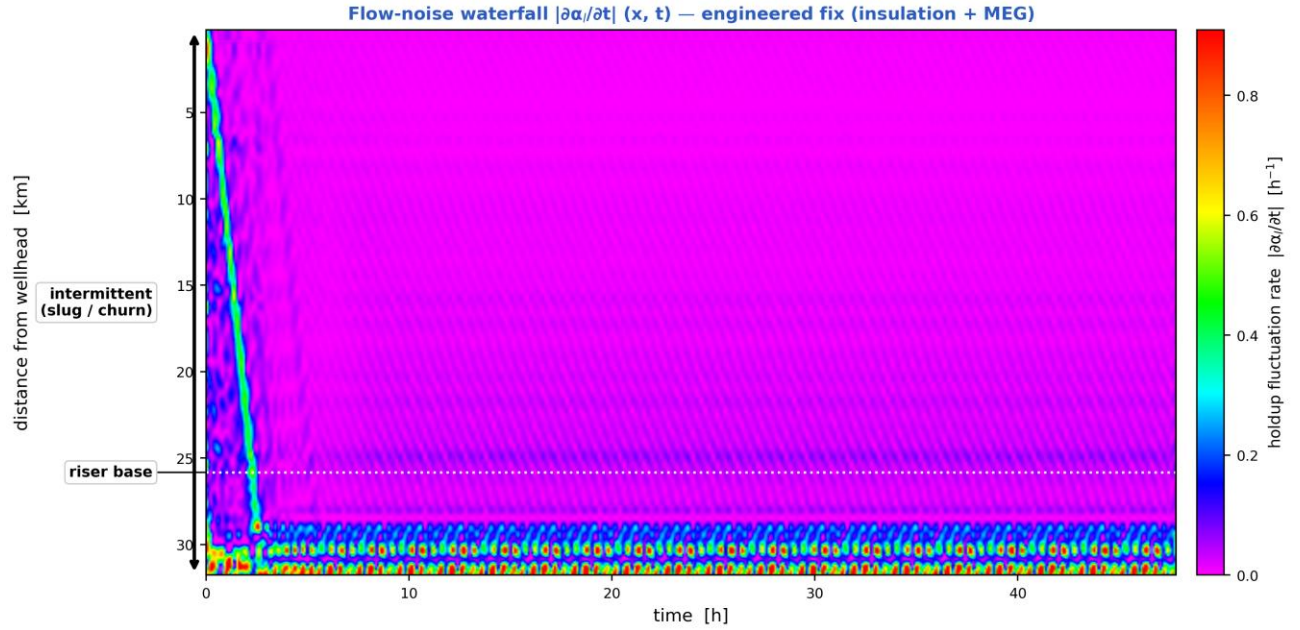


Distributed-temperature waterfall $T(x, t)$ — the thermal field over distance and time, with the monitored pressure overlaid, the operating stages marked and the hydrate-onset distance annotated. [mitigated]



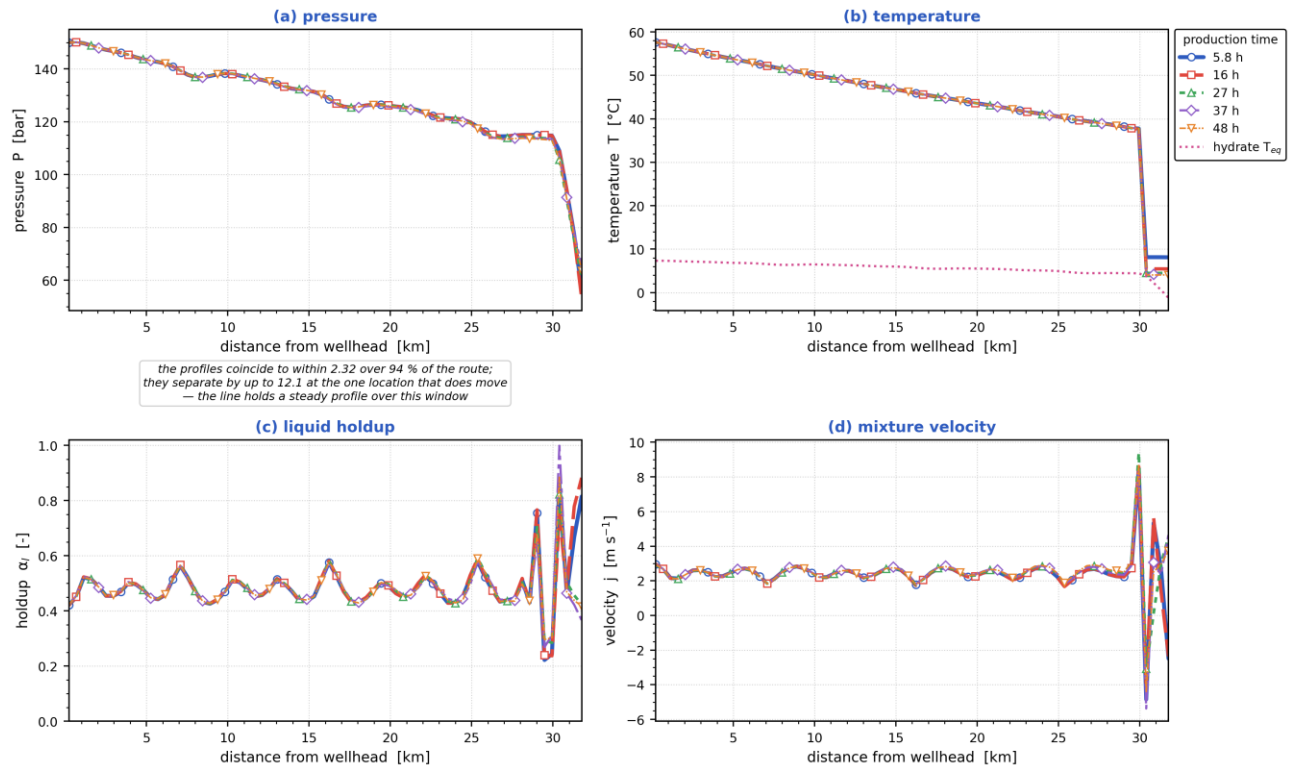
The dashed line tracks the steepest cooling at each instant. On this duty it does NOT travel: the hot inlet meeting a cold seabed holds the steepest gradient at 29.9 km, and stays within 0.0 km of it for 80 % of the window — the thermal front is stationary, not propagating.

Temperature-gradient waterfall $\partial T / \partial x (x, t)$. A travelling thermal front is a narrow band of steep gradient, so it is localised here even where the temperature map looks smooth; the dashed line tracks the steepest cooling. [mitigated]



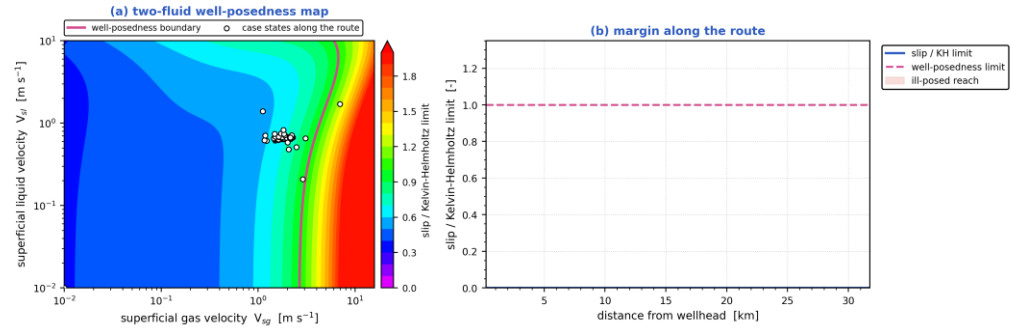
Flow-noise waterfall $|\partial\alpha_i/\partial t|(x, t)$ — where the liquid holdup changes fastest is where the flow is most unsteady, with the intermittent (slug/churn) reach and the riser base marked. [mitigated]

Parameters along the pipeline at successive times — engineered fix (insulation + MEG)



Pressure, temperature, liquid holdup and mixture velocity along the route, each at the same successive times. [mitigated]

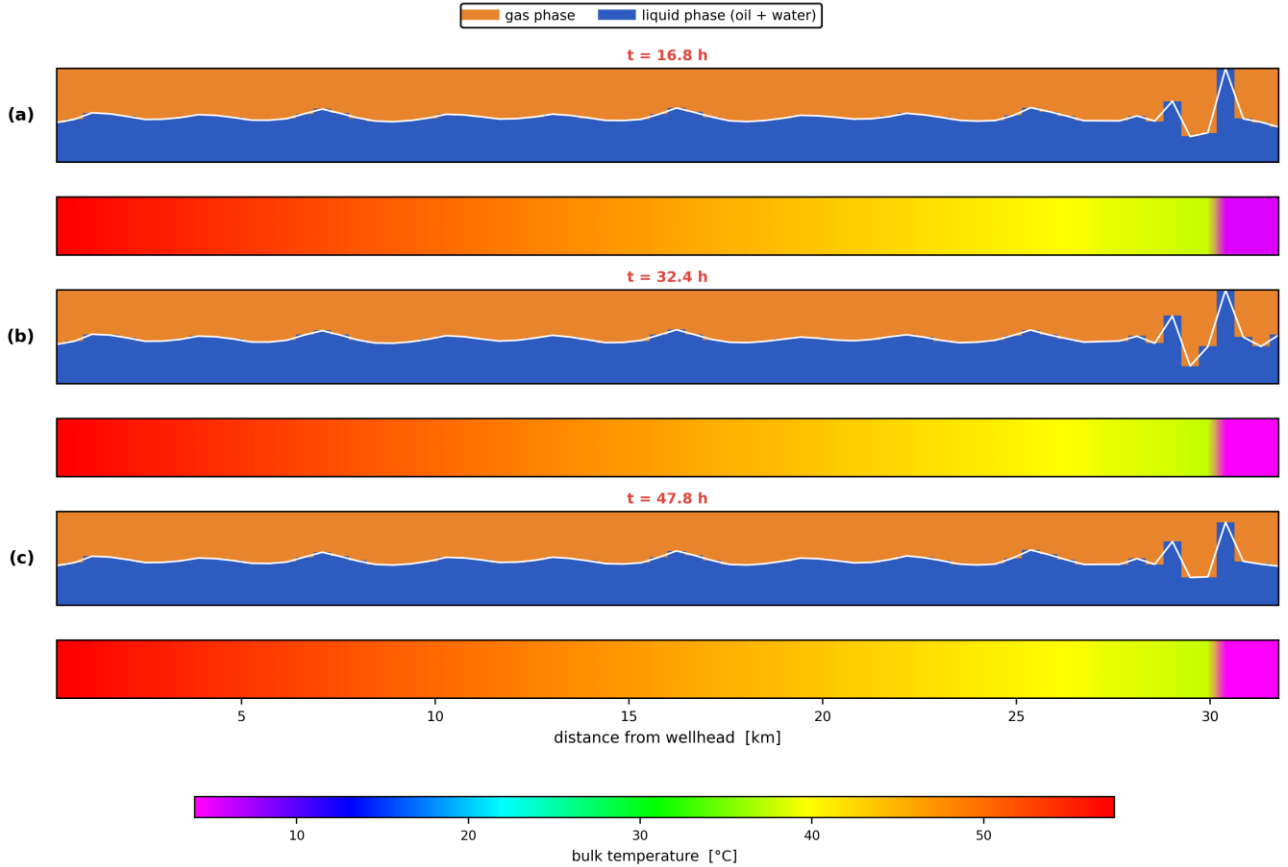
Well-posedness of the two-fluid description — engineered fix (insulation + MEG)



the interfacial-pressure term ($C = 1.20 \pm 1$) makes the two-fluid system unconditionally hyperbolic, so the initial-value problem is well posed over the whole route; against the INVISCID criterion ($C = 0$) 7 % of the route would be past the limit, and that is the reference the literature quotes

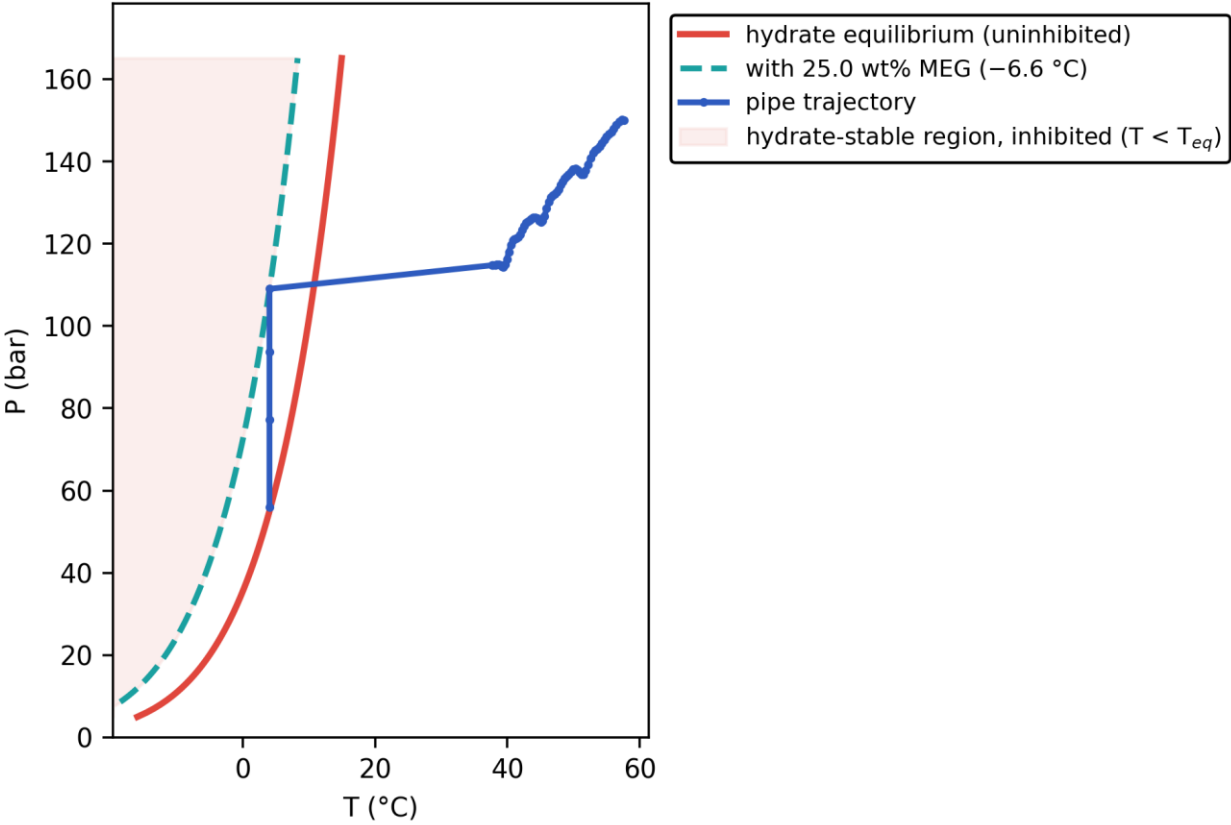
Well-posedness of the two-fluid description: (a) the inviscid Kelvin–Helmholtz boundary over the superficial-velocity plane with the case's own states, (b) the slip/KH margin along the route. Past the boundary the 1-D two-fluid initial-value problem is ill-posed and growth rates are grid-dependent. [mitigated]

Pipeline cloud maps at successive times — phase distribution (upper) and temperature (lower), engineered fix (insulation + MEG)

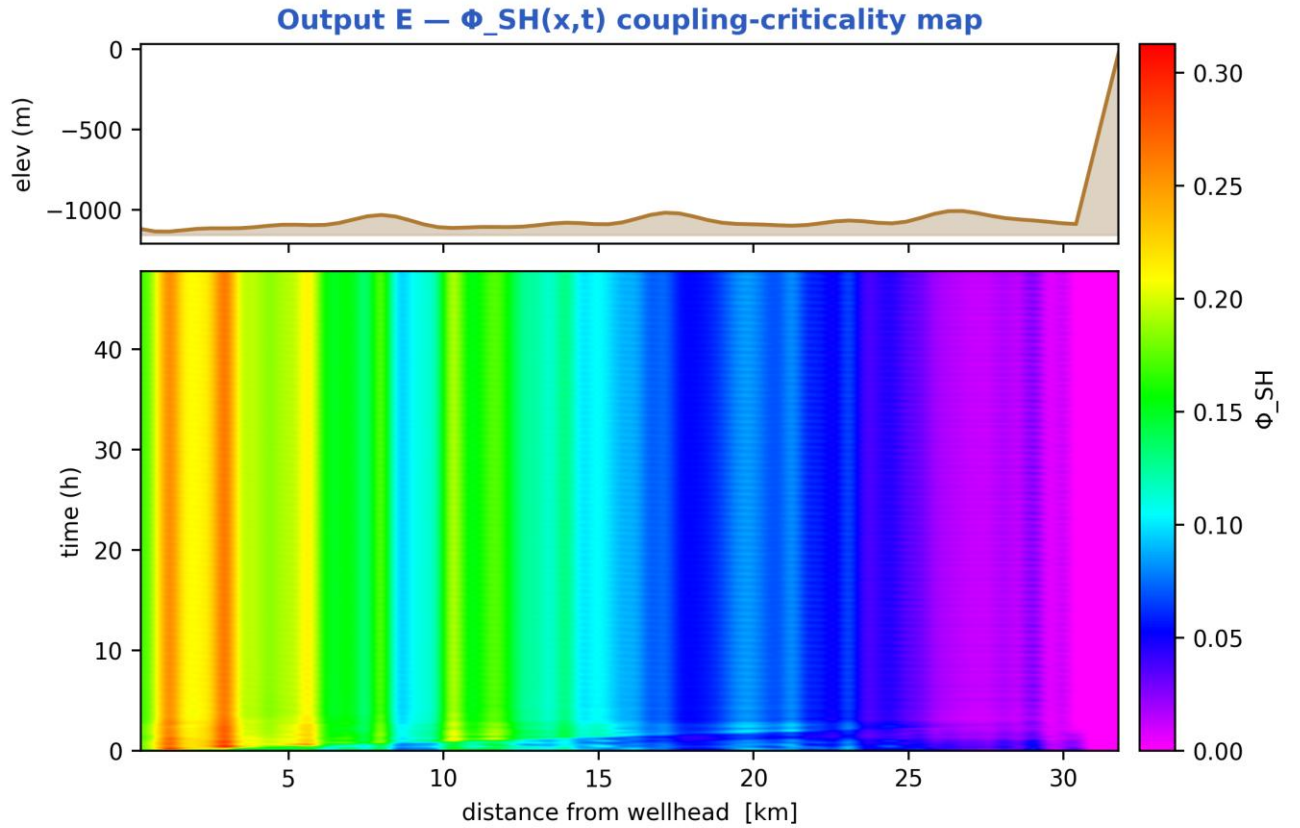


Pipeline cloud maps at successive times: the gas/liquid phase distribution inside the bore (upper strip of each pair) above the bulk-temperature field along the same reach (lower strip), on a shared temperature scale. [mitigated]

Output C — P-T trajectory vs hydrate envelope



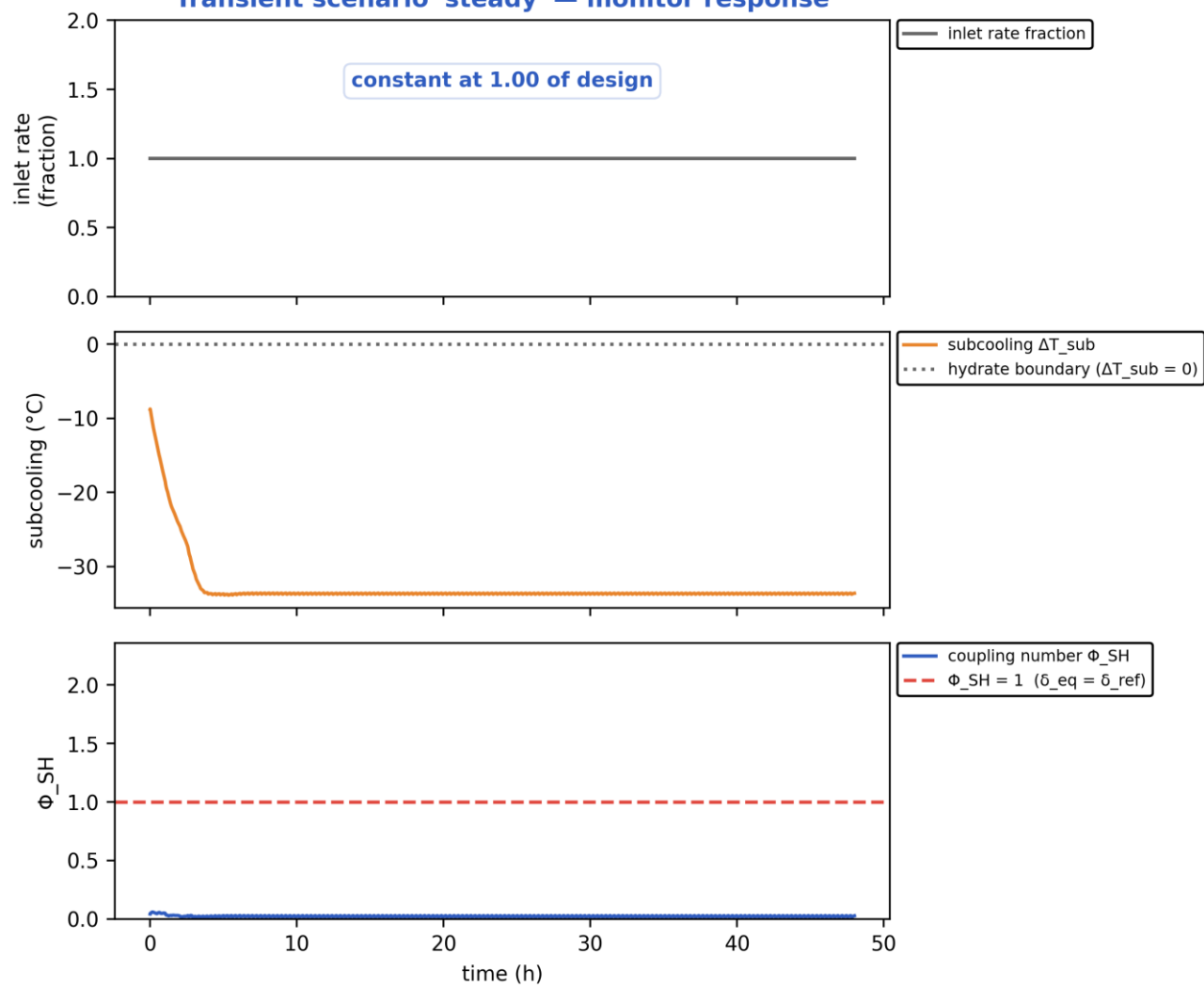
Production P-T trajectory against the hydrate-stability envelope (curve). [mitigated]



the field peaks at $\Phi_{SH} = 0.30$, everywhere below the critical $\Phi_{SH} = 1$ — the colour scale is the data's own range

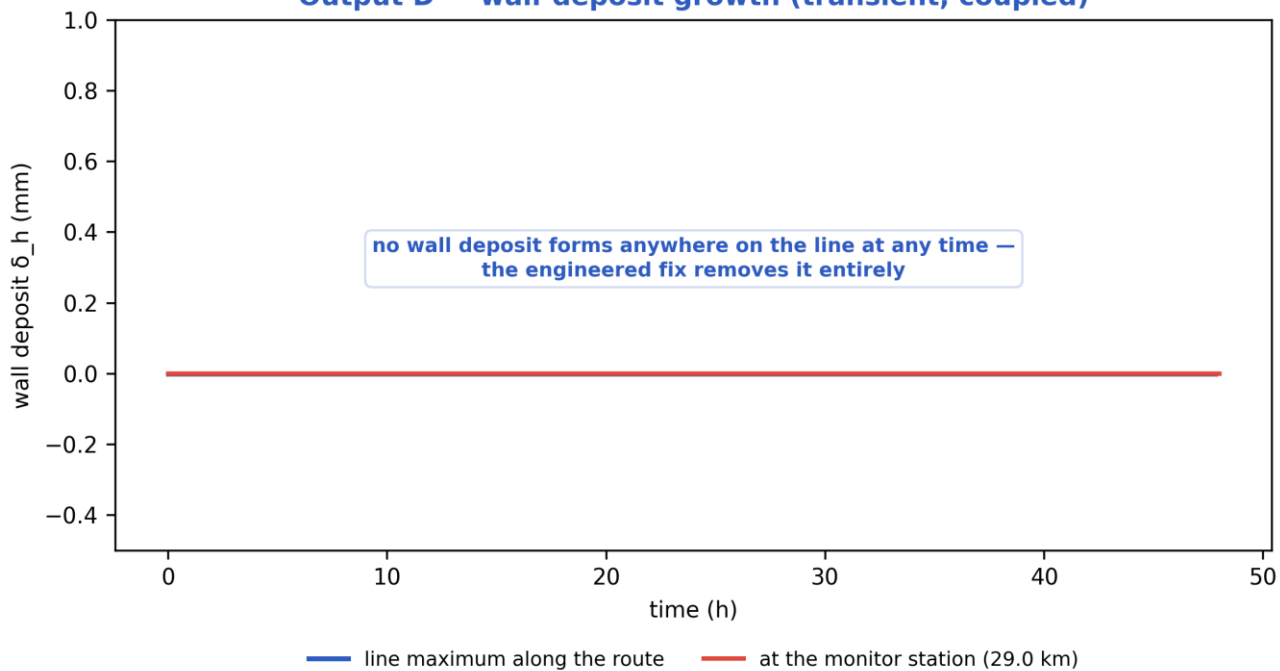
Slug-Hydrate coupling-criticality map $\Phi_{SH}(x,t)$. Φ_{SH} is the equilibrium deposit thickness in units of $\delta_{ref} = 21.2$ mm; the contour drawn is $\Phi_{SH} = 1$, just below the derived runaway threshold $\Phi_{crit} = 1.08$, which separates slug-scoured from plugging-critical (space-time map). [mitigated]

Transient scenario 'steady' — monitor response



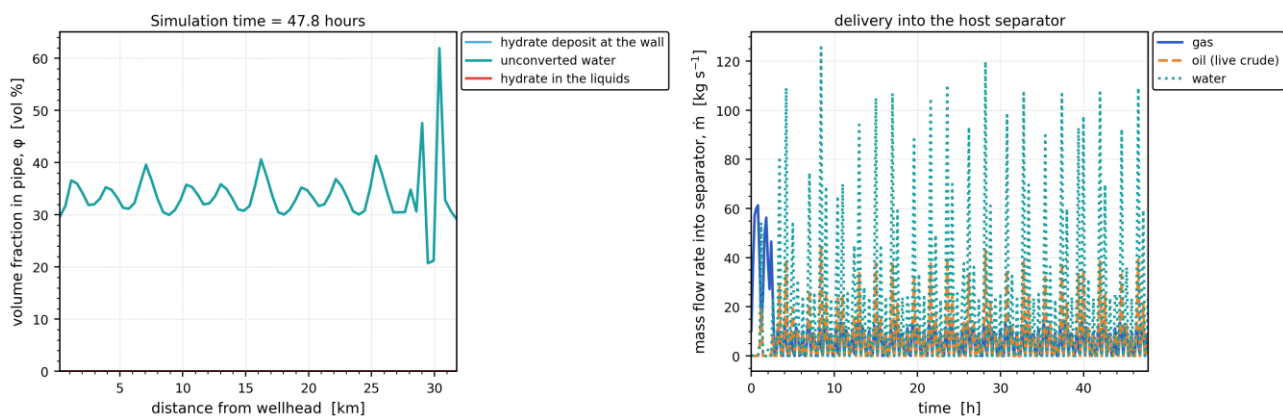
Monitored-station transient response vs time (curves). [mitigated]

Output D — wall-deposit growth (transient, coupled)



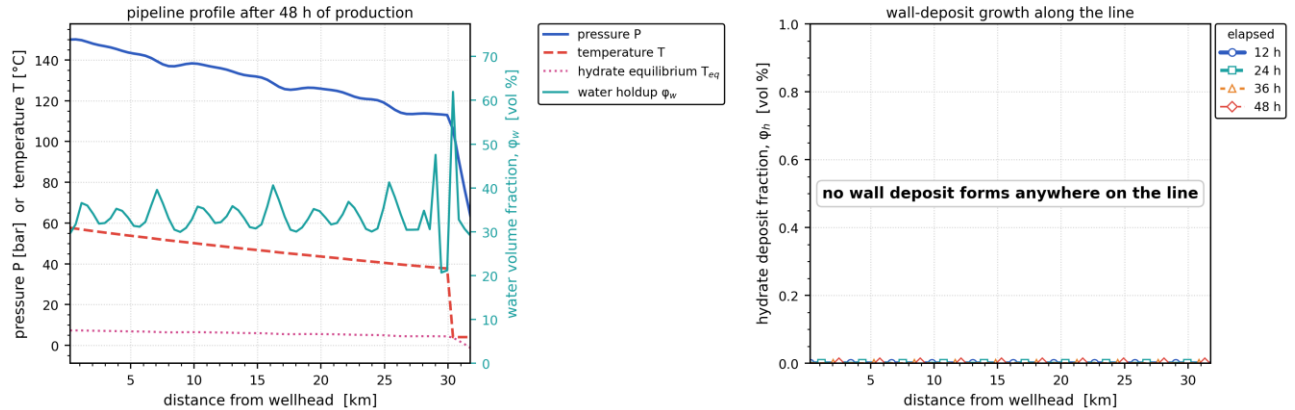
Wall-deposit growth at the monitor station vs time (curve). [mitigated]

In-pipe hydrate/water distribution and host delivery — engineered fix (insulation + MEG)

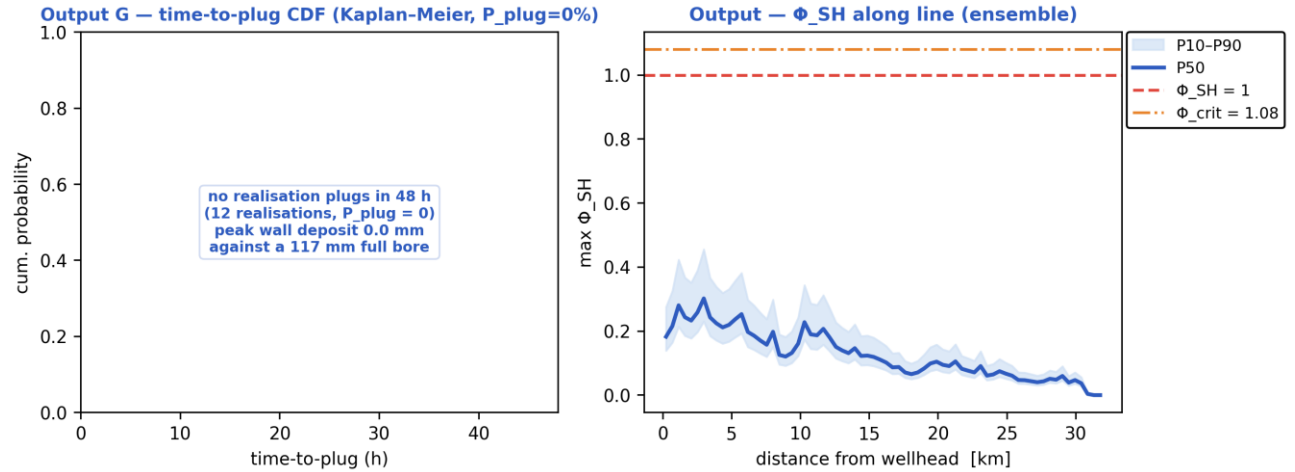


(a) In-pipe volume fractions along the route at the reported time — unconverted water, hydrate carried in the liquids, and the hydrate deposit standing on the wall; (b) the gas, oil and water mass rates delivered into the host separator against time. [mitigated]

Late-time pipeline state and deposit growth — engineered fix (insulation + MEG)

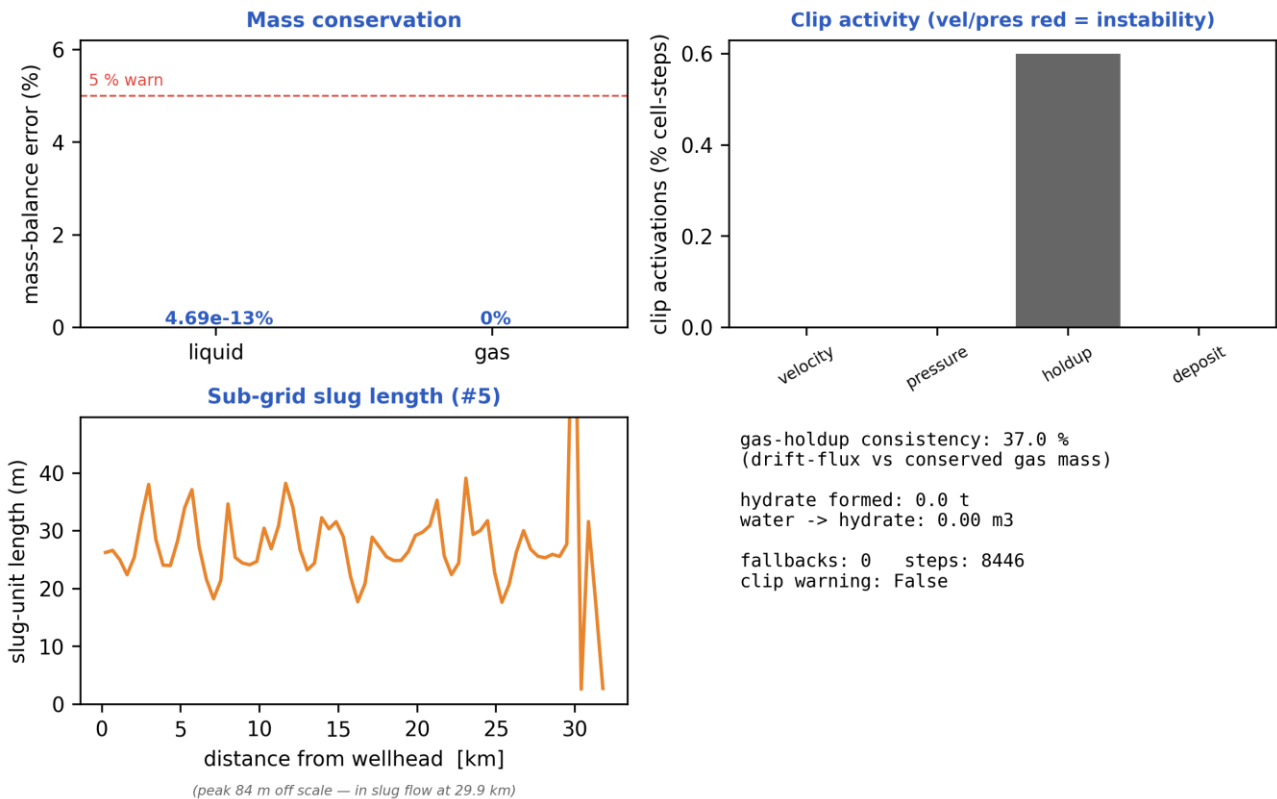


(a) The late-time pipeline profile — pressure, temperature against the hydrate equilibrium temperature, and the water volume fraction; (b) the wall-deposit volume fraction along the line at successive elapsed times. [mitigated]



Probabilistic time-to-plug CDF and the max- Φ_{SH} P10–P90 uncertainty band (Monte-Carlo ensemble). [mitigated]

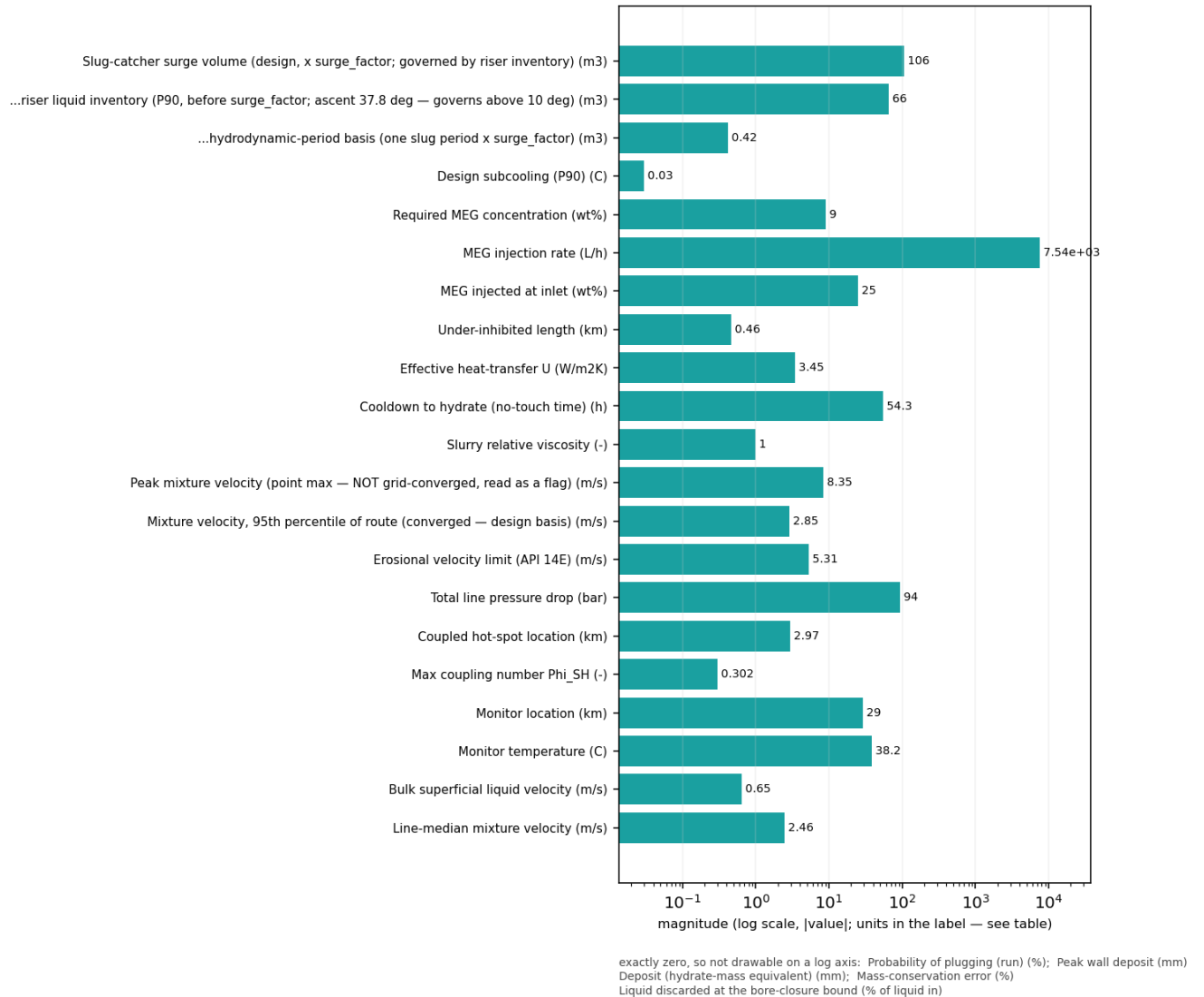
Output — solver diagnostics & balances



Solver diagnostics: liquid & gas mass balances, clip activity, slug length and numerical consistency. [mitigated]

11.3.3 Engineering deliverables (engineering_deliverables.csv)

engineering_deliverables.csv — numeric deliverables [mitigated]
(bar chart: these are unrelated named quantities, not a curve)

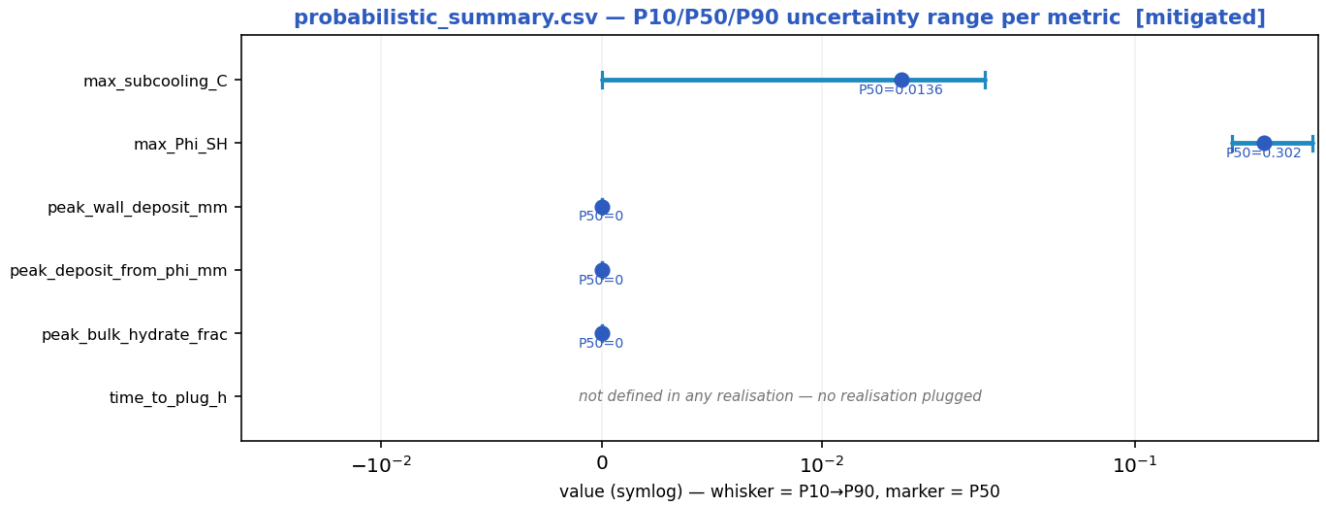


Numeric engineering deliverables as a magnitude chart [mitigated].

deliverable	value	units
Slug-catcher surge volume (design, x surge_factor; governed by riser inventory)	105.56	m3
...riser liquid inventory (P90, before surge_factor; ascent 37.8 deg — governs above 10 deg)	65.98	m3
...hydrodynamic-period basis (one slug period x surge_factor)	0.42	m3
Design subcooling (P90)	0.03	C
Required MEG concentration	9.0	wt%
MEG injection rate	7537.4	L/h
MEG injected at inlet	25.0	wt%
Under-inhibited length	0.46	km

Effective heat-transfer U	3.45	W/m2K
Cooldown to hydrate (no-touch time)	54.32	h
Slurry relative viscosity	1.00	-
Slurry transportable?	True	-
Peak mixture velocity (point max — NOT grid-converged, read as a flag)	8.35	m/s
Mixture velocity, 95th percentile of route (converged — design basis)	2.85	m/s
Slug-unit length (over the slugging reach)	27.3 mean / 84.3 max	m — over 100 % of the route
Erosional velocity limit (API 14E)	5.31	m/s
Total line pressure drop	94.0	bar
Probability of plugging (run)	0	%
Time-to-plug (P50)	n/a — nothing plugged	h
Coupled hot-spot location	2.97	km
Max coupling number Phi_SH	0.302	-
Phi_SH reported at plot cap (saturated)?	False	-
Phi_SH shear-limited (near-zero velocity, magnitude not resolved)?	False	-
Monitor location	29.03	km
Monitor temperature	38.2	C
Bulk superficial liquid velocity	0.65	m/s
Line-median mixture velocity	2.46	m/s
Peak wall deposit	0.0	mm
Wall deposit at full bore?	False	-
Deposit (hydrate-mass equivalent)	0.0	mm
No-touch time source	lumped	-
Mass-conservation error	0.00	%
Liquid discarded at the bore-closure bound	0.00	% of liquid in — a 1-D model limit once the bore shuts; not a solver error
Mass-balance reliability warning	False	-

11.3.4 Probabilistic summary (probabilistic_summary.csv)



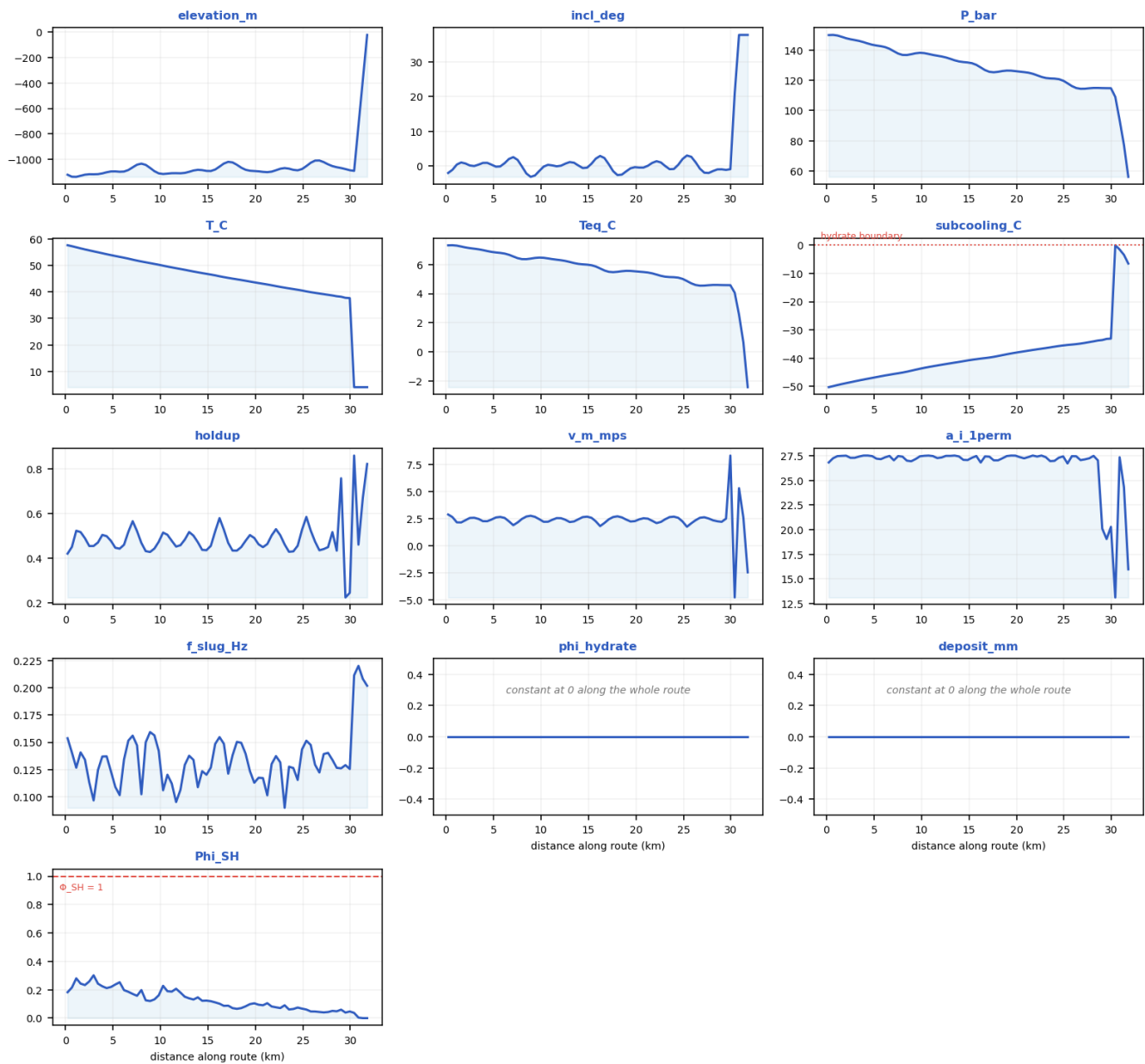
peak_wall_deposit_mm is what the transient actually deposits on the wall, capped at the full-bore radius once the bore shuts.
peak_deposit_from_phi_mm is the annulus equivalent to the peak SUSPENDED bulk hydrate fraction ϕ at one instant, $t = \phi \cdot D/4$.
These are different quantities, not a result and its bound: ϕ is instantaneous while wall deposit accumulates, so on a plugging duty the wall value exceeds the ϕ -equivalent (shut-in: 117 mm against 35 mm) and on a sub-critical one it falls well below it.

P10 → P90 uncertainty range with the P50 marker per metric [mitigated].

metric	P10	P50	P90
max_subcooling_C	0	0.0136	0.0278
max_Phi_SH	0.23	0.302	0.455
peak_wall_deposit_mm	0	0	0
peak_deposit_from_phi_mm	0	0	0
peak_bulk_hydrate_frac	0	0	0
time_to_plug_h	n/a	n/a	n/a

11.3.5 fields_profile.csv — steady-state spatial profile of every field along the route

fields_profile.csv — every column as a curve vs distance along route (km) [mitigated]



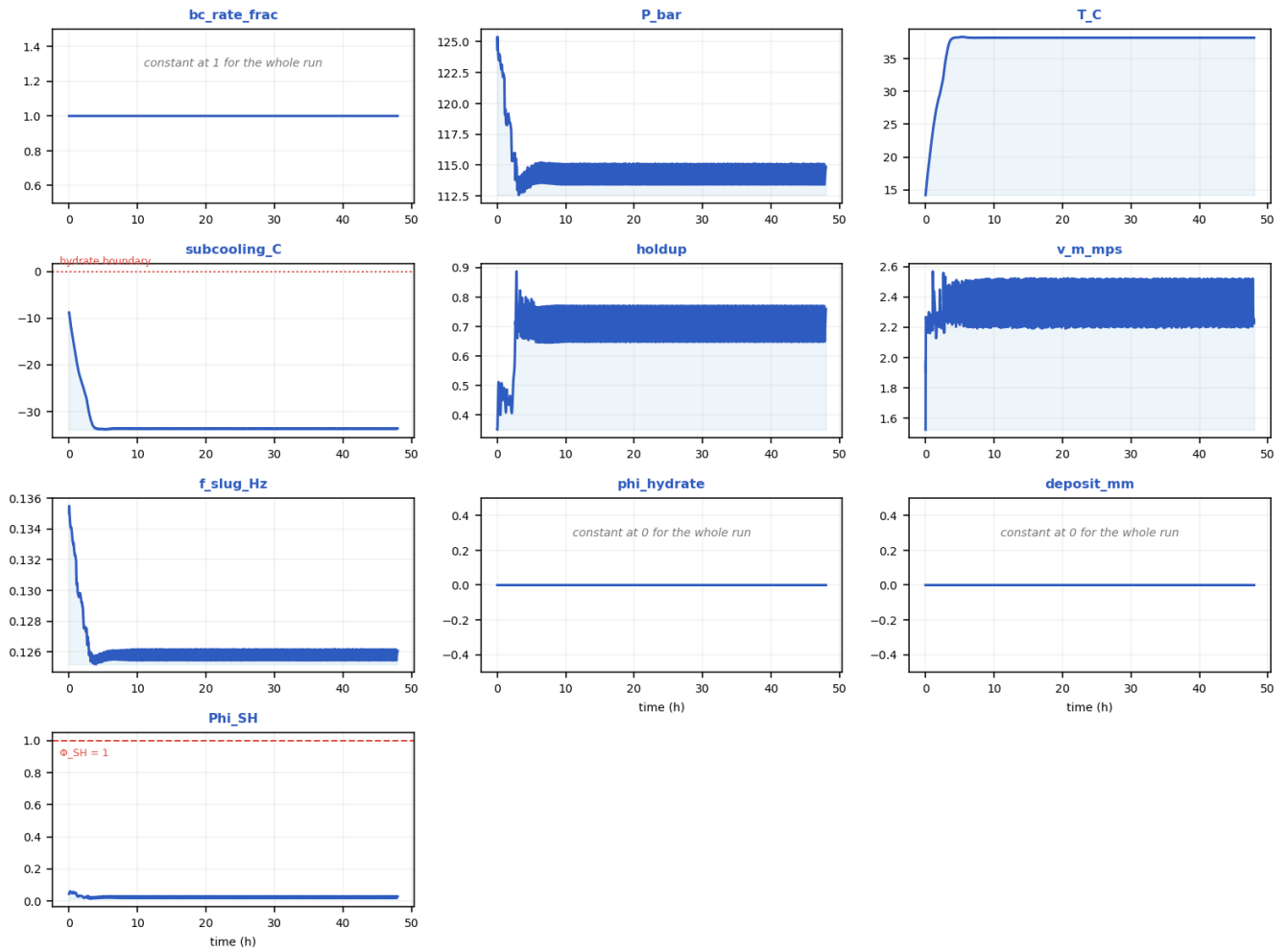
Every column of fields_profile.csv drawn as a curve [mitigated].

x_km	elevation_m	incl_deg	P_bar	T_C	Teq_C	subcooling_C	holdup	v_m_mps	regime	a_i_1perm	f_slug_Hz	phi_hydrate	deposit_mm	Phi_SH
0.22857	-1118.4	-1.9608	150	57.644	7.3136	-50.331	0.42008	2.9027	slug	26.802	0.15372	0	0	0.1825
2.5143	-1114.5	0.18865	147.23	55.702	7.1208	-48.581	0.45432	2.5842	slug	27.275	0.11313	0	0	0.25864
4.8	-1092.7	0.42362	143.61	53.911	6.8644	-47.046	0.47742	2.4375	slug	27.449	0.12318	0	0	0.21946

7.0857	-1060.5	2.601	139.28	52.195	6.5495	-45.644	0.5663	1.9128	slug	27.021	0.15606	0	0	0.17047
9.8286	-1107.7	-1.3732	138.25	50.269	6.473	-43.795	0.47355	2.4637	slug	27.428	0.14208	0	0	0.16078
12.114	-1107	0.13701	135.82	48.67	6.2915	-42.378	0.45798	2.5592	slug	27.311	0.10636	0	0	0.18014
14.4	-1082.3	-0.54271	132.12	47.128	6.0085	-41.119	0.43714	2.6623	slug	27.07	0.12349	0	0	0.12234
17.143	-1017.6	0.54352	125.65	45.316	5.4962	-39.82	0.46833	2.4571	slug	27.395	0.12114	0	0	0.087784
19.429	-1087.1	-0.61307	126.41	43.932	5.5573	-38.373	0.50369	2.2561	slug	27.503	0.12357	0	0	0.098764
21.714	-1094.2	0.93887	124.35	42.526	5.3902	-37.134	0.50187	2.3191	slug	27.505	0.13004	0	0	0.082216
24.457	-1083.8	0.40028	120.78	40.828	5.0949	-35.733	0.45574	2.5932	slug	27.289	0.1154	0	0	0.074657
26.743	-1007	-0.74006	114.39	39.409	4.5442	-34.866	0.43511	2.6004	slug	27.042	0.12226	0	0	0.043467
29.029	-1066	-0.91009	114.86	38.197	4.5859	-33.611	0.75945	2.2259	slug	20.099	0.12607	0	0	0.0598
31.771	-25	37.781	56.024	4.0505	-2.4258	-6.4922	0.82377	-2.4436	slug	15.972	0.20188	0	0	0

11.3.6 timeseries_monitor.csv — transient response at the monitored station

timeseries_monitor.csv — every column as a curve vs time (h) [mitigated]



Every column of timeseries_monitor.csv drawn as a curve [mitigated].

time_h	bc_rate_frac	P_bar	T_C	subcooling_C	holdup	v_m_mps	f_slug_Hz	phi_hydrate	deposit_mm	Phi_SH
0	1	124.37	14.205	-8.7816	0.35096	1.5238	0.13502	0	0	0.043612
2.2815	1	115.62	30.527	-25.893	0.48759	2.2255	0.12756	0	0	0.022385
5.5359	1	114.96	38.317	-33.723	0.76434	2.22	0.12596	0	0	0.028716
9.3084	1	113.57	38.204	-33.729	0.69246	2.5199	0.12552	0	0	0.018223
13.178	1	114.67	38.197	-33.628	0.73061	2.2435	0.12598	0	0	0.025442
17.047	1	113.54	38.206	-33.729	0.69284	2.5211	0.12553	0	0	0.018136
20.917	1	114.67	38.197	-33.629	0.72544	2.2412	0.12598	0	0	0.025222
24.786	1	114.04	38.205	-33.69	0.71637	2.4144	0.12571	0	0	0.021239
28.649	1	114.5	38.199	-33.645	0.70729	2.2367	0.12591	0	0	0.023764
32.521	1	114.73	38.199	-33.622	0.76021	2.336	0.12602	0	0	0.026462
36.387	1	114.27	38.203	-33.67	0.6875	2.2574	0.12581	0	0	0.021759
40.257	1	115.01	38.197	-33.599	0.76938	2.2151	0.12612	0	0	0.029004
44.122	1	114	38.206	-33.695	0.65557	2.2895	0.1257	0	0	0.019331

47.996	1	114.86	38.197	-33.611	0.75945	2.2259	0.12607	0	0	0.027626
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12. Cross-scenario summary and conclusions

12.1 As-operated vs engineered fix

Metric	As-operated	Mitigated
Max subcooling (°C)	6.34	0.0136
Peak deposit (mm)	8.81	0
Plug probability (frac)	0	0
Time-to-plug P50 (h)	n/a	n/a
MEG required (wt%)	30.7	8.96
Under-inhibited (km)	12.3	0.457
No-touch time (h)	0	54.3
Effective U (W/m ² K)	22	3.45

Shut-in no-touch time $\approx$ 0 h (source: already-subcooled).

12.2 Engineering conclusions

- As-operated the line is slug- AND hydrate-critical: sustained $\Phi_{SH} = 1.11$, 0% plug probability, P50 time-to-plug n/a h, peak deposit 9 mm.
- Inhibitor demand to clear it: MEG $\approx$ 31 wt% (33886 L/h); under-inhibited length 12.3 km.
- Shut-in gives effectively no safe window (no-touch $\approx$ 0 h).
- The engineered fix (U_{eff} 3.45 W/m²K + MEG) removes the subcooling, zeroes the plug probability and restores 54 h of no-touch time.
- Predictions are mass-consistent and stable (liquid mass error 1.76e-15, 0 fallbacks).

13. References (validation and calibration sources)

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Provenance note: the case is a representative industrial ARCHETYPE; geometry, fluid and operating values are realistic open-literature values, not proprietary operator data. The physics and predictions are produced by the real solver; the closures and hydrate thermodynamics are validated against the published data above.